Quantitative Research

Part 4: Regression

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About the Lecturer



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Professor of Mathematics, specialising in Business Data Science

- Former Technical Lead for Artificial Intelligence and supervisor of industrial doctoral projects at BASF
- Research: Data Science, Machine Learning and Artificial Intelligence
- Focus: Application of AI in business and industry

Course materials:
github.com/ChrisW09/Quantitative-Research-Methods

The lecture at a glance

	Part	Content
	1 – Descriptive Statistics	Basic concepts, levels of measurement, frequencies, measures of central tendency and dispersion
	2 – Graphical Representations	Empirical CDF, histogram, boxplot
	3 – Statistical Inference	Normal distribution, CLT, confidence intervals, sample size, hypothesis tests, p-values, one-sided tests, types of error
▶	**4 – Regression**	**Correlation, least squares, goodness of fit, inference for regression, multiple regression (appendix: residual analysis)**
	5 – Classification	Conditional probability, logistic regression, classification rules, confusion matrix, ROC/AUC, Bayes classifiers
	Python course	Separate course: Python basics, pandas, plots, machine learning (Colab notebooks)

All analyses in **Python** (NumPy, pandas, SciPy, statsmodels, matplotlib); every deck comes with a Python script and a Jupyter notebook – also in Google Colab.

Contents – Part 4

1. Relationship Between Two Metric Variables
2. Simple Linear Regression
3. Goodness of Fit: the Coefficient of Determination
4. Inference in Simple Regression
5. Multiple Linear Regression
6. Summary

▶ **Try it:** notebook, all examples and exercises (Colab)

The common thread of this part

One question runs through every section

How does a variable Y **depend** on one or several variables X – and how *certain* are we about this dependence?

Three steps:

- **Describe** – scatter plot, correlation, least-squares line, R^2 (residual and QQ-plot: appendix).
- **Infer** – standard errors, t-tests, confidence intervals and p-values for the coefficients (the tools of Part 3).
- **Extend** – several explanatory variables at once: multiple regression.

Running stories:

The lamp

How does the sales quantity react to the price? ($n = 9$ stores; simple regression and inference)

Advertising and sales

12 stores, one budget: building up least squares; later 200 markets, three media: multiple regression.

Learning objectives – Part 4

After this part you will be able to ...

- compute and interpret **covariance** and **correlation**,
- explain **least squares** and how the OLS estimators follow,
- fit a **simple regression** by hand and in Python; interpret it (units, extrapolation),
- assess the fit with R^2 and know its limits,
- carry out t-**tests**, **confidence intervals** and p-**values** for regression coefficients,
- distinguish **confidence** and **prediction intervals**,
- interpret a **multiple regression** ("holding the others fixed") and recognise **confounding**,
- use the F-**test**, adjusted R^2 and dummy variables.

Key takeaway

Common thread: a coefficient is an *estimate* – report it with its uncertainty.

→ *Appendix: derivation of OLS, residual analysis, leverage, QQ-plot, interactions, diagnostics*

List of symbols (1/2)

Symbol	Meaning	Symbol	Meaning
X, Y	Explanatory / dependent variable	(x_i, y_i)	Observation pair i
s_{xy}	Covariance of X and Y	r_{xy}	Correlation coefficient (Pearson)
β_0, β_1	Regression coefficients (true)	$\hat{\beta}_0$, $\hat{\beta}_1$	Estimated coefficients
u_i	Error term	$\hat{u}_i$	Residual $y_i - \hat{y}_i$
$\hat{y}_i$	Fitted value	h_i	Leverage value
S_{xx}, S_{yy}, S_{xy}	Sums of squares / cross-products	TSS, ESS, RSS	Total, explained, residual sum of squares
R^2	Coefficient of determination	$\bar{R}^2$	Adjusted R^2
$S(\beta_0, \beta_1)$	Objective function of least squares	p_i, z_i	Plotting position, normal quantile

List of symbols (2/2)

Symbol	Meaning	Symbol	Meaning
σ^2	Variance of the error term	$\hat{\sigma}$	Residual standard error
$\mathrm{SE}(\hat{\beta}_j)$	Standard error of $\hat{\beta}_j$	T	t-statistic of a coefficient
$t_{1-\alpha/2,\,n-2}$	Two-sided quantile, t_{n-2}	$F_{t,n-2}$	CDF of t_{n-2}
p	Number of explanatory variables	$n-p-1$	Degrees of freedom (residuals)
x_{ij}	Value of variable j for obs. i	F	F-statistic
D	Dummy variable (0/1)	VIF_j	Variance inflation factor
$x_{(i)}$, $\hat{u}_{(i)}$	i-th smallest value (sorted)	$s^2_{\hat{y}}$, $s^2_{\hat{u}}$	Explained and residual variance

As in Part 3: H_0, H_1, α, p-value and confidence level $1-\alpha$. (Context distinguishes the p-value from the number of explanatory variables p.)

Overview

1 Relationship Between Two Metric Variables
- Example and scatter plot
- Covariance and correlation

2 Simple Linear Regression

3 Goodness of Fit: the Coefficient of Determination

4 Inference in Simple Regression

5 Multiple Linear Regression

6 Summary

Relationship between two metric variables

Research question

The aim is to investigate the relationship between two metrically scaled variables.

- **Exogenous variable** X: given from outside, assumed to be independent – also called the *explanatory variable*.
- **Endogenous variable** Y: dependent variable – also called the *variable to be explained*.

Regression – an example

Scenario

A new lamp is offered for testing in several stores at different prices. The price (in euros) and the sales quantity are observed.

Price x_i (EUR)	80	85	90	95	100	105	110	115	120
Quantity y_i	20	17	15	9	11	8	4	4	2

Assignment of the variables

The price was fixed in every store and not negotiable. Therefore

- the **price** is the *exogenous* variable X,
- the **quantity** is the *endogenous* variable Y (result of buying behaviour, depending on price).

Price-sales and demand function

Price-sales function

The functional relationship $Y = f(X)$ we are looking for is called the *price-sales function*. It answers the question: *What quantity is sold at a given price?*

Demand function

The inverse function $X = f^{-1}(Y)$ is called the *demand function*. It answers the reverse question: *At what price can a fixed quantity be sold?*

In words: Regression always has a *direction*: we explain Y by X, not the other way round.

Regression – a warning

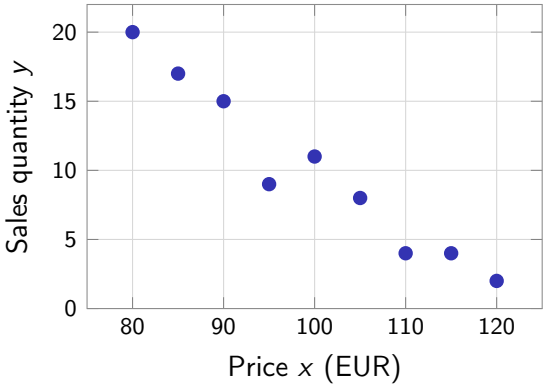
Price x_i (EUR)	80	85	90	95	100	105	110	115	120
Quantity y_i	20	17	15	9	11	8	4	4	2

Warning

- **Do not sort!**
  The values x_i and y_i must *not* be sorted independently of each other (unlike the quantiles of Part 1).
- **Pairs belong together.**
  Each pair (x_i, y_i) comes from *the same observation* (here: from the same store) – the link price $\leftrightarrow$ quantity must not be broken.

Scatter plot

Lamp example: price and sales quantity

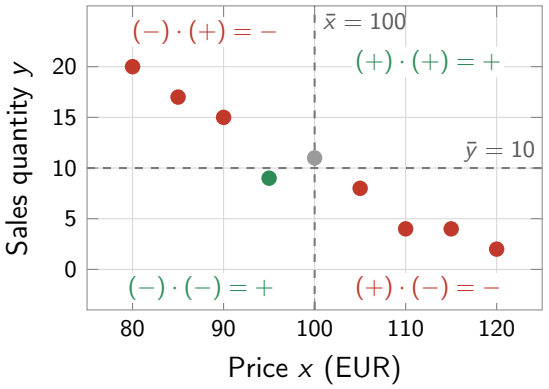


Idea

A *scatter plot* gives a first visual overview of the relationship between two metric variables X and Y.

- Each observation (x_i, y_i) is plotted as **a single point**.
- The **exogenous value** x_i (price) is on the x-axis, the **endogenous value** y_i (quantity) on the y-axis.
- The cloud shows *direction*, *strength* and *form* of the relationship – here: falling, strong, roughly linear.

Covariance: how do we measure the direction of a relationship?



Idea: four quadrants

The lines $\bar{x} = 100$ and $\bar{y} = 10$ split the lamp data into four quadrants. For every point form the product $(x_i - \bar{x})(y_i - \bar{y})$:

- top right, bottom left: product > 0
- top left, bottom right: product < 0

Key takeaway

Most products are negative (red): their sum – and hence the **covariance** – is negative.

Covariance

Idea

The *covariance* s_{xy} measures the **joint variability** of two metric variables X and Y.

Computation (sample covariance)

$$s_{xy} = \mathrm{Cov}(X, Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}), \qquad -\infty < s_{xy} < +\infty$$

Short names for the sums

$S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})$: sum of cross-products $\Rightarrow$ $s_{xy} = S_{xy}/(n-1)$
$S_{xx} = \sum(x_i - \bar{x})^2$: sum of squares $\Rightarrow$ $s_x^2 = S_{xx}/(n-1)$; S_{yy} likewise for y

In words: Same divisor $n-1$ as the sample variance of Part 1 – in fact $s_{xx} = s_x^2$: the covariance of a variable with itself is its variance.

Covariance – interpretation

Sign indicates the direction of the relationship

$s_{xy} > 0$ **Positive (concordant) relationship.** Larger values of X tend to go together with larger values of Y.

$s_{xy} < 0$ **Negative (discordant) relationship.** Larger values of X tend to go together with smaller values of Y.

$s_{xy} = 0$ **No linear relationship.** The cloud of points shows no recognisable linear structure.

Magnitude not directly interpretable

The covariance s_{xy} has the unit "unit of X · unit of Y" and depends on the scaling of the data.

Covariance – lamp example

Working table ($n = 9$; $\bar{x} = 900/9 = 100$, $\bar{y} = 90/9 = 10$)

Price x_i (EUR)	80	85	90	95	100	105	110	115	120	$\sum$
Quantity y_i	20	17	15	9	11	8	4	4	2	
$x_i - \bar{x}$	−20	−15	−10	−5	0	5	10	15	20	0
$y_i - \bar{y}$	10	7	5	−1	1	−2	−6	−6	−8	0
$(x_i - \bar{x})(y_i - \bar{y})$	−200	−105	−50	5	0	−10	−60	−90	−160	**−670**
$(x_i - \bar{x})^2$	400	225	100	25	0	25	100	225	400	1500
$(y_i - \bar{y})^2$	100	49	25	1	1	4	36	36	64	316

Covariance

$s_{xy} = S_{xy}/(n-1) = -670/8 = \mathbf{-83.75}$ (EUR · units): negative – price and quantity move in opposite directions

For later: the variances

$s_x^2 = S_{xx}/8 = 1500/8 = 187.5$

$s_y^2 = S_{yy}/8 = 316/8 = 39.5$

Exercise – online shop (1): scatter plot and covariance

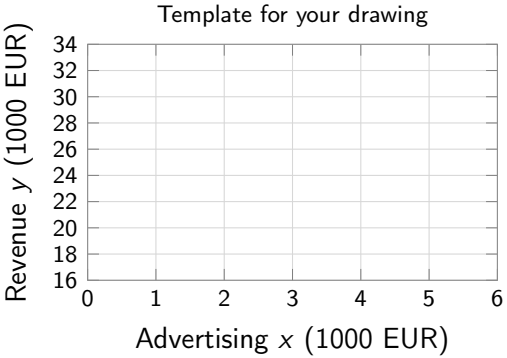
Running exercise: these data accompany the whole part – the exercises build on each other.

Problem statement

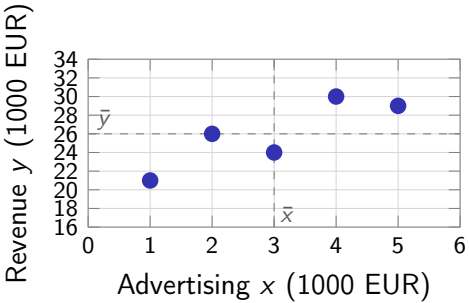
Advertising x and revenue y of an online shop in five months (both in 1000 EUR):

Month i	1	2	3	4	5
Advertising x_i	1	2	3	4	5
Revenue y_i	21	26	24	30	29

1. Draw the scatter plot; describe the relationship.
2. Compute the means $\bar{x}$ and $\bar{y}$.
3. Compute s_{xy} with a working table.



Solution – online shop (1): scatter plot and covariance



(1) More advertising tends to go with more revenue: a **positive**, roughly linear relationship.

(2) Means

$\bar{x} = 15/5 = \mathbf{3}, \qquad \bar{y} = 130/5 = \mathbf{26}$

(3) Rule: $s_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})/(n-1)$

i	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
1	1	21	-2	-5	10
2	2	26	-1	0	0
3	3	24	0	-2	0
4	4	30	1	4	4
5	5	29	2	3	6
$\sum$	15	130	0	0	$S_{xy} = \mathbf{20}$

$s_{xy} = 20/(5-1) = \mathbf{5}$ $(1000\ \text{EUR})^2$: positive.

Correlation coefficient

Idea

The correlation coefficient r_{xy} is a **normalised covariance**: it measures the strength and direction of the *linear* relationship between X and Y – *independent of the units of measurement*.

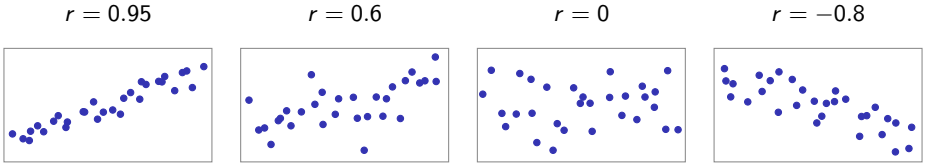
Computation

$$r_{xy} = \frac{s_{xy}}{s_x\, s_y} = \mathrm{Cov}\left(\frac{X-\bar{x}}{s_x}, \frac{Y-\bar{y}}{s_y}\right) \qquad -1 \leq r_{xy} \leq +1.$$

Reading

r_{xy} is the covariance of the *z-standardised* variables. Dividing by $s_x s_y$ cancels out the units.

Correlation coefficient – strength of the relationship



Rule of thumb

Range	Interpretation
$\lvert r_{xy}\rvert > 0.8$	strong correlation
$0.5 \leq \lvert r_{xy}\rvert \leq 0.8$	moderate correlation
$\lvert r_{xy}\rvert < 0.5$	weak correlation

$r_{xy} = \pm 1$: all points exactly on a line.

Lamp example

$s_x = \sqrt{187.5} = 13.6931, \quad s_y = \sqrt{39.5} = 6.2849$

$$r_{xy} = \frac{-83.75}{13.6931 \cdot 6.2849} = \mathbf{-0.9732}$$

strongly **negative**: the higher the price, the fewer lamps are sold.

Exercise – online shop (2): correlation coefficient

Known from exercise (1)

$n = 5$, $\bar{x} = 3$, $\bar{y} = 26$, $s_{xy} = 5$, $S_{xy} = 20$

$x_i - \bar{x}$: $-2,\ -1,\ 0,\ 1,\ 2$

$y_i - \bar{y}$: $-5,\ 0,\ -2,\ 4,\ 3$

Problem statement

1. Compute S_{xx} and S_{yy} from the deviations.
2. Compute s_x, s_y and r_{xy}; check the result with $S_{xy}/\sqrt{S_{xx}\,S_{yy}}$.
3. Assess the strength and direction of the relationship.

Tools

Sum of squares: $S_{xx} = \sum(x_i - \bar{x})^2$

Standard deviation: $s_x = \sqrt{S_{xx}/(n-1)}$

Correlation: $r_{xy} = \dfrac{s_{xy}}{s_x\, s_y}$

Rule of thumb: $|r_{xy}| > 0.8$ strong, 0.5 to 0.8 moderate

Solution – online shop (2): correlation coefficient

(1) Rule: square the deviations and add up

$S_{xx} = (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2 = \mathbf{10}$

$S_{yy} = (-5)^2 + 0^2 + (-2)^2 + 4^2 + 3^2 = \mathbf{54}$

(2) Rule: $r_{xy} = s_{xy}/(s_x\, s_y)$

$s_x = \sqrt{10/4} = 1.5811, \quad s_y = \sqrt{54/4} = 3.6742$

$$r_{xy} = \frac{5}{1.5811 \cdot 3.6742} = \mathbf{0.8607}$$

Check: $\dfrac{S_{xy}}{\sqrt{S_{xx}\, S_{yy}}} = \dfrac{20}{\sqrt{10 \cdot 54}} = 0.8607$ ✓

(3) Assessment

$|r_{xy}| = 0.86 > 0.8$: a **strong positive** linear relationship. Whether advertising *causes* the revenue cannot be read from r_{xy}.

In words: The covariance 5 depends on the units, the correlation 0.86 does not.

Correlation coefficient – limitations

No causal relationship

Correlation is **symmetric**, $r_{xy} = r_{yx}$: from r_{xy} alone one cannot tell which variable is X and which is Y.

Spurious correlation possible

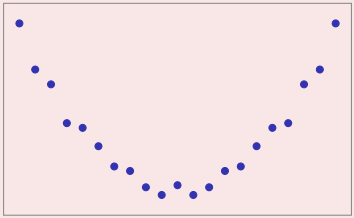
A common third variable can produce $r_{xy} \neq 0$ – e.g. ice-cream sales and sunburns, both driven by sunshine (later: *confounding* in multiple regression).

Remember

A high value of $|r_{xy}|$ is a *statistical finding*, not a *causal statement*.

Only *linear* relationships

$r = 0$



A clear U-shape – and yet $r_{xy} = 0$. Always look at the scatter plot first.

Covariance and correlation with Python

Data

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.DataFrame({"price":    [80, 85, 90, 95, 100, 105, 110, 115, 120],
                   "quantity": [20, 17, 15,  9,  11,   8,   4,   4,   2]})
plt.scatter(df["price"], df["quantity"])     # scatter plot
```

Covariance and correlation

```
df["price"].cov(df["quantity"])     # -83.75  (divides by n-1)
df["price"].corr(df["quantity"])    # -0.9732 (Pearson is the default)
```

▶ **Try it:** notebook, section 1.2 (Colab)

Overview

1 Relationship Between Two Metric Variables

2 Simple Linear Regression
- The best line
- Least squares
- Interpretation
- Exercises

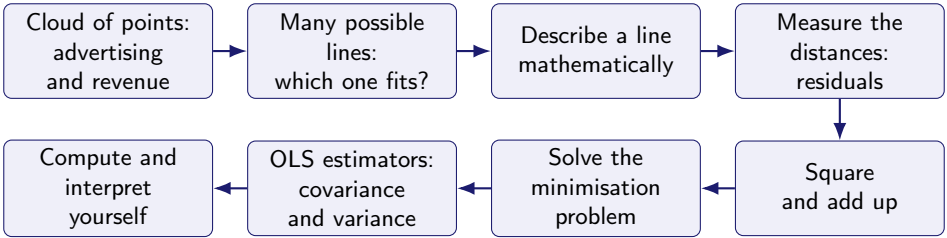
3 Goodness of Fit: the Coefficient of Determination

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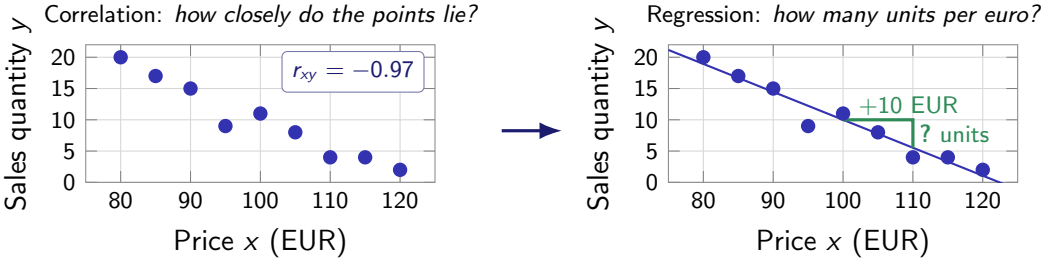
Roadmap: how do we find the best line through a cloud of points?



Key takeaway

The method of least squares is not an isolated formula but the logical answer to a simple question: *How do we find the best line through a cloud of points?*

From correlation to regression



	Correlation (so far)	**Regression** (now)
Question	How close is the relationship?	How does Y change with X? Which Y do we expect?
Roles	symmetric: $r_{xy} = r_{yx}$	directed: X explains Y
Result	one number *without* a unit	a line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, $\hat{\beta}_1$ in *units per euro*

As r_{xy} has no unit, it cannot answer "how many units more or less?" – only the slope $\hat{\beta}_1$ of a line can.

Starting question: how is advertising related to revenue?

Situation

Twelve stores: advertising budget and revenue of the last quarter (both in 1000 EUR). We develop the method on these data.

Store i	1	2	3	4	5	6	7	8	9	10	11	12
Advertising x_i (1000 EUR)	1	3	4	5	6	8	8	9	10	11	12	13
Revenue y_i (1000 EUR)	33	30	37	33	38	41	39	43	45	40	42	47

- x: exogenous, y: endogenous
- each pair (x_i, y_i): the *same* store
- $r_{xy} = 0.88$: strongly positive

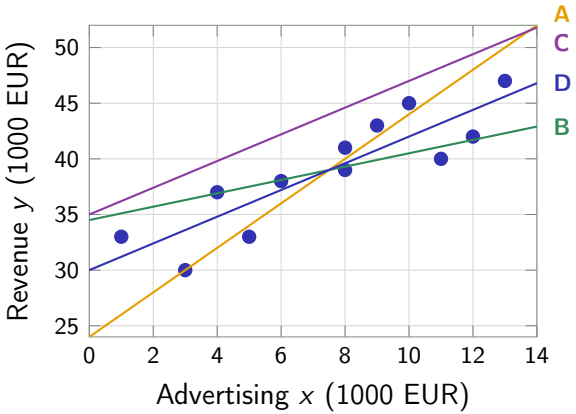
Helper values: $\bar{x} = 7.5$, $\bar{y} = 39$, $S_{xy} = 186$, $S_{xx} = 155$, $S_{yy} = 288$

Guiding question

How much more revenue goes along with 1000 EUR more advertising? Which revenue do we expect at 7000 EUR?

In "Multiple regression" the question returns – with 200 markets and three media.

Which line describes the data best?



Four candidates

A too steep

B too flat

C too high

D fits well – by eye

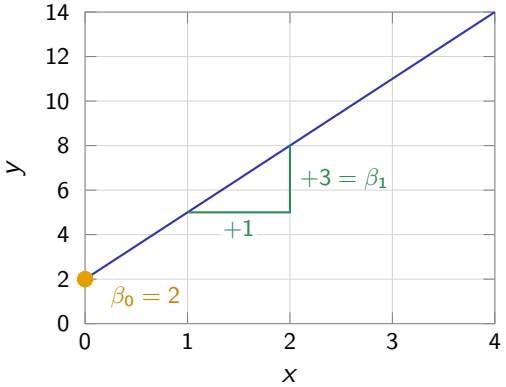
Question for you

Which line would you choose – and why?

Key takeaway

Judging by eye is **subjective**: we need a transparent **criterion**.

How do we describe a straight line mathematically?



Equation of a line

$$y = \beta_0 + \beta_1 x$$

Intercept β_0

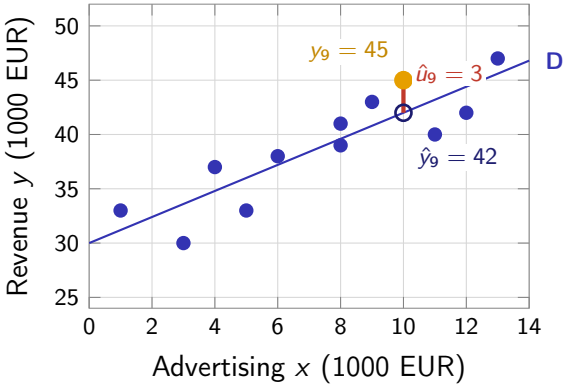
Value of y at $x = 0$, where the line crosses the y-axis (here: 2).

Slope β_1

Change of y when x increases by one unit: from $x = 1$ ($y = 5$) to $x = 2$ ($y = 8$), i.e. $+3$.

The coefficients enter with power 1 – hence **linear** regression. **Wanted:** the β_0 and β_1 that fit the twelve stores best.

Observed value, fitted value and residual



Store 9 ($x_9 = 10$), line D

- observed: $y_9 = 45$
- fitted: $\hat{y}_9 = 30 + 1.2 \cdot 10 = 42$
- residual: $\hat{u}_9 = 45 - 42 = 3$

In general

Fitted value: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$

Residual: $\hat{u}_i = y_i - \hat{y}_i$, the **vertical** distance between point and line: above the line $\hat{u}_i > 0$, below $\hat{u}_i < 0$.

The "hat" marks values computed from the data. Some textbooks write e_i for $\hat{u}_i$.

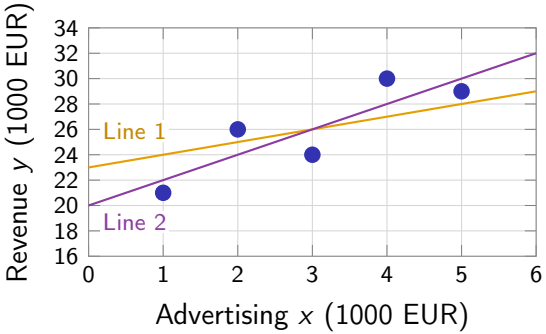
Exercise – online shop (3): residuals of two lines

Problem statement

Two employees propose a line for the online shop:

Line 1: $\hat{y} = 23 + x$
Line 2: $\hat{y} = 20 + 2\,x$

1. Compute $\hat{y}_i$ and $\hat{u}_i$ for both lines.
2. Add up the residuals of each line.
3. Which line fits better by eye? Does the sum of the residuals show it?



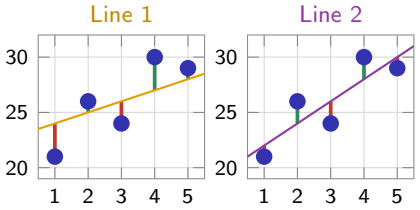
Month i	1	2	3	4	5
x_i	1	2	3	4	5
y_i	21	26	24	30	29

Solution – online shop (3): residuals of two lines

(1), (2) Rule: $\hat{u}_i = y_i - \hat{y}_i$, then add up

Month i	1	2	3	4	5	$\sum$
y_i	21	26	24	30	29	
Line 1: $\hat{y}_i = 23 + x_i$	24	25	26	27	28	
$\hat{u}_i = y_i - \hat{y}_i$	-3	1	-2	3	1	**0**
Line 2: $\hat{y}_i = 20 + 2x_i$	22	24	26	28	30	
$\hat{u}_i = y_i - \hat{y}_i$	-1	2	-2	2	-1	**0**

green: $\hat{u}_i > 0$, red: $\hat{u}_i < 0$ – they cancel out.



(3) Line 2 fits visibly better

Yet both sums are 0: the sum of the residuals does **not** detect the difference.

Why not simply add up the residuals?

Obvious idea

We look for the line whose residuals are as small as possible *overall*. "Small overall" could mean: the sum of the residuals $\sum_{i=1}^{n} \hat{u}_i$ should be as small as possible.

The problem

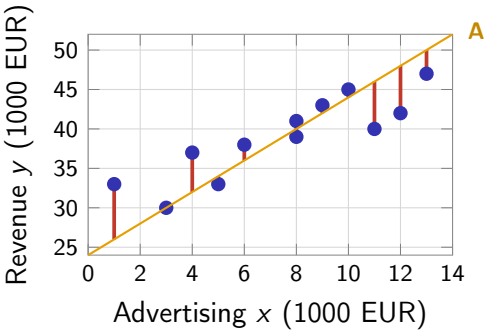
Positive and negative residuals **cancel out** – as for both lines of exercise (3). Line D of the stores: $1.8 - 3.6 + 2.2 - 3.0 + \ldots + 1.4 = 0$, although not a single point lies on the line.

Known from the variance (Part 1)

Deviations from the mean always add up to zero: $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$. Example: 2, 4, 9 with $\bar{x} = 5$, deviations -3, -1, $+4$.

This is why the variance **squares** the deviations. But first, let us check: does the problem occur for other lines, too?

The sum of residuals cannot tell good from bad lines



Line A: residuals 7, 0, 5, −1, 2, 1, −1, 1, 1, −6, −6, −3 – sum 0

Line		$\sum \hat{u}_i$
A	too steep	0
B	too flat	0
C	too high	−60
D	fits well	0

Two findings

- A, B and D: the same sum 0, although A and B fit badly.
- The higher a line lies, the more negative the sum: "smallest sum" would be absurd.

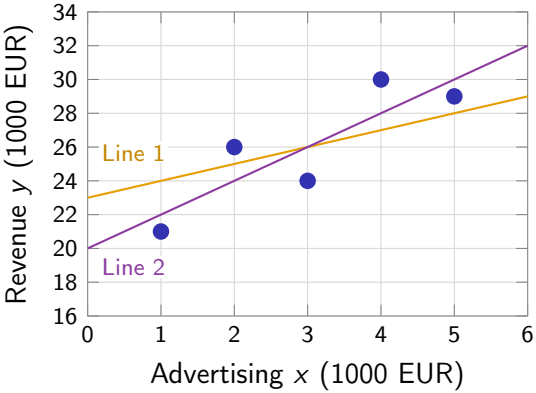
Exercise – online shop (4): squaring the residuals

Residuals $\hat{u}_i$ from exercise (3)

Month i	1	2	3	4	5
Line 1 $(23 + x)$	-3	1	-2	3	1
Line 2 $(20 + 2x)$	-1	2	-2	2	-1

Problem statement

1. Square the residuals: $\hat{u}_i^2$.
2. Add up per line: $S = \sum \hat{u}_i^2$.
3. Which line has the smaller sum? Does this agree with your judgement by eye?



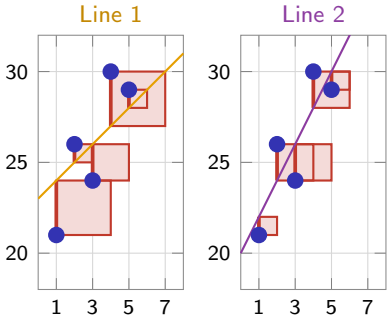
Solution – online shop (4): squaring the residuals

(1), (2) Rule: square, then add up: $S = \sum \hat{u}_i^2$

Month i		1	2	3	4	5	$\sum$
Line 1:	$\hat{u}_i$	-3	1	-2	3	1	0
	$\hat{u}_i^2$	9	1	4	9	1	**24**
Line 2:	$\hat{u}_i$	-1	2	-2	2	-1	0
	$\hat{u}_i^2$	1	4	4	4	1	**14**

(3) Line 2 has the smaller sum

$S_2 = 14 < 24 = S_1$ – as judged by eye. After squaring nothing cancels out: the better line has the **smaller sum of squares**.



Area of a square $= \hat{u}_i^2$

Gauss: the method of least squares



Carl Friedrich Gauss (1777–1855)
G. Biermann after C. A. Jensen, public domain

The idea – ordinary least squares (OLS)

1. **Square** the residuals: nothing cancels out.
2. **Add up** the squares: $\sum \hat{u}_i^2$.
3. Choose the line with the **smallest** sum.

Exactly what you did in exercise (4)

Line 2 has the smaller sum ($14 < 24$) – and it fits better.

A little history

Gauss (Göttingen) said he had used the method since 1795. The first to *publish* it was Legendre (1805), Gauss followed in 1809 – a famous priority dispute.

Squaring the residuals of the twelve stores

Squared residuals

$$\hat{u}_i^2 = \left(y_i - \hat{y}_i\right)^2$$

Line D, store	1	2	3	4	5	6	7	8	9	10	11	12	Sum
$\hat{u}_i$	1.8	−3.6	2.2	−3.0	0.8	1.4	−0.6	2.2	3.0	−3.2	−2.4	1.4	0
$\hat{u}_i^2$	3.24	12.96	4.84	9.00	0.64	1.96	0.36	4.84	9.00	10.24	5.76	1.96	**64.8**

Advantages of squaring

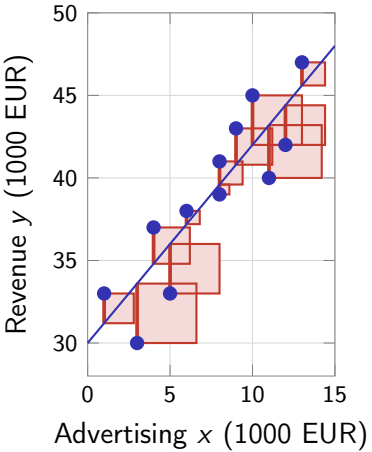
- All terms are ≥ 0: nothing cancels out.
- Large deviations count *disproportionately*: $3.6^2 = 12.96$, but $0.6^2 = 0.36$.
- The sum is easy to minimise (calculus).

The same trick as for the variance

$s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ uses *squared* deviations from the mean. Here: squared deviations from a *line*.

Flip side: a single point far away can pull the line strongly – see "leverage" in Appendix A.4.

Squared residuals as areas



Construction

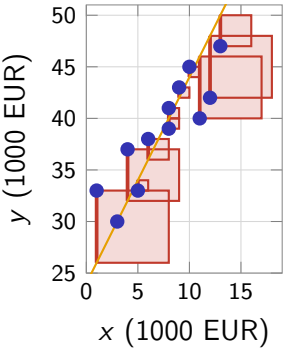
- Each red residual becomes the side of a square: side length $|\hat{u}_i|$, area $\hat{u}_i^2$.
- The sum of all areas is the sum of squared residuals, for line D: $\sum \hat{u}_i^2 = 64.8$.

Key takeaway

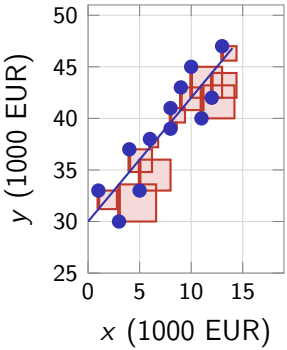
The best line is the one for which the **sum of these square areas** is as small as possible.

Scaling: both axes are in 1000 EUR and one unit has the same length on both axes. Only then are the squares really squares and their areas comparable.

Comparison: a poorly fitting line and the best line



A: $\sum \hat{u}_i^2 = \mathbf{164.0}$



D: $\sum \hat{u}_i^2 = \mathbf{64.8}$

Line	$\sum \hat{u}_i^2$
A	164.0
B	120.6
C	364.8
D	**64.8**

Key takeaway

The sum of squares separates good from bad lines. Is there an even better line than D?

The minimisation problem: the objective function

Sum of squared residuals for arbitrary values β_0 and β_1

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} \big(y_i - \beta_0 - \beta_1 x_i\big)^2 = \sum_{i=1}^{n} \Big(y_i - \underbrace{(\beta_0 + \beta_1 x_i)}_{=\hat{y}_i}\Big)^2 = \sum_{i=1}^{n} \hat{u}_i^2$$

E.g. online shop: $S(23,\, 1) = 24$, $S(20,\, 2) = 14$ (exercise (4)); stores: $S(24,\, 2) = 164.0$ (A), $S(30,\, 1.2) = 64.8$ (D).

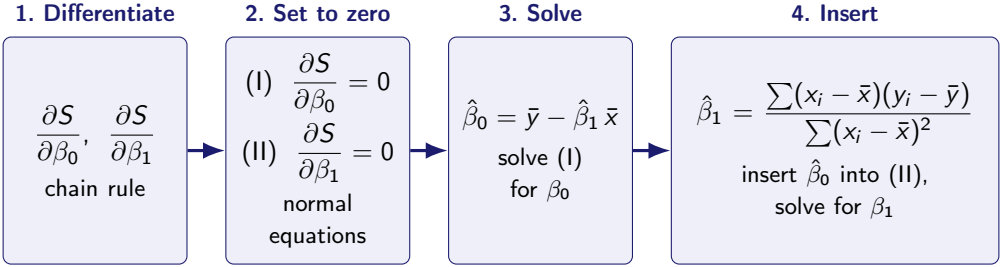
Wanted: the minimum

$$(\hat{\beta}_0, \hat{\beta}_1) \;=\; \underset{\beta_0,\, \beta_1}{\arg\min} \; \sum_{i=1}^{n} \big(y_i - \beta_0 - \beta_1 x_i\big)^2$$

In words: The coefficient values with the smallest sum of squared residuals.

At the minimum the derivative with respect to β_0 (β_1 fixed) and with respect to β_1 (β_0 fixed) is zero: the *partial* derivatives.

From the minimisation problem to the OLS estimator in four steps



Step 4 in detail: what does "insert into (II)" mean?

(II) written out and divided by -2: $\sum x_i (y_i - \beta_0 - \beta_1 x_i) = 0$ – two unknowns, β_0 and β_1.

Insert $\beta_0 = \bar{y} - \beta_1 \bar{x}$ from step 3: $\sum x_i \left(y_i - (\bar{y} - \beta_1 \bar{x}) - \beta_1 x_i\right) = 0$

Only β_1 is left $\Rightarrow$ solve for β_1: the formula top right.

▸ Every step: derivation in Appendix A.1

The OLS estimator of the slope

Solution of the minimisation problem

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

Numerator S_{xy}

Products of the deviations from the respective means, added up.

- both above or both below the mean: product > 0
- one above, one below the mean: product < 0

Positive if the cloud of points rises.

Denominator S_{xx}

Squared deviations of the x_i from their mean, added up.

- always > 0 (if the x_i are not all equal)
- measures the spread of the explanatory variable

Do the numerator and the denominator look familiar?

The link to covariance and variance

Sample covariance (Section 1)

$$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Sample variance of x (Part 1)

$$s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

The estimator, expanded by $\frac{1}{n-1}$

$$\hat{\beta}_1 = \frac{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{s_{xy}}{s_x^2}$$

The factor $\frac{1}{n-1}$ appears in the numerator *and* the denominator and cancels. What remains are two familiar statistics.

$$\boxed{\hat{\beta}_1 = \frac{s_{xy}}{s_x^2}}$$

Slope = covariance over variance

Key takeaway

The slope of the regression line follows directly from two statistics we already know: the **covariance** of x and y and the **variance** of x.

Intuition

- s_{xy} describes the *joint* variation of x and y: its sign = direction of the relationship.
- s_x^2 accounts for how much x itself varies.
- Their ratio: how much co-movement of y is there per unit of variation in x? That is the slope.

Twelve stores

$S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y}) = 186,$
$S_{xx} = \sum(x_i - \bar{x})^2 = 155$

$\hat{\beta}_1 = 186/155 = 1.2$

Via the statistics: $s_{xy} = 186/11 = 16.91$, $s_x^2 = 155/11 = 14.09$ and hence also $\hat{\beta}_1 = 16.91/14.09 = 1.2$.

For the slope, the plain sums are enough.

Exercise – online shop (5): the OLS line of the online shop

Known from exercises (1) and (2)

$\bar{x} = 3, \;\; \bar{y} = 26, \;\; s_{xy} = 5, \;\; s_x^2 = 2.5 \qquad (S_{xy} = 20, \;\; S_{xx} = 10)$

Problem statement

1. Compute the slope $\hat{\beta}_1 = s_{xy}/s_x^2$ and check it with S_{xy}/S_{xx}.
2. Compute the intercept $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ (step 3) and write down the OLS line.
3. Which of the two lines of exercise (3) is the OLS line? What does this mean for its sum of squared residuals 14?
4. Interpret the slope in words.

Solution – online shop (5): the OLS line of the online shop

(1) Rule: $\hat{\beta}_1 = s_{xy}/s_x^2 = S_{xy}/S_{xx}$

$$\hat{\beta}_1 = \frac{5}{2.5} = \mathbf{2}, \qquad \text{check: } \frac{20}{10} = 2 \checkmark$$

(2) Rule: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

$$\hat{\beta}_0 = 26 - 2 \cdot 3 = \mathbf{20} \;\Rightarrow\; \hat{y} = 20 + 2x$$

(3) Line 2 is the OLS line

Line 2 of exercise (3): its sum of squares 14 is the **smallest possible** (line 1: 24).

(4) The slope in words

+1000 EUR advertising: +2000 EUR revenue on average.

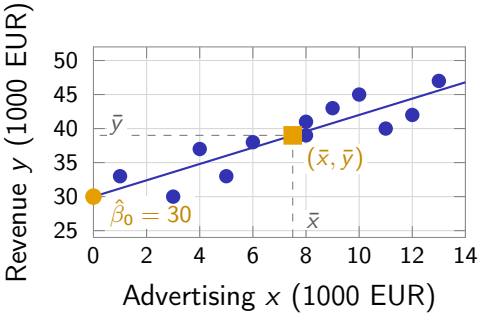
In words: Check: the line passes through the centroid $(\bar{x}, \bar{y}) = (3, 26)$, as $20 + 2 \cdot 3 = 26$.

Check with Python

```
x = pd.Series([1, 2, 3, 4, 5]); y = pd.Series([21, 26, 24, 30, 29])
((y - (20 + 2 * x))**2).sum(), ((y - (23 + x))**2).sum()   # 14  24: OLS line, line 1
```

▶ **Try it:** notebook, section 2.2 (Colab)

Intercept, OLS line and the answer



From the first normal equation

$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \iff \bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$: the line passes through the **centroid** $(\bar{x}, \bar{y})$.

Twelve stores: both estimators

$\hat{\beta}_1 = S_{xy}/S_{xx} = 186/155 = \mathbf{1.2}$,
$\hat{\beta}_0 = 39 - 1.2 \cdot 7.5 = \mathbf{30}$
Regression line: $\hat{y} = 30 + 1.2\,x$

Answer to the starting question

1000 EUR more advertising goes along with 1200 EUR more revenue on average. At 7000 EUR advertising: $\hat{y} = 30 + 1.2 \cdot 7 = \mathbf{38.4}$, i.e. 38 400 EUR.

Every line through $(\bar{x}, \bar{y})$ has residual sum 0 – also A, B and both lines of exercise (3).

Both estimators at a glance

The solution of the minimisation problem

$$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{S_{xy}}{S_{xx}} \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

The estimated regression line

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \qquad \text{for the twelve stores:} \quad \hat{y}_i = 30 + 1.2\, x_i$$

Line D *was* the OLS line: no other line has a smaller sum of squared residuals than 64.8.

Key takeaway

The initial problem is solved: from $\bar{x}$, $\bar{y}$, s_{xy} and s_x^2 we obtain the unique best line – best in the sense of the smallest sum of squared residuals.

Properties: through $(\bar{x}, \bar{y})$; $\sum \hat{u}_i = 0$ and $\sum x_i \hat{u}_i = 0$ (first-order conditions); slope with the sign of s_{xy}, 0 exactly if $s_{xy} = 0$.

▸ Derivation

Step by step to the regression line

Recipe

1. Means $\bar{x}$, $\bar{y}$
2. Covariance $s_{xy} = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$
3. Variance $s_x^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$
4. Slope $\hat{\beta}_1 = s_{xy}/s_x^2$
5. Intercept $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$
6. Line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

Means first: all further steps need them.

Working table for steps 2 and 3

i	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1						
2						
$\vdots$						
n						
$\sum$			0	0	$S_{xy} \to s_{xy}$	$S_{xx} \to s_x^2$

Check: the columns $x_i - \bar{x}$ and $y_i - \bar{y}$ add up to 0.

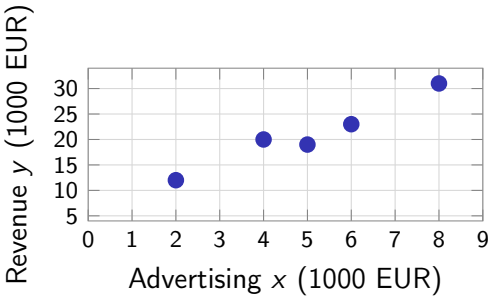
Exercise – online shop (6): the OLS line for a second shop

Problem statement

The retailer opens a second online shop: five months, both variables in 1000 EUR.

1. Compute $\bar{x}$, $\bar{y}$, S_{xy} and S_{xx} with a working table.
2. Determine $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$; write down the regression line.
3. Interpret the slope in words.
4. Compute $\hat{y}_i$ and $\hat{u}_i$; check $\sum \hat{u}_i = 0$.
5. Draw the line into the scatter plot.

Month i	1	2	3	4	5
Advertising x_i	2	4	5	6	8
Revenue y_i	12	20	19	23	31



Solution (1/2) – online shop (6): working table and OLS estimates

(1) Rule: means first, then deviations, products and squares

i	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1	2	12	-3	-9	27	9
2	4	20	-1	-1	1	1
3	5	19	0	-2	0	0
4	6	23	1	2	2	1
5	8	31	3	10	30	9
$\sum$	25	105	0	0	$S_{xy} = \mathbf{60}$	$S_{xx} = \mathbf{20}$

$\bar{x} = 25/5 = \mathbf{5}$, $\bar{y} = 105/5 = \mathbf{21}$; check: the deviations add up to 0.

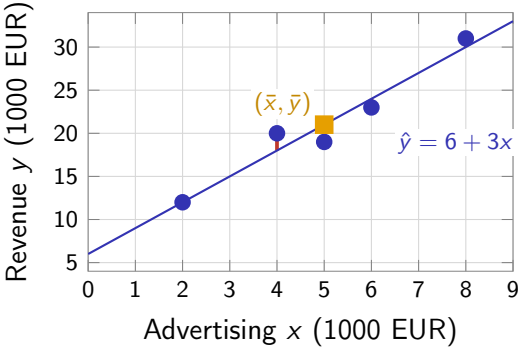
(2) Rule: $\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

$\hat{\beta}_1 = 60/20 = \mathbf{3}$, $\hat{\beta}_0 = 21 - 3 \cdot 5 = \mathbf{6} \Rightarrow \hat{y} = 6 + 3x$

(3) The slope in words

1000 EUR more advertising: 3000 EUR more revenue (on average).

Solution (2/2) – online shop (6): residuals and graph



(5) The line passes through the centroid (5, 21).

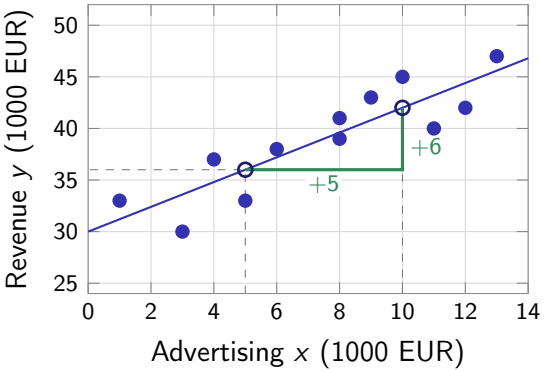
(4) Rule: $\hat{y}_i = 6 + 3\,x_i$, $\hat{u}_i = y_i - \hat{y}_i$

i	x_i	y_i	$\hat{y}_i$	$\hat{u}_i$
1	2	12	$6 + 3 \cdot 2 = 12$	0
2	4	20	$6 + 3 \cdot 4 = 18$	2
3	5	19	$6 + 3 \cdot 5 = 21$	-2
4	6	23	$6 + 3 \cdot 6 = 24$	-1
5	8	31	$6 + 3 \cdot 8 = 30$	1
$\sum$				**0** ✓

For comparison: the first shop

$\hat{y} = 20 + 2\,x$ (exercise (5)): there 1000 EUR more advertising goes with 2000 EUR, here with 3000 EUR more revenue.

How do we interpret the slope?



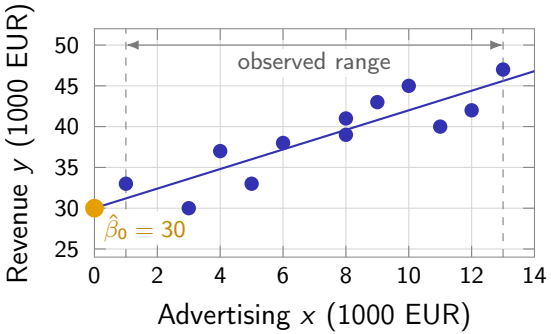
Slope $\hat{\beta}_1 = 1.2$

If x increases by one unit, the value of y on the line changes by $\hat{\beta}_1$ units. Here: advertising $5 \to 10$ (in 1000 EUR), $\hat{y}$ rises from 36 to 42, i.e. by $6 = 5 \cdot 1.2$. **Each additional** 1000 **euros of advertising goes along with** 1200 **euros more revenue on average.**

No causal statement

The line describes a statistical association. It does not follow that advertising *causes* the revenue – e.g. stores in busy locations might both advertise more and sell more.

How do we interpret the intercept?



Intercept $\hat{\beta}_0 = 30$

$\hat{\beta}_0 = \hat{y}\,\big|_{x=0}$: the value of the line at $x = 0$ – a quarterly revenue of 30 000 euros *without* advertising, according to the line.

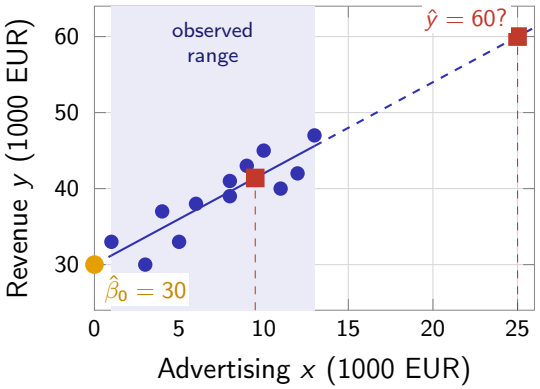
Careful

$x = 0$ was *not observed* (data: 1 to 13): the line is *extrapolated*. If $x = 0$ lies far from the data, $\hat{\beta}_0$ merely fixes the height of the line.

Key takeaway

Interpret the slope as an average change *within* the data range – and the intercept only if $x = 0$ is a meaningful value that lies near the data.

Interpretation at a glance – and predictions



$\hat{y} = 30 + 1.2\,x$ (twelve stores)

$\hat{\beta}_0$ height of the line at $x = 0$

$\hat{\beta}_1$ $\Delta\hat{y} = \hat{\beta}_1 \cdot \Delta x$ (Δ: change), e.g. $\Delta x = 5$: $\Delta\hat{y} = 6$

Prediction: insert x into the line

$x = 9.5$: $\hat{y} = 30 + 1.2 \cdot 9.5 = 41.4$ – **interpolation** (inside the observed range 1 to 13): plausible

$x = 25$: $\hat{y} = 30 + 1.2 \cdot 25 = 60$ – **extrapolation**: linearity was never observed there!

The same change Δx gives the same $\Delta\hat{y}$ everywhere – a line has only one slope.

Second example: the lamp – calculation

From the working table of Section 1 (slide "Covariance and correlation – lamp example"): $n = 9$, $\sum(x_i - \bar{x})(y_i - \bar{y}) = -670$, $\sum(x_i - \bar{x})^2 = 1500$.

Steps 1 to 3

$\bar{x} = 900/9 = 100, \quad \bar{y} = 90/9 = 10$

$s_{xy} = -670/8 = -83.75$

$s_x^2 = 1500/8 = 187.5$

Steps 4 to 6

$\hat{\beta}_1 = -83.75/187.5 = \mathbf{-0.4467}$
$(= -670/1500)$

$\hat{\beta}_0 = 10 - (-0.4467) \cdot 100 = \mathbf{54.67}$

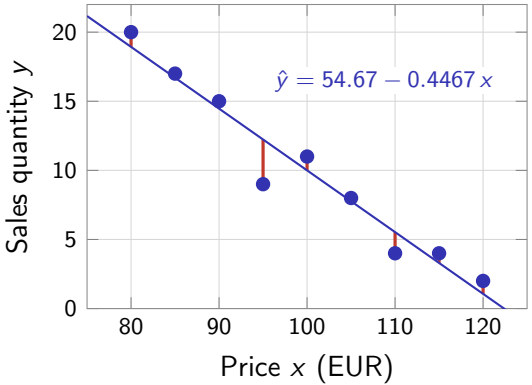
$\hat{y} = 54.67 - 0.4467\, x$

Estimated regression line for the lamp

$$\hat{y}_i = 54.67 - 0.4467\, x_i$$

In words: The negative covariance makes the slope negative – the higher the price, the lower the fitted sales quantity.

The lamp: interpreting the coefficients



Slope $\hat{\beta}_1 = -0.4467$

If the price rises by 1 euro, the sales quantity falls *on average* by 0.4467 units – or: a price increase of 10 euros costs about 4.5 lamps. A classical downward-sloping **price-sales function**.

Intercept $\hat{\beta}_0 = 54.67$

Formally the sales at a price of 0 euros. This value lies *far* outside the observed range (80 to 120 euros): here $\hat{\beta}_0$ has **no substantive meaning**. Statements outside the data range (*extrapolation*) are generally unreliable.

Linear regression with Python

Coefficients by hand (steps 2 to 5)

```
beta1 = df["price"].cov(df["quantity"]) / df["price"].var()    # -0.4467
beta0 = df["quantity"].mean() - beta1 * df["price"].mean()    # 54.6667
```

Regression with `smf.ols()` (ordinary least squares)

```
import statsmodels.formula.api as smf
reg = smf.ols("quantity ~ price", data=df).fit()  # quantity explained by price
reg.params                                        # Intercept 54.6667, price -0.4467
reg.resid.sum()                                   # 0 (up to rounding): first-order condition
plt.scatter(df["price"], df["quantity"]); plt.xlim(0, 125); plt.ylim(0, 60)
plt.axline((0, reg.params["Intercept"]), slope=reg.params["price"])
```

In words: `smf.ols()` solves the minimisation problem of this section; `"y ~ x"` reads "y explained by x".

▶ **Try it:** notebook, section 2.3 (Colab)

Can you answer these questions?

1. What does regression add to correlation – why a straight line?
2. How is a line described mathematically?
3. What do the intercept and the slope mean?
4. Why do we need an objective criterion for the best line?
5. What are residuals, and why are they squared?
6. Which minimisation problem does least squares solve?
7. Which four steps lead to the OLS estimators?
8. Why is the slope the ratio of covariance and variance?
9. How do you compute a regression line by hand for a few observations?
10. How do you interpret the estimated coefficients – and what must you not read into them?

Key takeaway

The OLS line is the line with the smallest sum of squared residuals – computed from $\bar{x}$, $\bar{y}$, s_{xy} and s_x^2.

Exercise – interpreting coefficients and making predictions

Data: furniture store, six weeks (both in 1000 EUR)

Advertising x_i	2	4	6	8	10	12
Weekly revenue y_i	44	49	58	62	61	71

Python gives the OLS line $\hat{y} = 40 + 2.5\,x$.

Tasks

1. Interpret $\hat{\beta}_0 = 40$ and $\hat{\beta}_1 = 2.5$ in context (with units). Is $\hat{\beta}_0$ meaningful?
2. What weekly revenue is expected for advertising of 5 and of 30 thousand euros? How reliable are the two predictions?
3. By how much does the expected revenue change with 2000 euros more advertising – from 4 to 6 and from 10 to 12?
4. Advertising is now measured in euros instead of thousands of euros: what are $\hat{\beta}_0$ and $\hat{\beta}_1$? And if revenue is also measured in euros?

Solution (1/3) – interpreting coefficients and making predictions

Rule: $\hat{\beta}_1$ is the change of $\hat{y}$ per unit of x; $\hat{\beta}_0$ is the value of $\hat{y}$ at $x = 0$; a change Δx gives $\Delta\hat{y} = \hat{\beta}_1 \cdot \Delta x$.

(1) Slope $\hat{\beta}_1 = 2.5$

Each additional 1000 euros of advertising goes along with 2500 **euros more weekly revenue** on average – a statistical association, not a proof of causality.

(1) Intercept $\hat{\beta}_0 = 40$

Revenue in a week *without* advertising according to the line: 40 000 **euros**. $x = 0$ was not observed (data: 2 to 12) but lies close: a rough orientation, strictly speaking an extrapolation.

(3) Change by $\Delta x = 2$

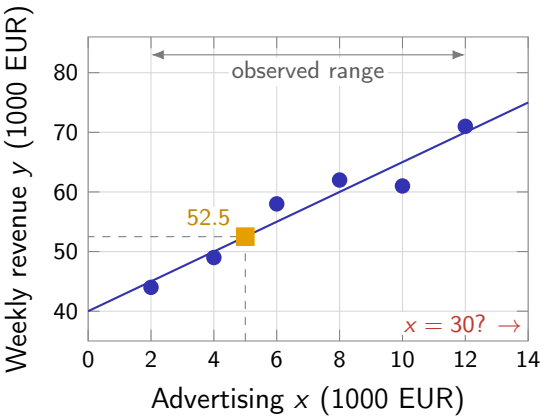
$\Delta\hat{y} = 2.5 \cdot 2 = \mathbf{5}$ (i.e. 5000 euros) – in both cases:

$4 \to 6$: $\hat{y}$ from $40 + 2.5 \cdot 4 = 50$ to 55

$10 \to 12$: $\hat{y}$ from 65 to 70

A line has the same slope everywhere: the increase does not depend on the starting level, and $\hat{\beta}_0$ plays no role.

Solution (2/3) – interpreting coefficients and making predictions



(2) Rule: insert x into the line

$x = 5$: $\hat{y} = 40 + 2.5 \cdot 5 = \mathbf{52.5}$ (52 500 euros)

$x = 30$: $\hat{y} = 40 + 2.5 \cdot 30 = \mathbf{115}$ (115 000 euros)

$x = 5$: interpolation

Inside the observed range (2 to 12): plausible – as an expected value, not as a certain value (residuals up to ± 4).

$x = 30$: extrapolation

Far outside the data: whether the relationship is still linear there (saturation?) was never observed. **Not reliable.**

Solution (3/3) – interpreting coefficients and making predictions

(4) Rule: the unit of x changes the slope, the unit of y both coefficients

Advertising in euros, $x_{\text{EUR}} = 1000\,x$: $\hat{y} = 40 + \frac{2.5}{1000}\,x_{\text{EUR}}$, so $\hat{\beta}_1 = \mathbf{0.0025}$, $\hat{\beta}_0 = \mathbf{40}$ unchanged.
Revenue also in euros (times 1000): $\hat{y}_{\text{EUR}} = \mathbf{40\,000} + \mathbf{2.5}\,x_{\text{EUR}}$ – 2.50 euros more revenue per euro of advertising: **the same statement**.

Check with Python

```
fs = pd.DataFrame({"adv": [2, 4, 6, 8, 10, 12], "revenue": [44, 49, 58, 62, 61, 71]})
reg = smf.ols("revenue ~ adv", data=fs).fit()
reg.params                                          # 40.0  2.5
reg.predict(pd.DataFrame({"adv": [5, 30]}))         # 52.5  115.0
smf.ols("revenue ~ I(1000 * adv)", data=fs).fit().params   # 40.0  0.0025
```

Key takeaway

The number $\hat{\beta}_1$ depends on the units – the substantive statement does not.

▶ **Try it:** notebook, section 2.4 (Colab)

Exercise – a falling price–demand line

Problem statement

A retailer studies how the weekly demand y (units sold) depends on the price x (in euros) in five weeks:

Price x_i (EUR)	10	12	14	16	18
Units sold y_i	64	50	46	42	38

Given: $\bar{x} = 14$, $\bar{y} = 48$, $S_{xx} = \sum (x_i - \bar{x})^2 = 40$, $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = -120$.

1. Compute $\hat{\beta}_1$ and $\hat{\beta}_0$ and write down the fitted line.
2. Interpret $\hat{\beta}_1$ in one sentence. Is its sign plausible?
3. Predict the demand at 15 euros and at 42 euros; judge both.
4. Compute the residual of week 1, $(x_1, y_1) = (10, 64)$. Above or below the line?
5. Which Python command predicts the demand at $x = 15$ from the fitted model `reg = smf.ols("units ~ price", data=dem).fit()`?

Solution (1/2) – a falling price–demand line

(1) Coefficients and fitted line

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-120}{40} = -\mathbf{3}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 48 - (-3) \cdot 14 = 48 + 42 = \mathbf{90}, \qquad \hat{y} = 90 - 3\,x$$

(2) Interpretation

If the price rises by 1 euro, the weekly demand falls **on average by** 3 **units**. The negative sign is economically plausible: a downward-sloping price–demand line, as for the lamp.

Common mistake

Sign errors in $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$: with a negative slope, *minus times minus* gives $48+42$, not $48 - 42$.

Solution (2/2) – a falling price–demand line

(3) Predictions

$\hat{y}(15) = 90 - 3 \cdot 15 = \mathbf{45}$ units – inside the observed range $[10, 18]$: plausible.

$\hat{y}(42) = 90 - 3 \cdot 42 = -\mathbf{36}$ units – **worthless**: a price of 42 euros lies far outside $[10, 18]$ (extrapolation), and a negative demand is impossible.

(4) Residual of week 1

$\hat{y}_1 = 90 - 3 \cdot 10 = 60$,
$\hat{u}_1 = y_1 - \hat{y}_1 = 64 - 60 = +\mathbf{4}$:
the point lies **above** the line.

(5) Prediction with Python

```
reg.predict(pd.DataFrame({"price": [15]}))      # 45.0
```

The new data frame needs a column with the *same name* as the explanatory variable in the formula.

▶ **Try it:** notebook, section 2.4 (Colab)

Overview

1 Relationship Between Two Metric Variables

2 Simple Linear Regression

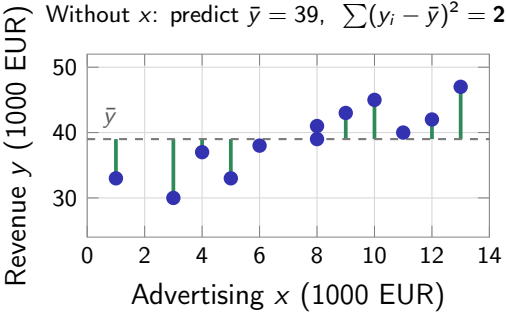
3 Goodness of Fit: the Coefficient of Determination
- Variance decomposition
- Coefficient of determination
- Useful transformations
- Exercises

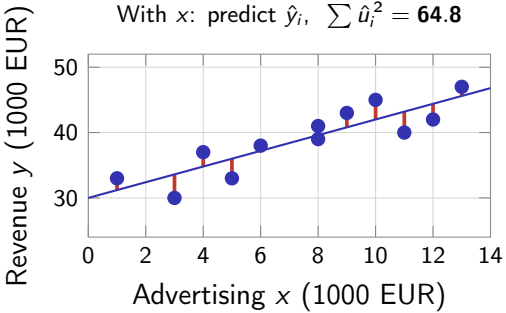
4 Inference in Simple Regression

5 Multiple Linear Regression

6 Summary

How much does the line explain?



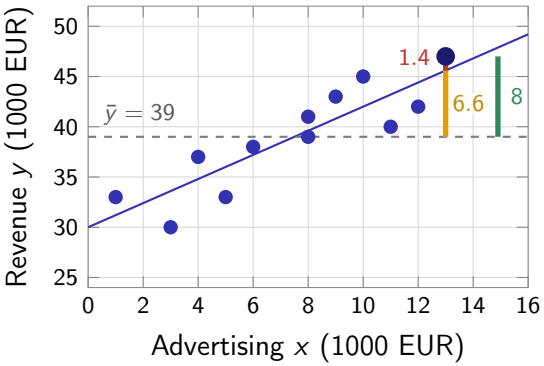


Guiding question: without x, $\bar{y}$ is the best guess for every store. How much better is the line?

Key takeaway

The sum of squares falls from 288 to 64.8: the line "explains" $223.2/288 = \mathbf{77.5}\,\%$ of the variation.

Splitting one deviation into two parts



Store 12: $x_{12} = 13$, $y_{12} = 47$

- total deviation: $y_{12} - \bar{y} = 47 - 39 = 8$
- explained by the line: $\hat{y}_{12} - \bar{y} = 45.6 - 39 = 6.6$
- not explained (residual): $y_{12} - \hat{y}_{12} = 47 - 45.6 = 1.4$

In general

$$\underbrace{y_i - \bar{y}}_{\text{total}} = \underbrace{\hat{y}_i - \bar{y}}_{\text{explained}} + \underbrace{y_i - \hat{y}_i}_{\text{residual}}$$

In words: Deviation from the average = part explained by advertising + rest.

Sum-of-squares decomposition: square and add up

Decomposition of the sums of squares (holds for the OLS line)

$$\underbrace{\sum_{i=1}^{n}(y_i-\bar{y})^2}_{\text{TSS (total)}} = \underbrace{\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2}_{\text{ESS (explained)}} + \underbrace{\sum_{i=1}^{n}\hat{u}_i^2}_{\text{RSS (residual)}}$$

Store i	1	2	3	4	5	6	7	8	9	10	11	12	Sum
$(y_i-\bar{y})^2$	36	81	4	36	1	4	0	16	36	1	9	64	**288**
$(\hat{y}_i-\bar{y})^2$	60.84	29.16	17.64	9.00	3.24	0.36	0.36	3.24	9.00	17.64	29.16	43.56	**223.2**
$\hat{u}_i^2$	3.24	12.96	4.84	9.00	0.64	1.96	0.36	4.84	9.00	10.24	5.76	1.96	**64.8**

Twelve stores

$288 = 223.2 + 64.8$ ✓

Also called **analysis of variance (ANOVA)**.

Careful

The decomposition holds for the *sums*, not for each store: store 1 has $36 \neq 60.84 + 3.24$.

Exercise – online shop (7): the sum-of-squares decomposition

Known

Exercise (1): $\bar{y} = 26, \quad y_i - \bar{y}$: $-5,\ 0,\ -2,\ 4,\ 3$

OLS line $\hat{y} = 20 + 2x$: $\hat{y}_i$: $22,\ 24,\ 26,\ 28,\ 30, \quad \hat{u}_i$: $-1,\ 2,\ -2,\ 2,\ -1$

Problem statement

1. Compute TSS, ESS and RSS.
2. Check: does $\text{TSS} = \text{ESS} + \text{RSS}$ hold?
3. Does the decomposition also hold month by month?

$$\text{TSS} = \sum (y_i - \bar{y})^2 \qquad \text{ESS} = \sum (\hat{y}_i - \bar{y})^2 \qquad \text{RSS} = \sum \hat{u}_i^2$$

Solution – online shop (7): the sum-of-squares decomposition

(1) Rule: square the deviations and add up

Month i	1	2	3	4	5	Sum
$y_i - \bar{y}$	-5	0	-2	4	3	
$(y_i - \bar{y})^2$	25	0	4	16	9	TSS $=$ **54**
$\hat{y}_i - \bar{y}$	-4	-2	0	2	4	
$(\hat{y}_i - \bar{y})^2$	16	4	0	4	16	ESS $=$ **40**
$\hat{u}_i$	-1	2	-2	2	-1	
$\hat{u}_i^2$	1	4	4	4	1	RSS $=$ **14**

(2) Check

TSS $=$ ESS $+$ RSS: $54 = 40 + 14$ ✓
(TSS $= S_{yy} = 54$ as in exercise (2))

(3) Only for the sums

Month 1: $25 \neq 16 + 1$. The decomposition holds for the *sums*, not for each observation.

The decomposition and least squares

Why does the decomposition work?

Squaring $a + b$ with $a = \hat{y}_i - \bar{y}$, $b = \hat{u}_i$ gives $a^2 + 2ab + b^2$. The mixed term vanishes in the sum:

$$\sum (\hat{y}_i - \bar{y})\,\hat{u}_i \;=\; \hat{\beta}_1 \sum (x_i - \bar{x})\,\hat{u}_i \;=\; 0,$$

because the OLS residuals satisfy $\sum \hat{u}_i = 0$ and $\sum x_i \hat{u}_i = 0$ (first-order conditions).

For other lines it does *not* hold – line A: $620 + 164 \neq 288$.

What least squares achieves

$$\underbrace{\text{TSS}}_{\text{fixed}} = \underbrace{\text{ESS}}_{\text{maximal}} + \underbrace{\text{RSS}}_{\text{minimal}}$$

- TSS depends only on the y_i.
- The OLS line makes RSS minimal.
- Hence the explained part TSS − RSS (= ESS for the OLS line) is maximal.

Key takeaway

No other line explains a larger share of the variation of y than the least-squares line.

The coefficient of determination R^2

Definition

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} \qquad 0 \leq R^2 \leq 1$$

In words: The *share* of the variation of y that the regression on x explains.

Twelve stores

$$R^2 = \frac{223.2}{288} = 1 - \frac{64.8}{288} = \mathbf{0.775}$$

77.5 % of the variation in revenue is explained (linearly) by advertising, 22.5 % remains unexplained.

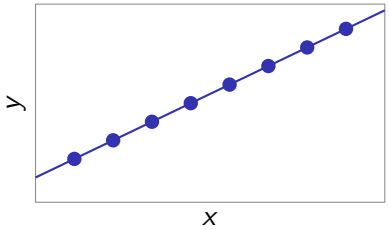
Also via variances

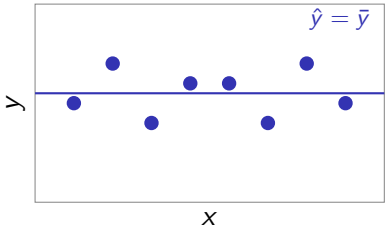
Dividing all sums of squares by $n - 1$:

$$s_y^2 = s_{\hat{y}}^2 + s_{\hat{u}}^2, \qquad R^2 = \frac{s_{\hat{y}}^2}{s_y^2}$$

(total variance = explained variance + residual variance; the factor cancels).

The extreme cases: $R^2 = 1$ and $R^2 = 0$





$R^2 = 1$: perfect fit

All points lie *on* the line: RSS $= 0$. The entire variation of y is explained.

$R^2 = 0$: nothing explained

$\hat{\beta}_1 = 0$: the line runs horizontally at the level $\bar{y}$, ESS $= 0$. Knowing x does not improve the prediction.

R^2 and the correlation coefficient

Step 1: the explained deviation – the intercept cancels

From $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$ (line through the centroid):
$\hat{y}_i - \bar{y} = (\hat{\beta}_0 + \hat{\beta}_1 x_i) - (\hat{\beta}_0 + \hat{\beta}_1 \bar{x}) = \hat{\beta}_1 (x_i - \bar{x})$, so $\text{ESS} = \hat{\beta}_1^2 \, S_{xx}$ (and $s_{\hat{y}}^2 = \hat{\beta}_1^2 \, s_x^2$).

Step 2: insert $\hat{\beta}_1 = S_{xy}/S_{xx}$

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = \frac{\hat{\beta}_1^2 \, S_{xx}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx} \, S_{yy}} = \left(\frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \right)^2 = r_{xy}^2$$

Key takeaway

In simple regression, R^2 is the **square of the correlation coefficient**. With several explanatory variables this no longer holds.

Twelve stores

$r_{xy} = \frac{186}{\sqrt{155 \cdot 288}} = 0.8803,$
$r_{xy}^2 = 0.775 = R^2$ ✓
Also: $\text{ESS} = 1.2^2 \cdot 155 = 223.2.$

What R^2 does – and does not – tell you

No direction

$\sqrt{R^2} = |r_{xy}|$: the sign is lost – only r_{xy} and $\hat{\beta}_1$ show it.

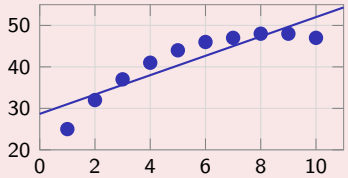
A low R^2 does *not* prove ...

that x is irrelevant (the slope can still be clearly $\neq 0$, Section 5) or that y does not depend on x – only that a *line* explains little.

A high R^2 does *not* prove ...

that x *causes* y, or that predictions outside the data range work.

High R^2, yet the wrong model



Ten regions, advertising and revenue: $R^2 = 0.81$ – yet the points clearly bend (residual analysis: Appendix A.3).

Key takeaway

R^2 does not replace the residual plot.

Assessing R^2

Rule of thumb (used in this course)

Value of R^2	Assessment
$R^2 < 0.6$	weak fit
$0.6 \leq R^2 < 0.8$	moderate fit
$R^2 \geq 0.8$	good fit

Because $R^2 = r_{xy}^2$, a correlation that looks high gives a much smaller R^2: $r_{xy} = 0.7$ yields only $R^2 = 0.49$.

Our examples

Twelve stores	$R^2 = 0.775$	moderate
Lamp	$R^2 = 0.947$	good
Ten regions	$R^2 = 0.81$	good – but curved!

Context matters

What counts as "good" depends on the field: in finance or the social sciences an R^2 of 0.1 can be valuable, in physics 0.95 may be poor.

Exercise – online shop (8): computing and assessing R^2

Known

Exercise (7): $\text{TSS} = 54$, $\text{ESS} = 40$, $\text{RSS} = 14$ Exercise (2): $r_{xy} = 0.8607$

Problem statement

1. Compute R^2 in two ways: via ESS and via RSS.
2. Check the result with r_{xy} from exercise (2).
3. Interpret R^2 in words and assess the fit with the rule of thumb.
4. True or false? "$R^2 = 0.74$ means that 74 % of the months lie exactly on the line." Justify briefly.

Solution (1/2) – online shop (8): computing and assessing R^2

(1) Rule: $R^2 = \text{ESS}/\text{TSS} = 1 - \text{RSS}/\text{TSS}$

$$R^2 = \frac{40}{54} = \mathbf{0.741} \qquad R^2 = 1 - \frac{14}{54} = 0.741$$

(2) Rule: $R^2 = r_{xy}^2$ (simple regression)

$0.8607^2 = 0.741$ ✓

(3) Interpretation and assessment

74 % of the variation of the monthly revenue is explained (linearly) by advertising, 26 % is not.

$0.6 \leq 0.741 < 0.8$: a **moderate** fit.

Note: $r_{xy} = 0.86$ counts as strong, $R^2 = 0.74$ only as moderate – two different rules of thumb.

Solution (2/2) – online shop (8): computing and assessing R^2

(4) False

R^2 is a share of *variation*, not a hit rate: here **not a single** month lies on the line (all residuals $\neq 0$), yet $R^2 = 0.74$.

Check with Python

```
shop = pd.DataFrame({"adv": [1, 2, 3, 4, 5], "revenue": [21, 26, 24, 30, 29]})
reg = smf.ols("revenue ~ adv", data=shop).fit()
reg.ssr, reg.centered_tss, reg.rsquared      # 14.0  54.0  0.7407 (RSS, TSS, R^2)
```

In words: The residuals sum to zero – a property of every least-squares line with intercept: $-1 + 2 - 2 + 2 - 1 = 0$.

▶ **Try it:** notebook, section 3.2 (Colab)

Coefficient of determination – lamp example

Known from the working table

$S_{xy} = -670, \quad S_{xx} = 1500, \quad \hat{\beta}_1 = -0.4467$

Sums of squares

- TSS $= \sum (y_i - \bar{y})^2 =$
  $10^2 + 7^2 + \ldots + (-8)^2 = 316$
- ESS $= \hat{\beta}_1^2 \, S_{xx} = \left(\frac{-670}{1500}\right)^2 \cdot 1500 = 299.27$
- RSS $= 316 - 299.27 = 16.73 \quad (= \sum \hat{u}_i^2)$

Coefficient of determination

$$R^2 = \frac{299.27}{316} = 1 - \frac{16.73}{316} = \mathbf{0.9470}$$

Check: $r_{xy} = \frac{-670}{\sqrt{1500 \cdot 316}} = -0.9732,$
$r_{xy}^2 = 0.9470$ ✓

Interpretation

About 95 % of the variation of the sales quantity across the stores is explained (linearly) by the price: a **good fit**. The sign (falling price-sales function) is in $\hat{\beta}_1 < 0$, not in R^2.

Coefficient of determination in Python

Python command

```
print(reg.summary())          # full output, incl. "R-squared: 0.947"
reg.rsquared                  # 0.9470
```

Three ways to the same value

```
RSS = (reg.resid**2).sum()                                  # 16.7333
TSS = ((df["quantity"] - df["quantity"].mean())**2).sum()   # 316
1 - RSS / TSS                                               # 0.9470
df["price"].corr(df["quantity"])**2                         # simple regression only
```

In words: `summary()` reports further quantities (standard errors, t-values, p-values) – they follow in Section "Inference".

▶ **Try it:** notebook, section 3.2 (Colab)

Exercise – from sums of squares to the correlation coefficient

Data: heating costs of a shop in five weeks

Week i	1	2	3	4	5
Mean outside temperature x_i (°C)	0	3	6	9	12
Heating costs y_i (EUR per day)	82	65	63	41	49

From Python: OLS line $\hat{y} = 78 - 3\,x$, $\quad$ TSS $= 1000$, $\quad$ RSS $= 190$.

Tasks

1. Compute ESS and R^2; interpret R^2.
2. Determine r_{xy} from R^2 – justify the sign.
3. Assess the fit with the rule of thumb.
4. Check: recompute ESS via $\hat{\beta}_1^2 S_{xx}$.

Solution (1/2) – from sums of squares to the correlation coefficient

(1) Rule: TSS = ESS + RSS

$\text{ESS} = \text{TSS} - \text{RSS} = 1000 - 190 = \mathbf{810}$

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = \frac{810}{1000} = 1 - \frac{190}{1000} = \mathbf{0.81}$$

81 % of the variation of the heating costs is explained (linearly) by the outside temperature, 19 % remains unexplained.

(2) Rule: $R^2 = r_{xy}^2$, sign of $\hat{\beta}_1$

$|r_{xy}| = \sqrt{R^2} = \sqrt{0.81} = 0.9$

r_{xy} and $\hat{\beta}_1$ always have the same sign; $\hat{\beta}_1 = -3 < 0$, hence

$$r_{xy} = \mathbf{-0.9}$$

(3) Assessment

$R^2 = 0.81 \geq 0.8$: a **good fit**. With only $n = 5$ weeks the verdict is uncertain, though.

Common mistake

R^2 alone gives only $|r_{xy}|$. The sign is in $\hat{\beta}_1$ (or s_{xy}): the warmer, the *lower* the heating costs – a strong negative relationship.

Solution (2/2) – from sums of squares to the correlation coefficient

(4) Rule: ESS $= \hat{\beta}_1^2 \, S_{xx}$

$\bar{x} = 30/5 = 6$; $x_i - \bar{x}$: $-6,\ -3,\ 0,\ 3,\ 6$; $S_{xx} = 36 + 9 + 0 + 9 + 36 = 90$;
ESS $= (-3)^2 \cdot 90 = \mathbf{810}$ ✓
Likewise TSS: $\bar{y} = 60$, $22^2 + 5^2 + 3^2 + (-19)^2 + (-11)^2 = 1000$ ✓

With Python

```
hc = pd.DataFrame({"temp": [0, 3, 6, 9, 12], "cost": [82, 65, 63, 41, 49]})
reg = smf.ols("cost ~ temp", data=hc).fit()
reg.rsquared                  # 0.81
hc["temp"].corr(hc["cost"])   # -0.9
reg.ssr, reg.ess              # 190.0  810.0  (careful: ssr = RSS, ess = ESS)
```

In words: TSS and RSS give ESS and R^2; with the sign of $\hat{\beta}_1$ also r_{xy}.

▶ **Try it:** notebook, section 3.2 (Colab)

Shortcut: three sums are enough

Guiding question

By hand we need s_{xy}, s_x, s_y, r_{xy}, $\hat{\beta}_1$, $\hat{\beta}_0$ and R^2. Can we do it with less work and fewer rounding errors?

The three basic sums (from the working table)

$S_{xx} = \sum(x_i - \bar{x})^2$	sum of squares of x	numerator of the variance s_x^2
$S_{yy} = \sum(y_i - \bar{y})^2$	sum of squares of y	$=$ TSS
$S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})$	sum of cross-products	numerator of the covariance s_{xy}

Key takeaway

All key figures of simple regression can be computed from S_{xx}, S_{yy} and S_{xy} – the factors $1/(n-1)$ cancel.

Why the factor $1/(n-1)$ cancels

Slope (known)

$$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{\frac{1}{n-1} S_{xy}}{\frac{1}{n-1} S_{xx}} = \frac{S_{xy}}{S_{xx}}$$

Correlation coefficient

$$r_{xy} = \frac{s_{xy}}{s_x \, s_y} = \frac{\frac{1}{n-1} S_{xy}}{\sqrt{\frac{1}{n-1} S_{xx}} \cdot \sqrt{\frac{1}{n-1} S_{yy}}}$$

Combining the roots

$$\sqrt{\tfrac{1}{n-1} S_{xx}} \cdot \sqrt{\tfrac{1}{n-1} S_{yy}} = \sqrt{\tfrac{1}{(n-1)^2} S_{xx} \, S_{yy}} = \tfrac{1}{n-1} \sqrt{S_{xx} \, S_{yy}} \quad \Longrightarrow \quad r_{xy} = \frac{S_{xy}}{\sqrt{S_{xx} \, S_{yy}}}$$

In words: Numerator and denominator contain the same factor, so it cancels – no matter whether one works with $1/(n-1)$ or with $1/n$.

All key figures from three sums

Quantity	via the statistics	via the sums
Slope	$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2}$	$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$
Intercept	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$	(the same)
Correlation	$r_{xy} = \frac{s_{xy}}{s_x \, s_y}$	$r_{xy} = \frac{S_{xy}}{\sqrt{S_{xx} \, S_{yy}}}$
Coefficient of determination	$R^2 = r_{xy}^2$	$R^2 = \frac{S_{xy}^2}{S_{xx} \, S_{yy}}$
Explained sum of squares	$\mathrm{ESS} = \hat{\beta}_1^2 \, S_{xx}$	$\mathrm{ESS} = \hat{\beta}_1 \, S_{xy}$
Residual sum of squares	$\mathrm{RSS} = \mathrm{TSS} - \mathrm{ESS}$	$\mathrm{RSS} = S_{yy} - \hat{\beta}_1 \, S_{xy}$

$\mathrm{ESS} = \hat{\beta}_1^2 S_{xx} = \hat{\beta}_1 \cdot \frac{S_{xy}}{S_{xx}} \cdot S_{xx} = \hat{\beta}_1 S_{xy}$. $R^2 = r_{xy}^2$ holds only in simple regression; $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS}$ also works in multiple regression (statsmodels: `rsquared`).

Example: the twelve stores from three sums

Basic sums

$$S_{xy} = 186$$
$$S_{xx} = 155$$
$$S_{yy} = 288$$
$$\bar{x} = 7.5, \quad \bar{y} = 39$$

Everything else in a few lines

$$\hat{\beta}_1 = 186/155 = \mathbf{1.2}$$
$$\hat{\beta}_0 = 39 - 1.2 \cdot 7.5 = \mathbf{30}$$
$$r_{xy} = 186/\sqrt{155 \cdot 288} = \mathbf{0.8803}$$
$$R^2 = 186^2/(155 \cdot 288) = 34\,596/44\,640 = \mathbf{0.775}$$
$$\text{ESS} = 1.2 \cdot 186 = \mathbf{223.2}$$
$$\text{RSS} = 288 - 223.2 = \mathbf{64.8}$$

Key takeaway

The same values as before – without $1/(n-1)$ factors and without roots of rounded variances.

Sums without a deviation table: the shift theorem

Shift theorem (computational formula; used in the derivation of the slope, Appendix A.1)

$$S_{xy} = \sum x_i y_i - n\,\bar{x}\,\bar{y} \qquad S_{xx} = \sum x_i^2 - n\,\bar{x}^2 \qquad S_{yy} = \sum y_i^2 - n\,\bar{y}^2$$

In words: Instead of forming every deviation, add up the raw values (products, squares) and subtract the mean part once.

Check: online shop ($n = 5$, $\bar{x} = 3$, $\bar{y} = 26$)

$\sum x_i y_i = 21 + 52 + 72 + 120 + 145 = 410$
$S_{xy} = 410 - 5 \cdot 3 \cdot 26 = 20$ ✓
$S_{xx} = 55 - 5 \cdot 3^2 = 10$ ✓
$S_{yy} = 3434 - 5 \cdot 26^2 = 54$ ✓

Careful: rounding

The difference of two large numbers is sensitive to rounded means. Use $n\,\bar{x}\,\bar{y} = \frac{\sum x_i \sum y_i}{n}$ or unrounded means.

Correlation as a standardised slope

Transformation

$$r_{xy} = \frac{s_{xy}}{s_x\, s_y} = \frac{s_{xy}}{s_x^2} \cdot \frac{s_x}{s_y} = \hat{\beta}_1 \cdot \frac{s_x}{s_y}$$

Checks

Twelve stores: $\frac{s_x}{s_y} = \sqrt{\frac{S_{xx}}{S_{yy}}} = \sqrt{\frac{155}{288}} = 0.7336$,
$r_{xy} = 1.2 \cdot 0.7336 = 0.8803$ ✓

Lamp:
$\hat{\beta}_1 = r_{xy}\, \frac{s_y}{s_x} = -0.9732 \cdot \frac{6.2849}{13.6931} = -0.4467$ ✓

In words

Measure x and y in standard deviations, and the slope $\hat{\beta}_1$ becomes the correlation coefficient: the **standardised regression coefficient**.

Consequences

- r_{xy} and $\hat{\beta}_1$ always have the same sign.
- $\hat{\beta}_1$ depends on the units, r_{xy} does not.
- Squaring: $R^2 = \hat{\beta}_1^2\, \frac{s_x^2}{s_y^2} = r_{xy}^2$.

Three sums with Python

Three sums, all key figures and the decomposition (twelve stores)

```
stores = pd.DataFrame({"adv":     [1, 3, 4, 5, 6, 8, 8, 9, 10, 11, 12, 13],
                       "revenue": [33, 30, 37, 33, 38, 41, 39, 43, 45, 40, 42, 47]})
x, y = stores["adv"], stores["revenue"]
Sxx = ((x - x.mean())**2).sum()                    # 155.0
Syy = ((y - y.mean())**2).sum()                    # 288.0 (= TSS)
Sxy = ((x - x.mean()) * (y - y.mean())).sum()      # 186.0
b1 = Sxy / Sxx; b0 = y.mean() - b1 * x.mean()      # 1.2  30.0
Sxy / np.sqrt(Sxx * Syy), b1 * Sxy                 # r = 0.8803, ESS = 223.2
(x * y).sum() - len(x) * x.mean() * y.mean()       # 186.0 = Sxy (shift theorem)
reg = smf.ols("revenue ~ adv", data=stores).fit() # TSS = ESS + RSS, R^2:
reg.centered_tss, reg.ess, reg.ssr, reg.rsquared  # 288.0 = 223.2 + 64.8; R^2 = 0.775
```

▶ **Try it:** notebook, section 3.3 (Colab)

Can you answer these questions?

1. Into which two parts does $y_i - \bar{y}$ split?
2. Why does $\text{TSS} = \text{ESS} + \text{RSS}$ hold only for the OLS line?
3. What does R^2 measure, and why is $0 \leq R^2 \leq 1$?
4. Why is $R^2 = r_{xy}^2$ in simple regression?
5. Why is a high R^2 not a seal of quality?
6. How do you get all key figures from S_{xx}, S_{yy}, S_{xy}?

Formulas

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} = r_{xy}^2$$

Key takeaway

R^2 is the share of the variation of y that the line explains – together with the residual plot (Appendix A.3) it forms the verdict on the line.

Exercise – reading a simple regression output in Python

For $n = 12$ students the study hours x and the exam points y (at most 100) were recorded; the observed study hours range from 5 to 24.

```
>>> reg = smf.ols("points ~ hours", data=st).fit()
>>> print(reg.summary())                                          # excerpt
Dep. Variable:                  points   R-squared:                       0.900
Model:                             OLS   Adj. R-squared:                  0.890
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     28.0000      3.962      7.068      0.000      19.173      36.827
hours          2.5000      0.264      9.487      0.000       1.913       3.087
```

Also known: $\bar{x} = 14$, $S_{xx} = 360$, $\text{TSS} = \sum(y_i - \bar{y})^2 = 2500$. (The columns `std err` to `0.975` follow in Section "Inference".)

1. Write down the fitted line and interpret $\hat{\beta}_1$ and $\hat{\beta}_0$ in context.
2. Predict the points for 16 and for 30 study hours; judge both predictions.
3. Determine the correlation coefficient r_{xy} from the output.
4. Compute ESS and RSS and check the result via $\hat{\beta}_1^2 S_{xx}$.
5. Name a third variable that could explain part of the association.

Solution (1/2) – reading a simple regression output in Python

(1) Fitted line and interpretation

The row `Intercept` gives $\hat{\beta}_0$, the row `hours` gives $\hat{\beta}_1$: $\hat{y} = 28 + 2.5\,x$.

- $\hat{\beta}_1 = 2.5$: each additional study hour goes along with 2.5 **more exam points on average** – not an exact prediction for every student.
- $\hat{\beta}_0 = 28$: 28 points at zero study hours – but $x = 0$ lies outside the observed range $[5, 24]$: an extrapolation, to be read with caution.

(2) Predictions

$\hat{y}(16) = 28 + 2.5 \cdot 16 = \mathbf{68}$ points – inside $[5, 24]$: plausible.
$\hat{y}(30) = 28 + 2.5 \cdot 30 = \mathbf{103}$ points – **not trustworthy** for two reasons: $x = 30$ lies outside $[5, 24]$ (extrapolation; a flattening curve is more plausible), and more than 100 points are impossible.

Solution (2/2) – reading a simple regression output in Python

(3) Correlation coefficient

In simple regression $R^2 = r_{xy}^2$, so $|r_{xy}| = \sqrt{0.9} = 0.9487$ (not 0.9!). The sign is that of $\hat{\beta}_1 = 2.5 > 0$:

$$r_{xy} = +\mathbf{0.9487}$$

(4) Sums of squares

$\text{ESS} = R^2 \cdot \text{TSS} = 0.9 \cdot 2500 = \mathbf{2250}$
$\text{RSS} = (1 - R^2) \cdot \text{TSS} = 0.1 \cdot 2500 = \mathbf{250}$
Check: $\hat{\beta}_1^2 S_{xx} = 2.5^2 \cdot 360 = 2250$ ✓,
$2250 + 250 = 2500$ ✓

(5) Third variable

E.g. prior knowledge or motivation: students with more of it study more *and* score higher – the association would then partly be produced by this third variable. A causal effect of studying could only be shown by an experiment (random allocation of study time).

Exercise – body heights and shoe sizes: overview

Data basis: $n = 15$ persons

Person i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Body height k_i (cm)	164	162	168	198	166	166	192	160	151	175	180	160	168	175	168
Shoe size s_i	39	38	39	46	40	39	45	37	36	42	43	38	41	40	39

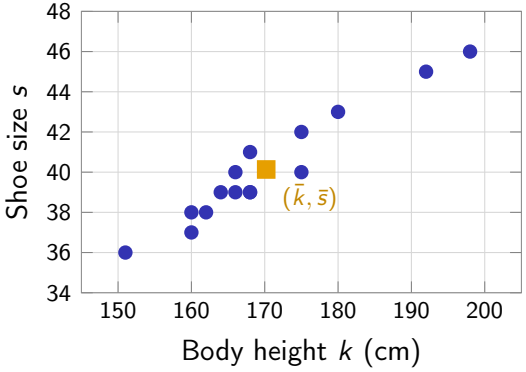
Auxiliary sums: $\sum k_i = 2553$, $\sum s_i = 602$, $\sum k_i s_i = 102\,932$, $\sum k_i^2 = 436\,643$, $\sum s_i^2 = 24\,272$. (Body heights as in Part 2.)

Tasks (model: $s_i = \beta_0 + \beta_1 k_i + u_i$)

Task 1 Scatter plot; **covariance** and **correlation coefficient**.

Task 2 Regression line and interpretation, coefficient of determination R^2.

Solution Task 1 – scatter plot and sums of squares



Description

Taller persons tend to have larger feet: a **positive**, almost **linear** relationship.

Means

$\bar{k} = 2553/15 = 170.2\,\text{cm}, \quad \bar{s} = 602/15 = 40.1\overline{3}$

Shift theorem (no rounded means)

$S_{ks} = 102\,932 - 2553 \cdot 602/15 = \mathbf{471.6}$
$S_{kk} = 436\,643 - 2553^2/15 = \mathbf{2122.4}$
$S_{ss} = 24\,272 - 602^2/15 = \mathbf{111.73}$

Solution Task 1 – covariance and correlation

Covariance and variances (divide by $n - 1 = 14$)

$$s_{ks} = \frac{471.6}{14} = \mathbf{33.6857}$$

$$s_k^2 = \frac{2122.4}{14} = 151.6 \;\Rightarrow\; s_k = 12.3126, \qquad s_s^2 = \frac{111.7333}{14} = 7.9810 \;\Rightarrow\; s_s = 2.8251$$

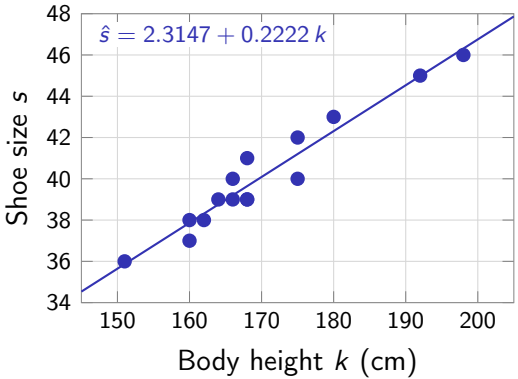
Correlation coefficient

$$r_{ks} = \frac{s_{ks}}{s_k\, s_s} = \frac{33.6857}{12.3126 \cdot 2.8251} = \mathbf{0.9684} \qquad \left(= \frac{S_{ks}}{\sqrt{S_{kk} S_{ss}}} = \frac{471.6}{\sqrt{2122.4 \cdot 111.73}} \;\checkmark \right)$$

Interpretation

Very strong positive linear relationship between body height and shoe size ($|r| > 0.8$). The covariance has the unit "cm $\times$ shoe size" and is not directly interpretable.

Solution Task 2 – regression line



Estimates

$\hat{\beta}_1 = S_{ks}/S_{kk} = 471.6/2122.4 = \mathbf{0.2222}$

$\hat{\beta}_0 = 40.1\overline{3} - 0.2222 \cdot 170.2 = \mathbf{2.3147}$

(computed with the unrounded $\hat{\beta}_1$)

Interpretation

- Per cm of body height the shoe size increases on average by 0.22 – about one size per 4.5 cm.
- $\hat{\beta}_0$: shoe size at 0 cm body height – far outside the data, **not interpretable**.

Solution Task 2 – coefficient of determination R^2

Via the sums of squares

$\text{ESS} = \hat{\beta}_1 \cdot S_{ks} = 0.2222 \cdot 471.6 = 104.79$

$\text{RSS} = S_{ss} - \text{ESS} = 111.73 - 104.79 = 6.94$

$R^2 = \frac{104.79}{111.73} = \mathbf{0.9379}$

Cross-check via r_{ks}^2

$R^2 = r_{ks}^2 = 0.9684^2 = 0.9379$ ✓

Interpretation and assessment

About 93.8 % of the variation of the shoe sizes is explained (linearly) by body height.
$R^2 \geq 0.8$: a **good fit**.

Key takeaway

A good R^2: the line explains most of the variation. Whether a line is adequate shows the residual plot (exercise in Appendix A.5).

Overview

1 Relationship Between Two Metric Variables

2 Simple Linear Regression

3 Goodness of Fit: the Coefficient of Determination

4 Inference in Simple Regression
- The regression model
- Sampling variability
- t-test for a coefficient
- Confidence intervals
- p-values and Python output
- Prediction
- Exercises

5 Multiple Linear Regression

6 Summary

From describing the data to the regression model

Now: an idea of how the data come about

So far the OLS line *describes* the cloud of points. From now on we assume that the data are generated as

$$y_i = \underbrace{\beta_0 + \beta_1 x_i}_{\text{systematic part}} + \underbrace{u_i}_{\text{error term}}, \qquad i = 1, \ldots, n.$$

$\beta_0,\ \beta_1$ *true*, unknown coefficients; $\hat{\beta}_0$, $\hat{\beta}_1$ are their estimates from the data.

u_i everything that x does not explain (other influences, chance) – *not observable*.

$\hat{u}_i = y_i - \hat{y}_i$ the residual: the estimate of u_i computed from the data.

Requirement for the error term

The u_i should contain **no systematic information** about Y – everything explainable belongs in the linear part. We can check this only with the residuals: the **residual analysis** (Appendix A.3–A.5).

What do we assume about the error term?

Model $y_i = \beta_0 + \beta_1 x_i + u_i$: the standard assumptions concern the *unobservable* error term u_i.

Assumption	**Formal**	**In words**
Expected value zero	$E(u_i) = 0$	No systematic error: on average the observations lie on the true line – for every x.
Homoscedasticity	$\mathrm{Var}(u_i) = \sigma^2$	The spread around the line is the same everywhere (independent of x_i).
No autocorrelation	$\mathrm{Cov}(u_i, u_j) = 0$ $(i \neq j)$	The errors are pairwise uncorrelated (e.g. from month to month).
Normal distribution	$u_i \sim N(0, \sigma^2)$	The errors scatter bell-shaped around zero – required for confidence intervals and tests (this section).

The problem

The assumptions are **not self-evident** and must be **checked** – with the **residuals** $\hat{u}_i$, our observable estimates of the unobservable u_i (residual plot and QQ-plot: Appendix A.3–A.5).

Why regression coefficients carry uncertainty

Intuition (as in Part 3)

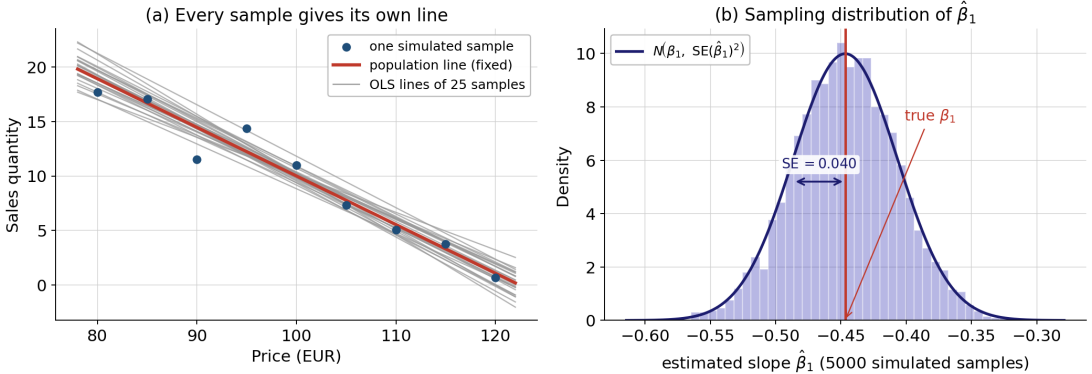
The nine stores are only a **sample**. Other stores – or the same stores in another week – would give slightly different sales, hence a slightly different line.

- The **true** coefficients β_0, β_1 (population line) are fixed but unknown.
- The **estimates** $\hat{\beta}_0, \hat{\beta}_1$ are computed from a random sample – they are **random variables** with a sampling distribution, exactly like $\bar{X}$ in Part 3.
- Questions: How much does $\hat{\beta}_1$ wobble? Is the true slope possibly 0? Which values of β_1 are compatible with the data?

Key takeaway

The tools are the same as for the mean: **standard error**, t-**test**, **confidence interval** and p-**value**.

Population line vs. sample lines



Simulation with the lamp prices: (a) every sample gives its own OLS line around the fixed population line; (b) over 5000 samples the slopes $\hat{\beta}_1$ are normally distributed around β_1 – their standard deviation is the **standard error**.

Sampling distribution of the slope

Result (under the four assumptions on u_i)

$$\hat{\beta}_1 \sim N\left(\beta_1,\ \frac{\sigma^2}{S_{xx}}\right), \qquad \mathrm{SE}(\hat{\beta}_1) = \frac{\sigma}{\sqrt{S_{xx}}} = \frac{\sigma}{\sqrt{\sum_i (x_i - \bar{x})^2}}.$$

- $E(\hat{\beta}_1) = \beta_1$: the estimator is **unbiased** – on average it hits the true slope.
- **Less noise** ($\sigma \downarrow$) $\Rightarrow$ smaller SE.
- **More spread in** x ($S_{xx} \uparrow$) $\Rightarrow$ smaller SE: a long "lever arm" fixes the slope precisely.
- **More observations** increase S_{xx} $\Rightarrow$ smaller SE.

In words: Compare Part 3: $\mathrm{SE}(\bar{X}) = \sigma/\sqrt{n}$. Here the role of n is taken by S_{xx}.

Estimating σ: the residual standard error

Problem

The error variance σ^2 is unknown – as σ in Part 3. We estimate it from the residuals.

Residual standard error

$$\hat{\sigma} = \sqrt{\frac{\text{RSS}}{n-2}} = \sqrt{\frac{1}{n-2}\sum_{i=1}^{n}\hat{u}_i^2}, \qquad \text{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}}, \qquad \text{SE}(\hat{\beta}_0) = \hat{\sigma}\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}.$$

- **Why $n-2$?** Two parameters (β_0, β_1) were estimated – two degrees of freedom are used up (Part 1: one parameter $\bar{x} \Rightarrow n-1$).
- $\hat{\sigma}$ is measured in the **units of** y: the typical size of a residual.

Standard errors – lamp example

Given: $n = 9$, RSS $= 16.733$, $S_{xx} = 1500$, $\bar{x} = 100$.

Calculation

$$\hat{\sigma} = \sqrt{\frac{16.733}{9-2}} = \sqrt{2.3905} = \mathbf{1.5461}$$

$$\mathrm{SE}(\hat{\beta}_1) = \frac{1.5461}{\sqrt{1500}} = \frac{1.5461}{38.730} = \mathbf{0.03992}$$

$$\mathrm{SE}(\hat{\beta}_0) = 1.5461\sqrt{\frac{1}{9} + \frac{100^2}{1500}} = 1.5461 \cdot 2.6034 = \mathbf{4.0252}$$

In words: A typical store deviates by about 1.5 lamps from the line; the slope -0.4467 is uncertain by about ± 0.04 (one standard error).

Exercise – online shop (9): residual standard error and standard error

Problem statement

Back to the first five months: $\hat{y} = 20 + 2\,x$ with $n = 5$, $S_{xx} = 10$ and RSS $= 14$ (exercises (2) and (7)).

1. Compute the residual standard error $\hat{\sigma}$.
2. Compute the standard error SE$(\hat{\beta}_1)$.
3. Interpret both numbers in words.

Solution – online shop (9): residual standard error and standard error

(1), (2) Rule: $\hat{\sigma} = \sqrt{\text{RSS}/(n-2)}$, $\text{SE}(\hat{\beta}_1) = \hat{\sigma}/\sqrt{S_{xx}}$

$$\hat{\sigma} = \sqrt{\frac{14}{5-2}} = \sqrt{4.6667} = \mathbf{2.1602}, \qquad \text{SE}(\hat{\beta}_1) = \frac{2.1602}{\sqrt{10}} = \frac{2.1602}{3.1623} = \mathbf{0.6831}$$

(3) In words

A typical month deviates by about 2160 EUR from the line. The slope $\hat{\beta}_1 = 2$ is uncertain by about ± 0.68 (one standard error): quite a lot for only five months.

In words: Divide by $n - 2 = 3$: two coefficients were estimated.

The first question: is there any relationship at all?

Hypotheses

$$H_0 : \beta_1 = 0 \qquad \text{vs.} \qquad H_1 : \beta_1 \neq 0$$

- H_0: the price has **no** (linear) influence on the sales quantity – the population line is flat.
- H_1: the price has *some* influence.
- As in Part 3: what we want to **establish** goes into H_1.

Key takeaway

Rejecting $H_0 : \beta_1 = 0$ is what licenses us to interpret the slope at all – a slope that could just as well be zero tells us nothing.

The t-statistic – the same recipe as in Part 3

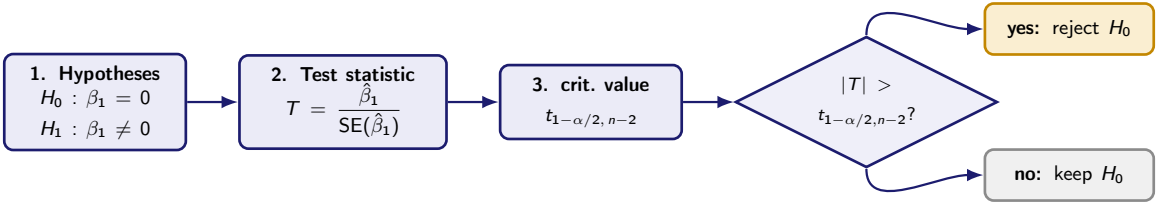
Test statistic

$$T = \frac{\hat{\beta}_1 - \beta_{1,0}}{\mathrm{SE}(\hat{\beta}_1)} \sim t_{n-2} \text{ under } H_0 : \beta_1 = \beta_{1,0} \qquad \text{(usually } \beta_{1,0} = 0\text{)}$$

	Part 3: mean	**Part 4: slope**
Estimate	$\bar{x}$	$\hat{\beta}_1$
Hypothesised value	μ_0	$\beta_{1,0}$ (mostly 0)
Standard error	$s/\sqrt{n}$	$\hat{\sigma}/\sqrt{S_{xx}}$
Distribution under H_0	t_{n-1}	t_{n-2}
Reject (two-sided) if	$\lvert T_n \rvert > t_{1-\alpha/2,\, n-1}$	$\lvert T \rvert > t_{1-\alpha/2,\, n-2}$

In words: "How many standard errors is the estimate away from the hypothesised value?"

The test recipe for a regression coefficient



- Fix α (typically 0.05) *before* looking at the data.
- Equivalent decision via the p-value: reject H_0 if $p < \alpha$, with $p = 2\,F_{t,n-2}(-|T|)$.

t-test – lamp example

Data: $\hat{\beta}_1 = -0.4467$, $\mathrm{SE}(\hat{\beta}_1) = 0.03992$, $n = 9$, $\alpha = 0.05$, $t_{0.975;\,7} = 2.3646$.

$$T = \frac{-0.4467 - 0}{0.03992} = -11.19$$

$$|T| = 11.19 \;>\; 2.3646 = t_{0.975;\,7}$$

Decision

$H_0 : \beta_1 = 0$ is **rejected**. The price has a significant influence on the sales quantity. The p-value is $p = 2\,F_{t,7}(-11.19) = 0.00001$.

In words: The estimated slope lies more than eleven standard errors away from zero – far too far to be a chance result.

Exercise – online shop (10): is the slope significant?

Problem statement

From exercise (9): $\hat{\beta}_1 = 2$, $\mathrm{SE}(\hat{\beta}_1) = 0.6831$, $n = 5$.

1. Test $H_0\colon \beta_1 = 0$ against $H_1\colon \beta_1 \neq 0$ at $\alpha = 0.05$.
2. $R^2 = 0.74$ looked quite good. Why is the test result nevertheless so weak?

Hint: $t_{0.975;\,3} = 3.182$.

Solution – online shop (10): is the slope significant?

(1) Rule: $T = (\hat{\beta}_1 - 0)/\,\mathrm{SE}(\hat{\beta}_1)$, reject if $|T| > t_{0.975;\,n-2}$

$$T = \frac{2}{0.6831} = \mathbf{2.928}, \qquad |T| = 2.928 < 3.182 = t_{0.975;\,3}$$

$\Rightarrow$ H_0 is **not** rejected at the 5 % level ($p = 0.061$).

(2) Only five months

Only $n - 2 = 3$ degrees of freedom: large critical value (3.182 instead of about 2) and large standard error. The data are *compatible* with $\beta_1 = 0$. R^2 describes the fit, it is no test.

Testing other hypothesised values

Question: A marketing study claims that each euro of price costs *half* a lamp. Is this compatible with our data?

$$H_0 : \beta_1 = -0.5 \quad \text{vs.} \quad H_1 : \beta_1 \neq -0.5$$

Test

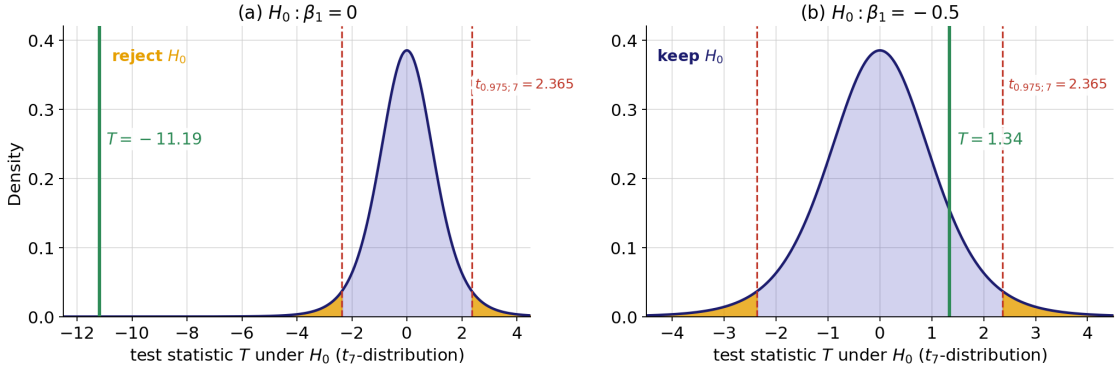
$$T = \frac{-0.4467 - (-0.5)}{0.03992} = \frac{0.0533}{0.03992} = 1.336, \qquad |T| = 1.336 < 2.3646 = t_{0.975;\,7}$$

$\Rightarrow$ **keep** H_0 $(p = 0.223)$.

Reading

The data cannot distinguish $\beta_1 = -0.5$ from the estimate -0.4467: the claim is *compatible* with the data – but not *proven*.

Two t-tests in one picture



Both tests use the same t_7-distribution and the same bars ± 2.365. (a) $T = -11.19$ lies far in the rejection region (orange); (b) $T = 1.34$ lies inside the acceptance region. The decision is nothing more than: **where does the statistic fall?**

One-sided test for the slope

Economic theory gives a direction

A price-sales function should *fall*. We want evidence for a negative slope:

$$H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_1 : \beta_1 < 0 \qquad \text{(left-sided)}$$

Lamp example

Reject if $T < -t_{1-\alpha,\, n-2} = -t_{0.95;\, 7} = -1.8946$:

$$T = -11.19 < -1.8946 \;\Rightarrow\; \text{reject } H_0, \qquad p = F_{t,7}(-11.19) = 0.000005.$$

As in Part 3

The direction must be fixed *before* looking at the data, and the one-sided p-value is half the two-sided one only if T points in the direction of H_1. statsmodels always reports the **two-sided** p-value (`P>|t|`, `reg.pvalues`).

Before you test – the assumptions checklist

1. **Linear model is correct** – residual plot without systematic pattern (Appendix A.3).
2. **Independent observations** – no autocorrelation (e.g. not the same store measured repeatedly).
3. **Constant variance** – no funnel in the residual plot.
4. **Normally distributed errors** – QQ-plot (Appendix A.5); for large n the central limit theorem makes the t-test approximately valid anyway.

Key takeaway

If the assumptions fail, the standard errors – and with them every t-test, confidence interval and p-value – are not reliable. The point estimates $\hat{\beta}_0, \hat{\beta}_1$ may still be fine.

Confidence interval for β_1

Construction – exactly as for μ in Part 3

$$\mathrm{CI}_{1-\alpha}(\beta_1) = \left[\hat{\beta}_1 - t_{1-\alpha/2,\,n-2}\ \mathrm{SE}(\hat{\beta}_1),\ \hat{\beta}_1 + t_{1-\alpha/2,\,n-2}\ \mathrm{SE}(\hat{\beta}_1)\right]$$

$$\mathrm{CI} = \Big[\ \underbrace{\hat{\beta}_1}_{\text{estimate}}\ \pm\ \underbrace{t_{1-\alpha/2,\,n-2}}_{\substack{\text{critical}\\ \text{value}}} \cdot\ \underbrace{\mathrm{SE}(\hat{\beta}_1)}_{\text{standard error}}\ \Big]$$

In words: For large n the critical value is close to $1.96 \approx 2$: $\hat{\beta}_1 \pm 2\ \mathrm{SE}(\hat{\beta}_1)$ is a quick 95 % interval.

Confidence interval – lamp example

Given: $\hat{\beta}_1 = -0.4467$, $\mathrm{SE}(\hat{\beta}_1) = 0.03992$, $t_{0.975;\,7} = 2.3646$.

$$\text{margin of error } e = 2.3646 \cdot 0.03992 = 0.0944$$

$$\mathrm{CI}_{0.95}(\beta_1) = [\,-0.4467 \pm 0.0944\,] = [\mathbf{-0.5411};\ \mathbf{-0.3523}]$$

Interpretation

The procedure covers the true slope in 95 % of all samples. Values of β_1 between -0.54 and -0.35 lamps per euro are *compatible* with the data.

Duality with the test

$0 \notin \mathrm{CI} \Leftrightarrow H_0 : \beta_1 = 0$ is rejected; $\quad -0.5 \in \mathrm{CI} \Leftrightarrow H_0 : \beta_1 = -0.5$ is kept.

Confidence intervals – interpretation pitfalls

Careful with the interpretation

"With 95 % probability β_1 lies in *this particular* interval $[-0.5411; -0.3523]$" is, strictly speaking, **wrong**.

- β_1 is a fixed (unknown) number; the *bounds* of the interval are random.
- Correct: "The **procedure** yields, in 95 % of samples, an interval that covers β_1."

Intercept

The CI for β_0 is [45.15; 64.18]. It is computed in the same way – but β_0 (sales at price 0) has no substantive meaning here.

Exercise – standard error, t and confidence interval

Problem statement

A simple linear regression on $n = 10$ observations gives $\hat{\beta}_1 = 2.5$, residual standard error $\hat{\sigma} = 4.0$ and $S_{xx} = \sum (x_i - \bar{x})^2 = 16$.

1. Compute the standard error $\mathrm{SE}(\hat{\beta}_1)$.
2. Compute the t-statistic for $H_0 : \beta_1 = 0$.
3. Compute the 95 % confidence interval for β_1.
4. Is the slope significant at the 5 % level? Justify in two ways.

Hint: $t_{0.975;\,8} = 2.306$.

Solution – standard error, t and confidence interval

$$\textbf{(1)} \quad \mathrm{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}} = \frac{4.0}{\sqrt{16}} = \mathbf{1.0}$$

$$\textbf{(2)} \qquad T = \frac{2.5 - 0}{1.0} = \mathbf{2.5}, \qquad \mathrm{df} = n - 2 = 8$$

$$\textbf{(3)} \qquad \mathrm{CI}_{0.95} = 2.5 \pm 2.306 \cdot 1.0 = [\mathbf{0.194}; \mathbf{4.806}]$$

(4) Decision

$|T| = 2.5 > 2.306$ – equivalently, 0 lies outside the CI. $H_0 : \beta_1 = 0$ is rejected at $\alpha = 0.05$ ($p \approx 0.037$).

In words: Common mistake: dividing $\hat{\sigma}$ by $\sqrt{n}$ – the standard error of the slope divides by $\sqrt{S_{xx}}$.

The p-value of a coefficient

Definition (as in Part 3)

The p-value is the probability, under $H_0 : \beta_1 = 0$, of a test statistic that is **at least as extreme** as the observed one:

$$p = P\big(|T| \geq |T_{\text{obs}}| \,\big|\, H_0\big) = 2\,F_{t,n-2}\big(-|T_{\text{obs}}|\big).$$

Hypothesis (lamp)	T	p (two-sided)	Decision at $\alpha = 0.05$
$H_0 : \beta_1 = 0$	-11.19	0.00001	reject
$H_0 : \beta_1 = -0.5$	1.34	0.223	keep

Key takeaway

$p < \alpha \Leftrightarrow |T| > t_{1-\alpha/2,\,n-2} \Leftrightarrow$ hypothesised value outside the CI.

Reading the regression output in Python

```
>>> print(reg.summary())                                        # excerpt
Dep. Variable:                quantity   R-squared:                       0.947
Model:                             OLS   Adj. R-squared:                  0.939
Method:                  Least Squares   F-statistic:                     125.2
                                         Prob (F-statistic):           1.02e-05
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     54.6667      4.025     13.581      0.000      45.149      64.185
price         -0.4467      0.040    -11.189      0.000      -0.541      -0.352
>>> print(np.sqrt(reg.scale))           # sigma-hat: not part of the summary
1.5461164867099084
```

`coef`, `std err`	$\hat{\beta}_j$, $\mathrm{SE}(\hat{\beta}_j)$ (rounded; exact: `reg.bse` gives 0.03992)
`t`, `P>|t|`	T for $H_0 : \beta_j = 0$, two-sided p-value (0.000: $p < 0.0005$; exact: `reg.pvalues`)
`[0.025 0.975]`	95 % confidence interval for β_j (`reg.conf_int()`)
`R-squared`, `Adj. R-squared`	R^2 and adjusted R^2 (see multiple regression)
`F-statistic`, `Prob (F-statistic)`	all slopes jointly: $F = T^2 = (-11.19)^2 = 125.2$, $p = 1.02 \cdot 10^{-5}$
`np.sqrt(reg.scale)`	$\hat{\sigma} = \sqrt{\mathrm{RSS}/(n-2)}$, $n - 2 = 7$ degrees of freedom (`Df Residuals`)

▶ **Try it:** notebook, section 4.5 (Colab)

Significant $\neq$ important

The p-value measures evidence, not size

With many observations even a tiny slope becomes highly significant. The size of the effect is read from $\hat{\beta}_1$ and its confidence interval.

Advertising data ($n = 200$): sales on TV budget

$\hat{\beta}_1 = 0.0475$ (sales in 1000 units, budget in 1000 \$), SE $= 0.0027$, $T = 17.67$, $p < 0.0001$, $\text{CI}_{0.95} = [0.0422; 0.0528]$.

- *Significance:* overwhelming evidence that TV advertising and sales are related.
- *Size:* 1000 \$ more TV budget goes along with 42 to 53 additional units sold – whether this pays off is a business question, not a statistical one.

Key takeaway

Report the estimate *and* its confidence interval – not just the p-value or the stars.

Two questions about a new x_0

At a new price x_0 the prediction is $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$. How precise is it? It depends on the question:

Confidence interval

for the **mean** sales $E(Y \mid x_0)$ of *all* stores with price x_0:

$$\hat{y}_0 \pm t_{1-\alpha/2,\, n-2}\, \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Prediction interval

for the sales Y_0 of **one** new store with price x_0:

$$\hat{y}_0 \pm t_{1-\alpha/2,\, n-2}\, \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Key takeaway

The "1+" adds the error u_0 of the single store: a prediction interval is **always wider** and never shrinks below $\pm t\,\hat{\sigma}$, however large n becomes.

Prediction – lamp example

Given: $\hat{\sigma} = 1.5461$, $n = 9$, $\bar{x} = 100$, $S_{xx} = 1500$, $t_{0.975;\,7} = 2.3646$, new price $x_0 = 110$: $\hat{y}_0 = 54.67 - 0.4467 \cdot 110 = 5.53$.

Calculation

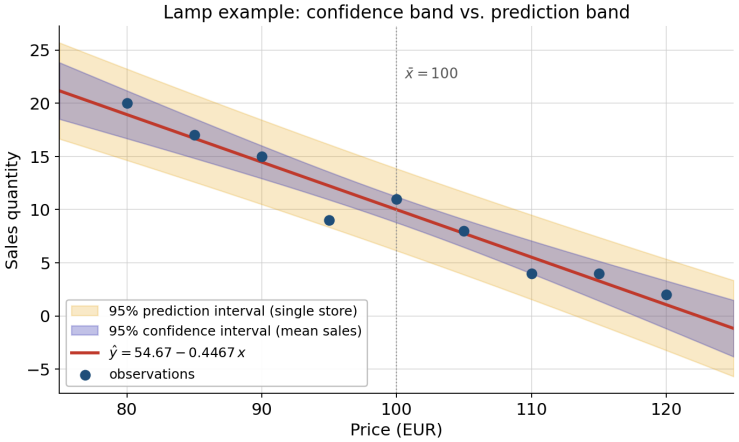
$$\text{CI: } 5.53 \pm 2.3646 \cdot 1.5461 \sqrt{\tfrac{1}{9} + \tfrac{100}{1500}} = 5.53 \pm 1.54 = [3.99;\, 7.07]$$

$$\text{PI: } 5.53 \pm 2.3646 \cdot 1.5461 \sqrt{1 + \tfrac{1}{9} + \tfrac{100}{1500}} = 5.53 \pm 3.97 = [1.57;\, 9.50]$$

Reading

At 110 euros, stores sell on *average* between 4 and 7 lamps (95 % CI); a *single* new store will sell between about 2 and 9 lamps (95 % PI).

Confidence band vs. prediction band



- Both bands are narrowest at $\bar{x} = 100$ and widen towards the edges.
- The prediction band contains most of the points; the confidence band does not need to.

Extrapolation

At $x_0 = 125$ euros the line predicts $\hat{y}_0 = -1.17$ lamps – impossible. Outside the data range the linear model breaks down.

Inference in simple regression with Python (1/2)

Tests, *p*-values and confidence intervals

```
reg = smf.ols("quantity ~ price", data=df).fit()
print(reg.summary())              # estimates, SE, t, p, R^2
reg.conf_int(alpha=0.05)          # CI for beta0 and beta1
```

By hand

```
from scipy import stats
se_b1 = reg.bse["price"]                          # 0.03992
T = (reg.params["price"] - (-0.5)) / se_b1        # H0: beta1 = -0.5
2 * stats.t.cdf(-abs(T), df=7)                    # p-value 0.223
stats.t.ppf(0.975, df=7)                          # critical value 2.3646
```

▶ **Try it:** notebook, section 4.6 (Colab)

Inference in simple regression with Python (2/2)

Prediction

```
new = pd.DataFrame({"price": [110]})
pred = reg.get_prediction(new).summary_frame(alpha=0.05)
pred[["mean_ci_lower", "mean_ci_upper"]]     # mean sales: [3.99, 7.07]
pred[["obs_ci_lower", "obs_ci_upper"]]       # one store:  [1.57, 9.50]
```

Note

`get_prediction()` needs a data frame whose column has *the same name* as the explanatory variable in the formula (here `price`). `mean_ci_*` is the confidence interval for the mean, `obs_ci_*` the prediction interval for one observation.

▶ **Try it:** notebook, section 4.6 (Colab)

Exercise – inference for the shoe-size regression

Problem statement

For the body heights and shoe sizes ($n = 15$) we found $\hat{s} = 2.3147 + 0.2222\,k$. In addition: $\text{RSS} = 6.9432$, $S_{kk} = 2122.4$, $\bar{k} = 170.2$.

1. Compute $\hat{\sigma}$ and $\text{SE}(\hat{\beta}_1)$.
2. Test $H_0 : \beta_1 = 0$ at $\alpha = 0.05$.
3. Compute the 95 % confidence interval for β_1.
4. A rule of thumb says "one shoe size per 4 cm", i.e. $\beta_1 = 0.25$. Test this claim.
5. Give a 95 % prediction interval for the shoe size of a person with $k_0 = 180$ cm.

Hint: $t_{0.975;\,13} = 2.1604$.

Solution (1/3) – inference for the shoe-size regression

(1) Residual standard error and standard error

$$\hat{\sigma} = \sqrt{\frac{6.9432}{15-2}} = \sqrt{0.5341} = 0.7308, \qquad \mathrm{SE}(\hat{\beta}_1) = \frac{0.7308}{\sqrt{2122.4}} = \frac{0.7308}{46.069} = 0.01586$$

(2) t-test for $H_0 : \beta_1 = 0$

$$T = \frac{0.2222}{0.01586} = 14.01 \; > \; 2.1604 = t_{0.975;\,13} \quad \Rightarrow \quad \text{reject } H_0 \quad (p = 3.2 \cdot 10^{-9}).$$

In words: Body height is a highly significant predictor of shoe size.

Solution (2/3) – inference for the shoe-size regression

(3) Confidence interval

$$0.2222 \pm 2.1604 \cdot 0.01586 = 0.2222 \pm 0.0343 = [\mathbf{0.1879}; \mathbf{0.2565}]$$

Between 0.19 and 0.26 shoe sizes per cm are compatible with the data.

(4) $H_0 : \beta_1 = 0.25$

$$T = \frac{0.2222 - 0.25}{0.01586} = -1.75, \qquad |T| = 1.75 < 2.1604 \quad \Rightarrow \quad \text{keep } H_0 \ (p = 0.103).$$

Consistent with the CI: $0.25 \in [0.1879; 0.2565]$.

Careful

"Keep H_0" does not prove the rule of thumb – the data are merely *compatible* with it.

Solution (3/3) – inference for the shoe-size regression

(5) Prediction interval at $k_0 = 180$

$$\hat{s}_0 = 2.3147 + 0.2222 \cdot 180 = 42.31$$

$$42.31 \pm 2.1604 \cdot 0.7308 \sqrt{1 + \tfrac{1}{15} + \tfrac{(180-170.2)^2}{2122.4}} = 42.31 \pm 2.1604 \cdot 0.7308 \cdot 1.0546 = 42.31 \pm 1.66$$

$$\text{PI}_{0.95} = [\mathbf{40.65}; \mathbf{43.98}]$$

In words: A person of 180 cm will, with 95 % prediction probability, wear a shoe size between about 40.5 and 44. The CI for the *average* shoe size of all such persons is much narrower: [41.78; 42.84].

Exercise – used cars from A to Z

Problem statement

A dealer records the resale value y (in 1000 EUR) of six used cars against their age x (in years):

x_i (age)	1	2	3	4	5	6
y_i (value)	19	18	17	15	12	9

1. Estimate $\hat{\beta}_0, \hat{\beta}_1$ and write down the fitted line.
2. Compute the residuals, RSS, $\hat{\sigma}$ and R^2.
3. Compute $\mathrm{SE}(\hat{\beta}_1)$, the t-statistic and the 95 % confidence interval for β_1.
4. Predict the value of a 7-year-old car with a 95 % prediction interval.

Hint: $t_{0.975;\,4} = 2.776$.

Solution (1/3) – used cars from A to Z

Summaries: $n = 6$, $\bar{x} = 3.5$, $\bar{y} = 15$, $x_i - \bar{x} = (-2.5, -1.5, -0.5, 0.5, 1.5, 2.5)$, $y_i - \bar{y} = (4, 3, 2, 0, -3, -6)$.

$$S_{xy} = -35, \qquad S_{xx} = 17.5, \qquad S_{yy} = 74$$

(1) Coefficients and fitted line

$$\hat{\beta}_1 = \frac{-35}{17.5} = -\mathbf{2}, \qquad \hat{\beta}_0 = 15 - (-2) \cdot 3.5 = \mathbf{22}, \qquad \hat{y} = 22 - 2\,x$$

Each additional year of age lowers the value by 2000 euros on average.

Solution (2/3) – used cars from A to Z

(2) Fit

Fitted values $\hat{y}_i = (20, 18, 16, 14, 12, 10)$, residuals $\hat{u}_i = (-1, 0, 1, 1, 0, -1)$:

$$\text{RSS} = 4, \qquad \hat{\sigma} = \sqrt{\tfrac{4}{6-2}} = 1.0, \qquad R^2 = 1 - \tfrac{4}{74} = 0.946$$

(3) Inference on the slope

$$\text{SE}(\hat{\beta}_1) = \frac{1.0}{\sqrt{17.5}} = 0.239, \qquad T = \frac{-2}{0.239} = -8.37, \qquad |T| > 2.776 \;\Rightarrow\; \text{reject } H_0$$

$$\text{CI}_{0.95} = -2 \pm 2.776 \cdot 0.239 = [-2.66;\ -1.34]$$

In words: The annual loss of value lies between 1340 and 2660 euros; the interval excludes 0, in line with the t-test.

Solution (3/3) – used cars from A to Z

(4) Prediction at $x_0 = 7$

$\hat{y}_0 = 22 - 2 \cdot 7 = 8$ and $(x_0 - \bar{x})^2 = 12.25$:

$$8 \pm 2.776 \cdot 1.0\sqrt{1 + \tfrac{1}{6} + \tfrac{12.25}{17.5}} = 8 \pm 3.79 \;\Rightarrow\; \mathrm{PI}_{0.95} = [4.21;\, 11.79]$$

In words: The interval is wide: n is small, $x_0 = 7$ lies outside the observed ages (extrapolation), and it targets *one* car, not the average.

Common mistake

Dropping the "1+" under the root: 8 ± 2.58 is the confidence interval for the *average* 7-year-old car – far too narrow for a single car.

Overview

1 Relationship Between Two Metric Variables

2 Simple Linear Regression

3 Goodness of Fit: the Coefficient of Determination

4 Inference in Simple Regression

5 Multiple Linear Regression
- From one to many explanatory variables
- Interpreting the coefficients
- Inference for single coefficients
- F-test
- Goodness of fit and model choice
- Qualitative explanatory variables
- Exercise: reading a full output

6 Summary

Why one explanatory variable is rarely enough

Intuition

Sales do not depend on the price alone – advertising, season, location and competitors matter as well. Real questions involve **several** explanatory variables at once.

- Multiple regression studies the effect of one variable **while holding the others fixed** (*ceteris paribus*).
- Separate simple regressions cannot do this: if the explanatory variables move together, each simple slope quietly absorbs the effects of the others.

Key takeaway

One joint model disentangles the effects – this is what makes multiple regression the workhorse of quantitative research.

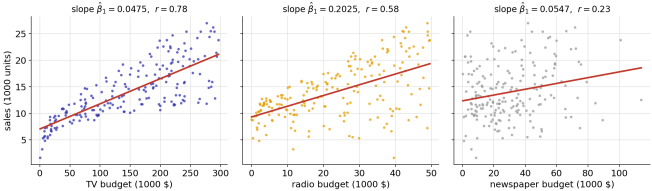
Our second story: the advertising data

The brief

A company has spent advertising budgets on TV, radio and newspaper in $n = 200$ markets and recorded the sales. *"Where should next year's budget go?"* (Section 2 started this story with twelve stores and one budget.)

Variables

- Y: `sales` (in 1000 units)
- X_1, X_2, X_3: `TV`, `radio`, `newspaper` budgets (in 1000 \$)



Source: Advertising data from James, Witten, Hastie, Tibshirani: An Introduction to Statistical Learning, www.statlearning.com

The multiple linear regression model

Model with p explanatory variables

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + u_i, \qquad i = 1, \ldots, n,$$

with the same four assumptions on the error terms u_i as before.

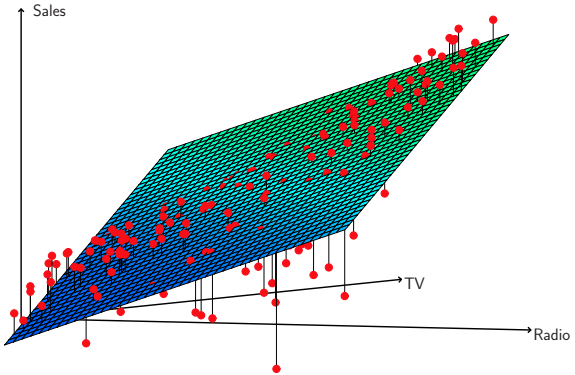
Interpretation of a coefficient

β_j is the expected change in Y when X_j increases by one unit **and all other explanatory variables are held fixed**.

Advertising

$$\texttt{sales}_i = \beta_0 + \beta_1 \, \texttt{TV}_i + \beta_2 \, \texttt{radio}_i + \beta_3 \, \texttt{newspaper}_i + u_i$$

Geometry: from a line to a plane



- With two explanatory variables the fitted model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ is a **plane**.
- The residuals are again the *vertical* distances of the points from the plane.
- With p variables: a "hyperplane" – no longer drawable, but the same idea.

Source: ISLR, Figure 3.5 (sales on TV and radio).

Least squares with several variables

Same principle

Choose $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$ such that the sum of squared residuals is minimal:

$$\min_{\hat{\beta}_0,\ldots,\hat{\beta}_p} \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}\right)^2.$$

- The solution is unique as long as no variable is an exact linear combination of the others.
- There is no simple "covariance over variance" formula any more – in matrix notation $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, which software computes for us.
- Residuals, RSS, TSS and $R^2 = 1 - \text{RSS}/\text{TSS}$ are defined exactly as before.

Multiple regression with Python

Fitting the model

```
ad = pd.read_csv("../data/Advertising.csv", index_col=0)
m3 = smf.ols("sales ~ TV + radio + newspaper", data=ad).fit()  # sales explained by ...
print(m3.summary()); print(np.sqrt(m3.scale))     # excerpt; sigma-hat in last line
```

```
Dep. Variable:                    sales   R-squared:                       0.897
Model:                              OLS   Adj. R-squared:                  0.896
Method:                   Least Squares   F-statistic:                     570.3
                                          Prob (F-statistic):           1.58e-96
                  coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      2.9389      0.312      9.422      0.000       2.324       3.554
TV             0.0458      0.001     32.809      0.000       0.043       0.049
radio          0.1885      0.009     21.893      0.000       0.172       0.206
newspaper     -0.0010      0.006     -0.177      0.860      -0.013       0.011
1.685510373414744
```

▶ **Try it:** notebook, section 5.1 (Colab)

Interpreting the coefficients – ceteris paribus

Advertising (budgets in 1000 \$, sales in 1000 units)

- **TV:** $\hat{\beta}_1 = 0.0458$ – holding radio and newspaper fixed, 1000 \$ more TV budget goes along with about $0.0458 \cdot 1000 = 46$ additional units sold.
- **Radio:** $\hat{\beta}_2 = 0.1885$ – holding TV and newspaper fixed, 1000 \$ more radio budget goes along with about 189 additional units.
- **Newspaper:** $\hat{\beta}_3 = -0.0010$ – essentially zero once TV and radio are in the model.
- **Intercept:** $\hat{\beta}_0 = 2.94$ – expected sales without any advertising (2940 units).

The clause that matters

"Holding the others fixed" – a coefficient in a multiple regression is a *partial* effect, not the total association of X_j with Y.

Simple vs. multiple slopes

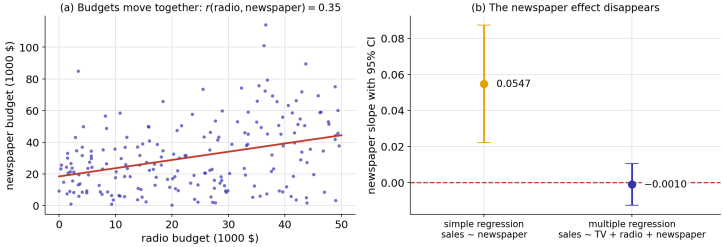
	Simple regression (one variable each)	**Multiple regression** (all three)
TV	0.0475	0.0458
Radio	0.2025	0.1885
Newspaper	0.0547 ($p = 0.001$)	−0.0010 ($p = 0.86$)

What just happened?

On its own, newspaper looks like a significant driver of sales – but its effect *disappears* once TV and radio are in the model.

In words: TV and radio hardly change – TV is almost uncorrelated with the other budgets, and radio is the variable that drives sales. Newspaper changes completely.

Confounding: why the newspaper slope collapses



- Markets with a large radio budget also spend more on newspapers ($r = 0.35$) – and radio really drives sales.
- In the simple regression, newspaper acts as a **proxy** for radio and "borrows" its effect.

Key takeaway

Confounding: a simple association can be produced by a third variable – include all relevant variables.

Exercise – why newspaper "works" alone

Problem statement

A simple regression of sales on the newspaper budget gives a positive, significant slope (0.0547, $p = 0.001$). In the multiple regression the newspaper coefficient is -0.0010 with $p = 0.86$. The correlation between the newspaper and radio budgets is 0.35.

1. Explain how newspaper can look useful on its own but useless in the joint model.
2. Which variable is the actual driver?
3. State the general lesson for interpreting simple correlations.

Solution – why newspaper "works" alone

1. **Confounding:** markets with high radio budgets also tend to have higher newspaper budgets. Radio raises sales. In the simple regression, the newspaper budget carries information about the radio budget that "travels with it" and inherits a positive slope. Once radio is included, radio absorbs this effect.
2. **Driver:** radio (and TV). Newspaper is a stand-in for radio in the simple regression.
3. **Lesson:** a simple association can be produced entirely by an omitted variable. To assess the effect of X_j *holding the others fixed*, all relevant variables must be in the model.

Common mistake

Concluding "newspaper advertising can never work". The model only says that newspaper adds nothing *given TV and radio, in these data* – an observational, not a causal statement.

t-test for a single coefficient

Same recipe – new degrees of freedom

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_1 : \beta_j \neq 0, \qquad T = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)} \sim t_{n-p-1} \text{ under } H_0$$

- $n - p - 1$ degrees of freedom: $p + 1$ coefficients were estimated (simple regression: $p = 1 \Rightarrow n - 2$).
- $\hat{\sigma} = \sqrt{\mathrm{RSS}/(n - p - 1)}$; the standard errors are computed by software.
- **Reading of H_0:** "X_j adds nothing *given the other variables in the model*".

Advertising ($n = 200$, $p = 3$, df $= 196$, $t_{0.975;\,196} = 1.972$)

Newspaper: $T = -0.0010/0.00587 = -0.18$, $|T| < 1.972 \Rightarrow$ keep H_0 ($p = 0.86$). TV ($T = 32.8$) and radio ($T = 21.9$): reject.

Confidence intervals in multiple regression

Construction

$$\mathrm{CI}_{1-\alpha}(\beta_j) = \hat{\beta}_j \pm t_{1-\alpha/2,\, n-p-1} \cdot \mathrm{SE}(\hat{\beta}_j)$$

Radio

$$0.18853 \pm 1.972 \cdot 0.00861 = 0.18853 \pm 0.01698 = [0.1715;\, 0.2055]$$

Holding TV and newspaper fixed, 1000 \$ more radio budget goes along with 172 to 206 additional units sold.

	TV	radio	newspaper
95 % CI	[0.0430; 0.0485]	[0.1715; 0.2055]	[−0.0126; 0.0105]

In words: The newspaper interval contains 0 – the same message as its t-test.

Is *any* explanatory variable useful? – the F-test

Hypotheses

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \quad \text{vs.} \quad H_1 : \text{at least one } \beta_j \neq 0$$

F-statistic

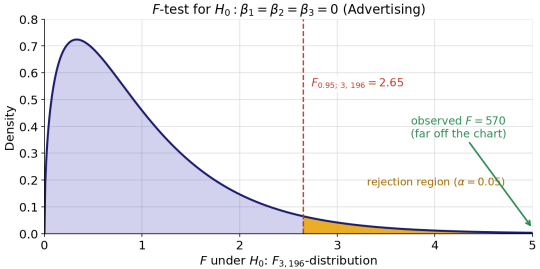
$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)} \sim F_{p,\, n-p-1} \text{ under } H_0$$

- Numerator: explained variation *per explanatory variable*.
- Denominator: unexplained variation per residual degree of freedom ($= \hat{\sigma}^2$).
- Large $F \Rightarrow$ the variables explain more than chance would $\Rightarrow$ reject H_0 (right-sided test).

F-test – advertising example

Given: TSS $= 5417.15$, RSS $= 556.83$, $n = 200$, $p = 3$.

$$F = \frac{(5417.15 - 556.83)/3}{556.83/196} = \frac{1620.11}{2.8410} = \mathbf{570.3} \gg 2.65 = F_{0.95;\,3,\,196}$$



In words: H_0 is rejected: at least one advertising medium is related to sales (`Prob (F-statistic)` $= 1.58 \cdot 10^{-96}$, practically 0).

Why not just several t-tests?

Multiple testing

With p explanatory variables and $\alpha = 0.05$ per test, even if *no* variable matters we expect about $0.05 \cdot p$ "significant" t-tests by chance alone – with $p = 20$ that is one false alarm per regression.

The F-test

One single test for all slopes together – its Type I error stays at α, however many variables are in the model. Consult it **first**, then look at the individual t-tests.

Special case

In simple regression ($p = 1$) the F-test and the two-sided t-test are identical: $F = T^2$ (lamp: $125.2 = (-11.19)^2$).

The F-test with Python

From the fitted model

```
m3 = smf.ols("sales ~ TV + radio + newspaper", data=ad).fit()
m3.fvalue, m3.f_pvalue        # 570.3  1.58e-96 ("F-statistic", "Prob (F-statistic)")
```

By hand

```
TSS, RSS = m3.centered_tss, m3.ssr                  # 5417.15  556.83
n, p = len(ad), 3                                   # 200  3
F = ((TSS - RSS) / p) / (RSS / (n - p - 1))         # 570.3
stats.f.ppf(0.95, p, n - p - 1)                     # critical value 2.65
stats.f.sf(F, p, n - p - 1)                         # p-value 1.58e-96
```

In words: `stats.f.sf()` is the area to the *right* of F: the F-test is right-sided.

▶ **Try it:** notebook, section 5.4 (Colab)

Exercise – the F-test

Problem statement

A multiple regression with $p = 3$ explanatory variables is fitted on $n = 100$ observations, giving $\text{TSS} = 1000$ and $\text{RSS} = 100$.

1. Compute R^2 and the F-statistic.
2. Compare with $F_{0.95;\,3,\,96} = 2.70$. What do you conclude?
3. Suppose instead 100 explanatory variables were all truly useless. About how many would appear "significant" at $\alpha = 0.05$ in individual t-tests?

Solution – the F-test

(1) R^2 and F

$$R^2 = 1 - \frac{100}{1000} = 0.9, \qquad F = \frac{(1000 - 100)/3}{100/(100 - 3 - 1)} = \frac{300}{1.0417} = \mathbf{288}$$

(2) $288 \gg 2.70 \Rightarrow$ reject $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$: at least one variable is related to Y.

(3) Each t-test rejects a true H_0 with probability 0.05, so about $100 \cdot 0.05 = 5$ variables would look significant purely by chance – which is why the F-test is consulted first.

Common mistake

Reading a significant F as "all three variables matter". It only says that *at least one* does; the t-tests show which.

R^2 and adjusted R^2

R^2 never decreases

Adding *any* variable – even pure noise – can only lower RSS, so R^2 never decreases. Comparing models by R^2 always favours the larger model.

Adjusted R^2

$$\bar{R}^2 = 1 - \frac{\mathrm{RSS}/(n-p-1)}{\mathrm{TSS}/(n-1)} = 1 - (1-R^2)\,\frac{n-1}{n-p-1}$$

Penalises each additional variable: $\bar{R}^2$ only rises if the new variable reduces RSS by more than chance would.

In words: The residual standard error $\hat{\sigma} = \sqrt{\mathrm{RSS}/(n-p-1)}$ is penalised in the same way.

Model choice – dropping newspaper

Model	$\hat{\beta}_{\text{TV}}$	$\hat{\beta}_{\text{radio}}$	R^2	$\bar{R}^2$	$\hat{\sigma}$
TV + radio + newspaper	0.0458	0.1885	0.89721	0.89564	1.6855
TV + radio	0.0458	0.1880	0.89719	0.89615	1.6814

Decision

The slopes barely move, R^2 is practically unchanged, $\bar{R}^2$ rises and $\hat{\sigma}$ falls: **keep the simpler model** `sales ~ TV + radio`.

Common mistake

Arguing with the unchanged R^2 – plain R^2 can only fall when a variable is dropped. Adjusted R^2 and $\hat{\sigma}$ are the fair judges.

Prediction in multiple regression

Same two intervals as before

For a market with TV $= 100$ and radio $= 20$ (in 1000 \$), the model `sales ~ TV + radio` predicts

$$\hat{y}_0 = 2.921 + 0.04575 \cdot 100 + 0.18799 \cdot 20 = 11.26 \text{ (1000 units)}.$$

	95 % interval	Question
Confidence interval	[10.99; 11.53]	mean sales of all such markets
Prediction interval	[7.93; 14.58]	sales of *one* such market

Key takeaway

With $n = 200$ the mean is estimated precisely – but a single market still varies by about $\pm 2\hat{\sigma} \approx \pm 3.3$.

What if an explanatory variable is categorical?

Many relevant variables are qualitative (student yes/no, region, brand) – but regression needs numbers.

Solution: dummy variable (0/1 coding)

$$D_i = \begin{cases} 1 & \text{if person } i \text{ is a student,} \\ 0 & \text{otherwise (baseline).} \end{cases}$$

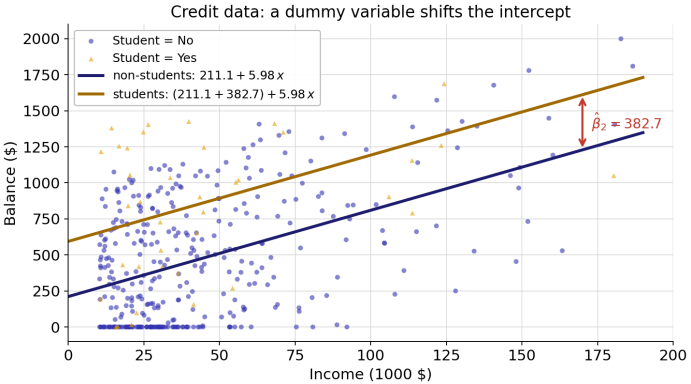
Credit data ($n = 400$): balance on student status

$$\widehat{\texttt{Balance}} = 480.37 + 396.46 \cdot D$$

Non-students: 480.37 \$ on average; students: $480.37 + 396.46 = 876.83$ \$. The coefficient of D is the **difference in means** between the groups ($p < 0.0001$).

Source: Credit data from James et al.: An Introduction to Statistical Learning.

A dummy shifts the intercept



$$\widehat{\texttt{Balance}} = 211.1 + 5.98\,\texttt{Income} + 382.7\,D$$

- Two **parallel** lines: same slope for income, intercept shifted by $\hat{\beta}_2 = 382.7$.
- Reading: at the *same* income, students carry on average 382.7 $ more balance.

K categories: use $K - 1$ dummies

Rule

A variable with K categories is coded with $K - 1$ dummies; the remaining category is the **baseline** represented by the intercept.

Why not K dummies? (dummy variable trap)

The K dummies always sum to 1 – exactly the column of the intercept. One variable would be an exact linear combination of the others, and least squares has no unique solution.

Credit data: region $\in$ {East, South, West}, baseline East

$$\widehat{\texttt{Balance}} = 531.00 - 12.50\, D_{\text{South}} - 18.69\, D_{\text{West}}$$

Both dummy coefficients are *differences to East* ($p = 0.83$ and $p = 0.77$): no evidence for regional differences.

Dummy variables with Python

A text column becomes a dummy automatically

```
cr = pd.read_csv("../data/Credit.csv")
smf.ols("Balance ~ Student", data=cr).fit().params
# Intercept 480.37, Student[T.Yes] 396.46
cr.groupby("Student")["Balance"].mean()      # No 480.37, Yes 876.825
```

Dummy plus metric variable; $K = 3$ categories

```
smf.ols("Balance ~ Income + Student", data=cr).fit().params
# Intercept 211.14, Student[T.Yes] 382.67, Income 5.98
smf.ols("Balance ~ Region", data=cr).fit().params
# Intercept 531.00, Region[T.South] -12.50, Region[T.West] -18.69
```

In words: Baseline: the first category in alphabetical order (No, East).

▶ **Try it:** notebook, section 5.6 (Colab)

Exercise – dummy coding

Problem statement

1. From $\widehat{\texttt{Balance}} = 480.37 + 396.46\, D_{\text{Student}}$, give the predicted mean balance of students and of non-students. Interpret 396.46.
2. From $\widehat{\texttt{Balance}} = 531.00 - 12.50\, D_{\text{South}} - 18.69\, D_{\text{West}}$ (baseline East), give the predicted mean balance in each region. How large is the difference between South and West?
3. Why are only $K - 1$ dummies used for a variable with K categories?

Solution – dummy coding

(1) Non-students ($D = 0$): 480.37 \$; students ($D = 1$): $480.37 + 396.46 = 876.83$ \$. The coefficient 396.46 is the **difference in average balance** between students and non-students.

(2) East $= 531.00$, South $= 531.00 - 12.50 = 518.50$, West $= 531.00 - 18.69 = 512.31$. South vs. West: $-12.50 - (-18.69) = +6.19$ \$.

(3) With K dummies plus intercept, the dummies sum to the intercept column (dummy variable trap) – the coefficients would not be identified. The dropped category becomes the baseline.

Common mistake

Reading -12.50 as "the mean of the South". Dummy coefficients are *contrasts* against the baseline, not group means.

Exercise – reading a multiple regression output

Credit data ($n = 400$): balance ($) explained by income (1000 $), credit limit ($) and student status.

```
>>> mc = smf.ols("Balance ~ Income + Limit + Student", data=cr).fit()
>>> print(mc.summary())                                          # excerpt
Dep. Variable:                 Balance   R-squared:                       0.949
Model:                             OLS   Adj. R-squared:                  0.948
Method:                  Least Squares   F-statistic:                     2447.
                                         Prob (F-statistic):          4.00e-255
                    coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept       -432.3374     12.431    -34.778      0.000    -456.777    -407.898
Student[T.Yes]   427.0233     17.416     24.519      0.000     392.784     461.263
Income            -7.9016      0.243    -32.495      0.000      -8.380      -7.424
Limit              0.2675      0.004     72.069      0.000       0.260       0.275
```

1. Interpret the coefficients of `Limit` and `Student[T.Yes]`.
2. Compute the 95 % CI for the income coefficient by hand ($t_{0.975;\,396} = 1.966$, unrounded `std err` 0.2432) and compare it with the columns `[0.025  0.975]`.
3. Alone, `Balance ~ Income` has slope $+6.05$. Why is it negative here? ($r(\text{Income}, \text{Limit}) = 0.79$.)
4. What do the F-test and R^2 say? Predict the balance of a non-student with income 50 and limit 5000.

Solution (1/2) – reading a multiple regression output

(1) Holding income and student status fixed, 1 \$ more credit limit goes along with 0.27 \$ more balance (1000 \$ limit $\rightarrow$ 267.5 \$). At the same income and limit, students (`Student[T.Yes]`: dummy = 1 for "Yes", baseline "No") carry on average 427 \$ more balance than non-students.

(2)

$$-7.902 \pm 1.966 \cdot 0.2432 = -7.902 \pm 0.478 = [-8.380;\ -7.424]$$

1000 \$ more income goes along with 7.4 to 8.4 \$ *less* balance – holding limit and student status fixed. This is exactly the interval in the columns `[0.025  0.975]`; 0 lies far outside it.

(3) Confounding: people with higher incomes get higher credit limits ($r = 0.79$), and a high limit means a high balance. Alone, income borrows the limit's effect (+6.05); at a *fixed* limit, higher income goes along with a *lower* balance.

Solution (2/2) – reading a multiple regression output

(4) F-test and R^2

`F-statistic` $= 2447$ with `Prob (F-statistic)` $= 4.00 \cdot 10^{-255}$, i.e. $p \approx 0$: at least one variable is related to the balance. The model explains 94.9 % of the variance (`R-squared`); `Adj. R-squared` $= 0.948$ is almost identical with only $p = 3$ variables.

(4) Prediction (non-student, income 50, limit 5000)

$$\hat{y}_0 = -432.3 - 7.902 \cdot 50 + 0.2675 \cdot 5000 + 427.0 \cdot 0 = -432.3 - 395.1 + 1337.5 \approx \mathbf{510}\,\$$$

Python (unrounded, `mc.get_prediction(new).summary_frame()`): 510.3, 95 % prediction interval (`obs_ci_*`) [304.7; 715.8]. A student with the same income and limit: 937.3 $.

In words: Every number in a regression output has a meaning – but only together with the clause "holding the other variables fixed".

Multiple regression with Python – overview

Fitting, testing, comparing

```
m3 = smf.ols("sales ~ TV + radio + newspaper", data=ad).fit()
print(m3.summary())          # t-tests, R^2, adjusted R^2, F-test
m3.conf_int()                # confidence intervals
m2 = smf.ols("sales ~ TV + radio", data=ad).fit()
sm.stats.anova_lm(m2, m3)    # F-test: does newspaper add anything? (p = 0.86)
```

Dummy variables and prediction

```
cr = pd.read_csv("../data/Credit.csv")
smf.ols("Balance ~ Income + Student", data=cr).fit()  # text -> dummy Student[T.Yes]
new = pd.DataFrame({"TV": [100], "radio": [20]})
m2.get_prediction(new).summary_frame()      # obs_ci_*: prediction interval
plt.scatter(m2.fittedvalues, m2.resid); plt.axhline(0)   # residual plot
```

→ *Appendix: interactions, diagnostics in multiple regression, collinearity*

▶ **Try it:** notebook, section 5.7 (Colab)

Overview

1 Relationship Between Two Metric Variables
2 Simple Linear Regression
3 Goodness of Fit: the Coefficient of Determination
4 Inference in Simple Regression
5 Multiple Linear Regression
6 Summary

Summary 1 – correlation and simple regression

Quantity	Formula
Covariance	$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$
Correlation coefficient	$r_{xy} = \frac{s_{xy}}{s_x\, s_y}, \qquad -1 \le r_{xy} \le 1$
Model	$y_i = \beta_0 + \beta_1 x_i + u_i, \qquad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$
Least squares	$\min_{\beta_0,\beta_1} S(\beta_0, \beta_1) = \sum(y_i - \beta_0 - \beta_1 x_i)^2$
First-order conditions	$\sum \hat{u}_i = 0, \quad \sum x_i \hat{u}_i = 0$ (line through $(\bar{x}, \bar{y})$)
Slope	$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{S_{xy}}{S_{xx}} = r_{xy}\frac{s_y}{s_x}$
Intercept	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
Coefficient of determination	$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$ $\left(= r_{xy}^2 \text{ in simple regression}\right)$

Summary 2 – inference for regression coefficients

Standard error and residual standard error

$$\hat{\sigma} = \sqrt{\frac{\text{RSS}}{n-p-1}}, \qquad \text{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}} \quad \text{(simple regression, } p = 1\text{)}$$

t-test, confidence interval, p-value

$$T = \frac{\hat{\beta}_j - \beta_{j,0}}{\text{SE}(\hat{\beta}_j)} \sim t_{n-p-1}, \qquad \hat{\beta}_j \pm t_{1-\alpha/2,\, n-p-1}\, \text{SE}(\hat{\beta}_j), \qquad p = 2\, F_{t,n-p-1}(-|T|)$$

Prediction at x_0 (simple regression)

$$\hat{y}_0 \pm t_{1-\alpha/2,\, n-2}\, \hat{\sigma} \sqrt{\underbrace{1}_{\text{PI only}} + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Summary 3 – multiple regression

- β_j: expected change in Y per unit of X_j, **holding all other variables fixed**.
- **Confounding**: simple and multiple slopes can differ completely (newspaper: $0.055 \to -0.001$).
- F-**test** for $H_0 : \beta_1 = \cdots = \beta_p = 0$: $F = \dfrac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)}$ – first check, then the t-tests.
- **Adjusted** $\bar{R}^2 = 1 - \dfrac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}$ for comparing models of different size.
- **Dummies**: K categories $\to K-1$ dummies; coefficients are differences to the baseline.

$\to$ *Appendix: interactions, diagnostics, collinearity (VIF)*

Summary 4 – important Python commands

Description and estimation

```
plt.scatter(df["x"], df["y"])    # scatter
df["x"].cov(df["y"])             # covariance
df["x"].corr(df["y"])            # correlation
reg = smf.ols("y ~ x", data=df).fit()
b0, b1 = reg.params              # estimates
plt.axline((0, b0), slope=b1)    # line
reg.predict(new)                 # prediction
print(reg.summary())     # SE, t, p, R^2, F
```

Inference and diagnostics

```
reg.conf_int()                          # CIs
reg.get_prediction(new).summary_frame()
plt.scatter(reg.fittedvalues, reg.resid)
sm.qqplot(reg.resid, line="q", a=0.5)   # QQ
stats.norm.ppf(p)                       # z_i
reg.get_influence().hat_matrix_diag
smf.ols("y ~ x1 + x2 + x3", data=df).fit()
sm.stats.anova_lm(small, large) # F-test
```

▶ **Try it:** notebook, Key Python commands (Colab)

Five classic pitfalls – final checklist

Forbidden sentence ×	**Correct version** ✓
"$r_{xy} = -0.97$, so price *causes* the sales to fall."	Correlation and regression show association, not causation.
"R^2 is high, so the model is correct."	Check the residual plot; a curved relationship can have a high R^2.
"The slope is not significant, so X has no effect."	No evidence found – the CI shows which effects are still compatible.
"In the multiple regression, $\hat{\beta}_j$ is the total effect of X_j."	It is the effect *holding the other variables fixed*.
"The model predicts -1.17 lamps at 125 euros."	No extrapolation far beyond the observed x-range.

Outlook: Part 5 – Classification

What we have so far

Regression explains a **metric** variable Y (sales, shoe size, balance) by one or several explanatory variables – with full inference.

Next part

What if Y is **categorical** – default yes/no, spam or not? A straight line would predict "probabilities" below 0 or above 1. **Logistic regression** keeps the linear model on the log-odds scale and reuses everything from this part: coefficients, standard errors, tests, confidence intervals and confounding.

Key takeaway

Classification = regression for categories – plus the question of how to turn a probability into a decision.

Overview

7 Appendix: Optional and Advanced Material
- A.1 Derivation of the OLS estimators
- A.2 Minimum property of OLS
- A.3 Residual analysis
- A.4 Outliers and leverage
- A.5 QQ-plot
- A.6 Interactions
- A.7 Diagnostics in multiple regression

Appendix – what is in here, and why

Optional material

The main part is self-contained. The appendix holds the derivations, the model checks and material for projects with real data.

Topic	Content
A.1 Derivation of OLS	Every calculation step from the minimisation problem to the OLS estimators, with an exercise
A.2 Minimum property	Why the OLS line is the unique minimum – without calculus
A.3 Residual analysis	Does a line fit? Residual plot, typical patterns, exercises
A.4 Outliers and leverage	Unusual points and the leverage value h_i
A.5 QQ-plot	Are the errors normally distributed? QQ-plot by hand and in Python
A.6 Interactions	When the effect of one variable depends on another
A.7 Diagnostics	Multiple regression: residual plots, leverage, collinearity, VIF

Appendix: derivation (1/7) – starting point and plan

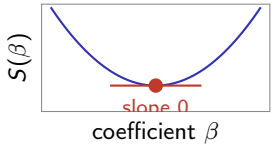
Where do $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$ come from? Starting point:

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i\right)^2 \;\to\; \min \qquad \Longrightarrow \qquad \frac{\partial S}{\partial \beta_0} = 0, \quad \frac{\partial S}{\partial \beta_1} = 0$$

β_0, β_1: arbitrary candidates; the solution: $\hat{\beta}_0$, $\hat{\beta}_1$. *Partial*: differentiate with respect to β_0 and treat β_1 as a fixed number (and vice versa).

Plan – the four steps of the main part

(2), (3)	step 1: form both partial derivatives
(4), (5)	steps 2–4: set to zero, solve for β_0, insert
(6)	bring the result into deviation form
(7)	check that it really is a minimum



back

Appendix: derivation (2/7) – derivative with respect to β_0

A single term: chain rule

$$\frac{\partial}{\partial \beta_0}\left(y_i - \beta_0 - \beta_1 x_i\right)^2 = \underbrace{2\left(y_i - \beta_0 - \beta_1 x_i\right)}_{\text{outer}} \cdot \underbrace{(-1)}_{\text{inner}} = -2\left(y_i - \beta_0 - \beta_1 x_i\right)$$

Inner function $y_i - \beta_0 - \beta_1 x_i$: differentiated with respect to β_0 it leaves -1, because y_i and $\beta_1 x_i$ do not depend on β_0.

All terms: sum rule

$$\frac{\partial S}{\partial \beta_0} = \sum_{i=1}^{n} -2\left(y_i - \beta_0 - \beta_1 x_i\right) = -2\sum_{i=1}^{n}\left(y_i - \beta_0 - \beta_1 x_i\right)$$

The derivative of a sum is the sum of the derivatives; the constant factor -2 moves in front of the sum.

Appendix: derivation (3/7) – derivative with respect to β_1

A single term: chain rule

$$\frac{\partial}{\partial \beta_1}\big(y_i - \beta_0 - \beta_1 x_i\big)^2 = \underbrace{2\big(y_i - \beta_0 - \beta_1 x_i\big)}_{\text{outer}} \cdot \underbrace{\big(-x_i\big)}_{\text{inner}} = -2\, x_i\big(y_i - \beta_0 - \beta_1 x_i\big)$$

Differentiated with respect to β_1: $\partial(-\beta_1 x_i)/\partial\beta_1 = -x_i$; y_i and β_0 do not depend on β_1.

All terms: sum rule

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^{n} x_i\big(y_i - \beta_0 - \beta_1 x_i\big)$$

Interim result

Both derivatives contain the residuals $y_i - \beta_0 - \beta_1 x_i$ of the respective line.

Appendix: derivation (4/7) – first condition: the intercept

$\partial S/\partial \beta_0 = 0$ step by step

$$
\begin{aligned}
-2\sum(y_i - \beta_0 - \beta_1 x_i) &= 0 && \mid \text{divide by } (-2)\\
\sum(y_i - \beta_0 - \beta_1 x_i) &= 0 && \mid \text{split the sum}\\
\sum y_i - \sum \beta_0 - \sum \beta_1 x_i &= 0 && \mid \textstyle\sum_{i=1}^{n} \beta_0 = n\beta_0,\ \ \sum \beta_1 x_i = \beta_1 \sum x_i\\
\sum y_i - n\beta_0 - \beta_1 \sum x_i &= 0 && \mid +n\beta_0\\
n\beta_0 &= \sum y_i - \beta_1 \sum x_i && \mid \text{divide by } n\\
\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} && \mid \text{as } \tfrac{1}{n}\sum y_i = \bar{y},\ \tfrac{1}{n}\sum x_i = \bar{x}
\end{aligned}
$$

Line 4 rearranged is the **first normal equation** $\sum y_i = n\beta_0 + \beta_1 \sum x_i$. Line 2 with the OLS values: $\sum \hat{u}_i = 0$ – the residuals of the OLS line sum to zero.

Appendix: derivation (5/7) – second condition: the slope

$\partial S/\partial \beta_1 = 0$ and insert the intercept

$$-2\sum x_i(y_i - \beta_0 - \beta_1 x_i) = 0 \quad | \text{ divide by } (-2)$$

$$\sum x_i(y_i - \beta_0 - \beta_1 x_i) = 0 \quad | \text{ multiply out, split the sum}$$

$$\sum x_i y_i - \beta_0 \sum x_i - \beta_1 \sum x_i^2 = 0 \quad | \text{ rearranged: 2nd normal equation}$$

$$\sum x_i y_i - (\bar{y} - \beta_1 \bar{x})\, n\bar{x} - \beta_1 \sum x_i^2 = 0 \quad | \ \beta_0 = \bar{y} - \beta_1 \bar{x} \text{ (4/7)}, \ \sum x_i = n\bar{x}$$

$$\sum x_i y_i - n\bar{x}\bar{y} + \beta_1 n\bar{x}^2 - \beta_1 \sum x_i^2 = 0 \quad | \text{ expand the bracket}$$

$$\sum x_i y_i - n\bar{x}\bar{y} = \beta_1\left(\sum x_i^2 - n\bar{x}^2\right) \quad | \ \beta_1 \text{ terms to the right, factor out}$$

$$\hat{\beta}_1 = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} \quad | \text{ divide by the bracket } (= S_{xx} > 0)$$

Line 2 with the OLS values: $\sum x_i \hat{u}_i = 0$ (see "Both estimators at a glance"). 2nd normal equation: $\sum x_i y_i = \beta_0 \sum x_i + \beta_1 \sum x_i^2$.

Exercise – online shop (A): solving the normal equations

Sums of the online shop

$n = 5, \quad \sum x_i = 15, \quad \sum y_i = 130$

$\sum x_i^2 = 1 + 4 + 9 + 16 + 25 = 55$

$\sum x_i y_i = 21 + 52 + 72 + 120 + 145 = 410$

Normal equations ((4/7) and (5/7))

I: $\sum y_i = n\,\beta_0 + \beta_1 \sum x_i$

II: $\sum x_i y_i = \beta_0 \sum x_i + \beta_1 \sum x_i^2$

Problem statement

1. Write down both normal equations with these sums.
2. Solve the system of equations for β_0 and β_1.
3. Compare the result with exercise (5).

Solution – online shop (A): solving the normal equations

(1) Insert the sums

I: $130 = 5\,\beta_0 + 15\,\beta_1$

II: $410 = 15\,\beta_0 + 55\,\beta_1$

(3) Comparison; Python check

$\hat{y} = 20 + 2\,x$: the OLS line of exercise (5) – obtained here without s_{xy} and s_x^2, directly from the sums. ✓

```
A = [[5, 15], [15, 55]]  # I, II
np.linalg.solve(A, [130, 410])
# array([20., 2.]): beta0, beta1
```

(2) Rule: solve I for β_0, insert into II

I divided by 5: $\beta_0 = 26 - 3\,\beta_1$

into II: $410 = 15\,(26 - 3\,\beta_1) + 55\,\beta_1$

$410 = 390 - 45\,\beta_1 + 55\,\beta_1 = 390 + 10\,\beta_1$

$10\,\beta_1 = 20 \;\Rightarrow\; \hat{\beta}_1 = \mathbf{2}$

$\hat{\beta}_0 = 26 - 3 \cdot 2 = \mathbf{20}$

In words: Solve (I) for β_0, insert into (II): exactly what (4/7) and (5/7) do in general.

▶ **Try it:** notebook, section A.1 (Colab)

Appendix: derivation (6/7) – back to the deviation form

Numerator from (5/7): multiply out the cross-products

$$\begin{aligned}
\sum(x_i-\bar{x})(y_i-\bar{y}) &= \sum\left(x_iy_i-\bar{y}\,x_i-\bar{x}\,y_i+\bar{x}\bar{y}\right) && \text{multiply out}\\
&= \sum x_iy_i-\bar{y}\sum x_i-\bar{x}\sum y_i+n\,\bar{x}\bar{y} && \text{split the sum, } \textstyle\sum_{i=1}^{n}\bar{x}\bar{y}=n\,\bar{x}\bar{y}\\
&= \sum x_iy_i-n\bar{x}\bar{y}-n\bar{x}\bar{y}+n\bar{x}\bar{y} && \textstyle\sum x_i=n\bar{x},\ \sum y_i=n\bar{y}\\
&= \sum x_iy_i-n\,\bar{x}\bar{y} && \text{combine}
\end{aligned}$$

Denominator: likewise (binomial formula)

$$\sum(x_i-\bar{x})^2=\sum\left(x_i^2-2\bar{x}\,x_i+\bar{x}^2\right)=\sum x_i^2-2n\bar{x}^2+n\bar{x}^2=\sum x_i^2-n\,\bar{x}^2$$

Result (shift theorem)

$$\hat{\beta}_1=\frac{\sum x_iy_i-n\bar{x}\bar{y}}{\sum x_i^2-n\bar{x}^2}=\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}=\frac{S_{xy}}{S_{xx}}$$

Appendix: derivation (7/7) – is it really a minimum?

Second partial derivatives: differentiate (2) and (3) once more

$$\frac{\partial^2 S}{\partial \beta_0^2} = -2\sum(-1) = 2n \qquad \frac{\partial^2 S}{\partial \beta_1^2} = -2\sum x_i(-x_i) = 2\sum x_i^2 \qquad \frac{\partial^2 S}{\partial \beta_0\,\partial \beta_1} = -2\sum(-x_i) = 2\sum x_i$$

Minimum in two variables: both conditions hold

$$\text{(1)} \quad \frac{\partial^2 S}{\partial \beta_0^2} = 2n > 0$$

$$\text{(2)} \quad \frac{\partial^2 S}{\partial \beta_0^2} \cdot \frac{\partial^2 S}{\partial \beta_1^2} - \left(\frac{\partial^2 S}{\partial \beta_0\,\partial \beta_1}\right)^2 = 2n \cdot 2\sum x_i^2 - \left(2\sum x_i\right)^2$$

$$= 4\left(n\sum x_i^2 - n^2\bar{x}^2\right) = 4n\left(\sum x_i^2 - n\bar{x}^2\right) = 4n\,S_{xx} > 0 \qquad \Big| \ \sum x_i = n\bar{x}$$

In words: S curves upwards in every direction – a "bowl" with exactly one minimum, as long as the x_i are not all equal ($S_{xx} > 0$). A check without calculus: A.2.

back

Why no other line beats the OLS line

That the OLS solution is the *unique* minimum can also be seen without calculus. For any line $\beta_0 + \beta_1 x$ let $a = \beta_0 - \hat{\beta}_0$, $b = \beta_1 - \hat{\beta}_1$; its residuals are $\hat{u}_i - (a + b x_i)$.

Decomposition of the objective function

$$S(\beta_0, \beta_1) = \sum \hat{u}_i^2 - 2a \underbrace{\sum \hat{u}_i}_{=0} - 2b \underbrace{\sum x_i \hat{u}_i}_{=0} + \sum (a + b x_i)^2 \qquad \triangleright \text{ first-order conditions}$$

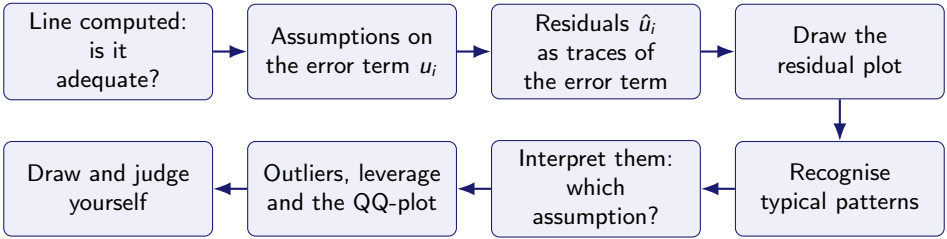
$$= \text{RSS} + \sum \big((a + b\bar{x}) + b\,(x_i - \bar{x})\big)^2 = \text{RSS} + n\,(a + b\bar{x})^2 + b^2\, S_{xx} \qquad \triangleright \sum (x_i - \bar{x}) = 0$$

In words: Every other line pays two penalties – for missing the centroid and for the wrong slope. Both vanish only for $a = b = 0$ (if $S_{xx} > 0$).

Twelve stores (RSS $= 64.8$, $n = 12$, $\bar{x} = 7.5$, $S_{xx} = 155$)

A ($a = -6$, $b = 0.8$), B ($a = 4.5$, $b = -0.6$) pass through the centroid ($a + b\bar{x} = 0$): $64.8 + 0.8^2 \cdot 155 = \mathbf{164.0}$, $64.8 + 0.6^2 \cdot 155 = \mathbf{120.6}$. C ($a = 5$, $b = 0$) is parallel to D: $64.8 + 12 \cdot 5^2 = \mathbf{364.8}$.

Roadmap: does a straight line fit the data at all?

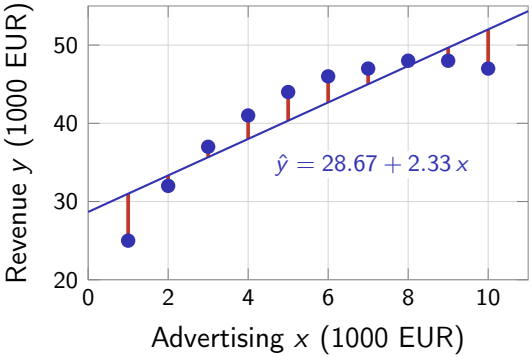


Key takeaway

The OLS line can be computed for *every* cloud of points. Whether a line describes the data adequately is revealed only by the **residuals**.

Model $y_i = \beta_0 + \beta_1 x_i + u_i$ and the assumptions on u_i: Section 4 ("The regression model").

A line can always be computed – but does it fit?



Ten regions, OLS line computed as usual. $R^2 = 0.81$.
Does it describe the data well?

Observation

At the edges the points lie *below* the line, in the middle *above* it. Residuals: -6.0, -1.3, 1.3, 3.0, 3.7, 3.3, 2.0, 0.7, -1.7, -5.0.

Cause

Saturation: additional advertising has less and less effect. The relationship is not linear.

Guiding question

How do we detect such patterns *systematically* – even when they are less obvious?

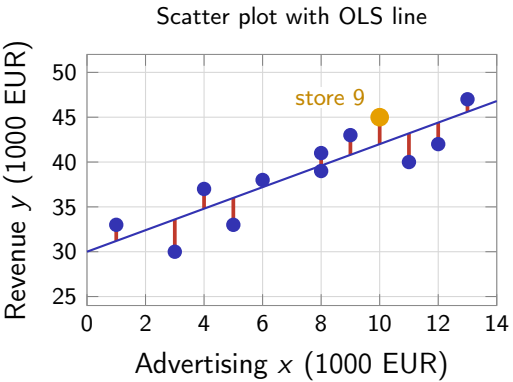
What happens if an assumption is violated?

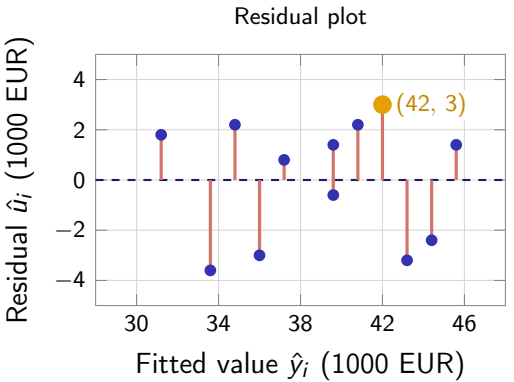
Assumption	Typically violated with	Consequence	Check
Expected value zero	a non-linear relationship, a missing explanatory variable	line systematically wrong, estimates **biased**	residual plot: **arc**
Homoscedasticity	spread growing with x (large stores fluctuate more)	estimates unbiased, but standard errors and tests **unreliable**	residual plot: **funnel**
No autocorrelation	time series (monthly revenues)	as for heteroscedasticity	residuals over time: **waves**
Normal distribution	skewed distributions, outliers	confidence intervals and tests unreliable for small n	**QQ-plot**

Key takeaway

The OLS line can always be computed. The assumptions decide how reliable the conclusions about the *true* coefficients β_0, β_1 are.

From the scatter plot to the residual plot





Each point moves to $(\hat{y}_i, \hat{u}_i)$, store 9 to (42, 3). **The line becomes the zero line**; the deviations are magnified. Here: no arc, no funnel, no outlier (largest $|\hat{u}_2| = 3.6$) – the line is adequate.

The residual plot

Layout

x-axis	fitted values $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$
y-axis	residuals $\hat{u}_i = y_i - \hat{y}_i$
zero line	corresponds to the OLS line

The points $(\hat{y}_i, \hat{u}_i)$, $i = 1, \ldots, n$, are plotted.

Why $\hat{y}_i$ and not x_i?

$\hat{y}_i$ is just a linear transformation of x_i: the pattern stays the same (mirrored if $\hat{\beta}_1 < 0$). With $\hat{y}_i$ the plot also works with several explanatory variables.

What always holds

The OLS residuals add up to zero (first-order condition): $\sum \hat{u}_i = 0$. So the points always lie around the zero line *as a whole* – only their **structure** is informative.

Key takeaway

Ideal: a **structureless** cloud of points in a band of constant width around the zero line. Then the line has captured everything systematic.

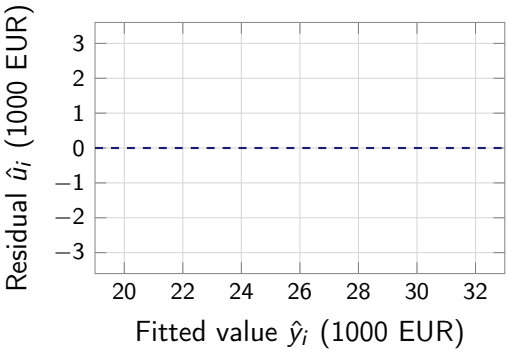
Exercise – online shop (B): draw the residual plot

Online shop: OLS line $\hat{y} = 20 + 2\,x$

Month i	1	2	3	4	5
x_i	1	2	3	4	5
y_i	21	26	24	30	29

Tasks

1. Compute the fitted values $\hat{y}_i$.
2. Compute the residuals $\hat{u}_i$; check their sum.
3. Draw the residual plot.
4. Judge: is there a pattern?



Template for your drawing

Solution – online shop (B): draw the residual plot

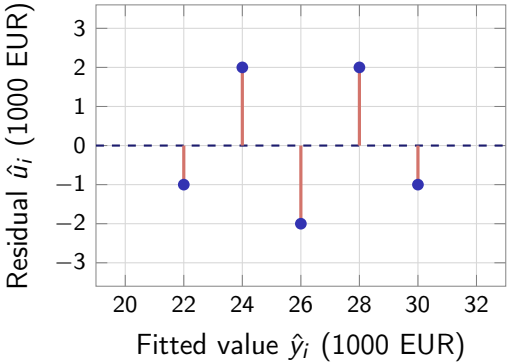
(1) and (2) Working table

i	x_i	y_i	$\hat{y}_i = 20 + 2x_i$	$\hat{u}_i = y_i - \hat{y}_i$
1	1	21	22	-1
2	2	26	24	2
3	3	24	26	-2
4	4	30	28	2
5	5	29	30	-1
$\sum$				**0**

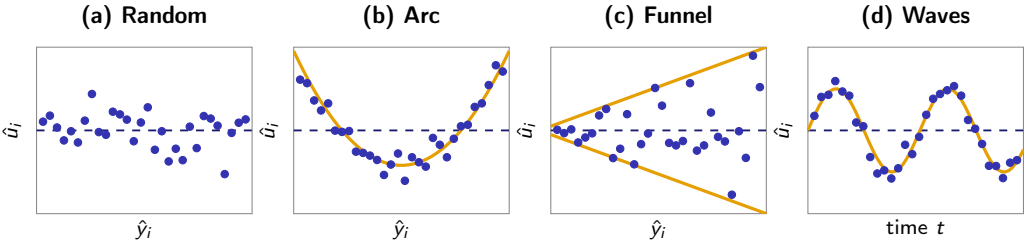
(4) Assessment

No arc, no funnel. The signs alternate – with only five values **no reliable pattern**.

(3) Residual plot



Four typical patterns in the residual plot



even band

model fits

first positive, then negative, then positive

wrong functional form

spread grows with $\hat{y}$

heteroscedasticity

long runs of the same sign

autocorrelation

For time series, the residuals are also plotted in time order: only then do waves become visible.

Interpreting patterns: what is behind them?

Pattern	What you see	Points to	Possible remedy (outlook)
Random	irregular cloud, equally wide everywhere	model adequate	–
Arc	residuals in groups: first above, then below zero (or vice versa)	non-linear relationship, missing explanatory variable	other functional form, e.g. $\log x$ or a quadratic term x^2
Funnel	spread grows (or shrinks) with $\hat{y}$	heteroscedasticity	transform y (e.g. $\log y$), robust standard errors
Waves	long runs of the same sign in time order	autocorrelation	time-series methods

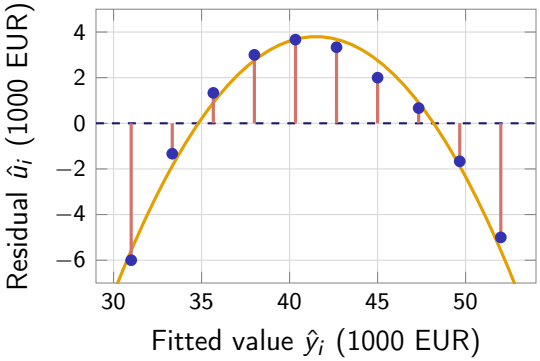
Single outliers

Check the cause (typo? special promotion?) – do *not* delete automatically.

Few observations

With small n, apparent patterns arise by chance: take only clear structures seriously.

Back to the saturation example



Finding

The residuals form a clear **arc**: negative, positive, negative.

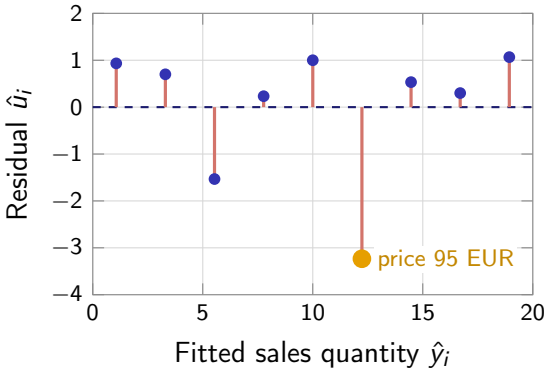
Interpretation

The line underestimates the middle and overestimates the edges: **wrong functional form**. A curve with decreasing increments would fit better.

Key takeaway

The residual plot shows what the line alone – and a high $R^2 = 0.81$ – do not reveal.

Lamp example: assessing the residual plot



$\hat{y} = 54.67 - 0.4467\,x$; high prices $\Rightarrow$ small $\hat{y}_i$ (left)

Checklist

- Arc or funnel? No clear structure.
- Outlier? One **conspicuous residual**: at a price of 95 euros, 9 instead of the expected 12.2 lamps were sold ($\hat{u}_4 = -3.23$).

Conclusion

No clear systematic pattern. Check the conspicuous store substantively (supply shortage? recording error?). With $n = 9$ every verdict is uncertain.

Creating a residual plot with Python

Python commands

```
reg = smf.ols("quantity ~ price", data=df).fit()
plt.scatter(reg.fittedvalues, reg.resid)
plt.axhline(0, linestyle="--")      # zero line for orientation
plt.xlabel("fitted values"); plt.ylabel("residuals"); plt.title("Residual plot")
```

Attributes of the fitted model

- `reg.fittedvalues`: fitted values $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ (x-axis)
- `reg.resid`: residuals $\hat{u}_i = y_i - \hat{y}_i$ (y-axis)

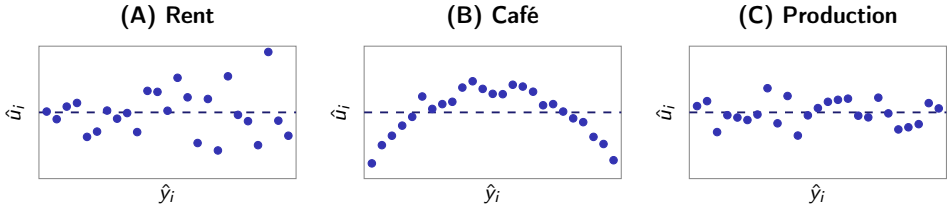
Check

```
reg.resid.sum()      # practically 0 (rounding): first-order condition
```

▶ **Try it:** notebook, section A.3 (Colab)

Exercise – diagnosing residual plots

Three regressions ($n = 25$ each): **(A)** monthly rent on floor space; **(B)** daily revenue of a café on outside temperature; **(C)** production costs on number of units.



Tasks

1. Which pattern does each residual plot show?
2. Which assumption on u_i is violated – if any? Give a reason.
3. What does this mean for the OLS line, and what should be done?

Solution – diagnosing residual plots

Rule: ideal = structureless band; arc $\rightarrow E(u_i) = 0$ violated; funnel $\rightarrow$ homoscedasticity violated.

Plot	(1) Pattern	(2) Violated assumption, reason	(3) Consequence, remedy
(A) Rent	**funnel**: spread grows with $\hat{y}$	**homoscedasticity**: rents of large flats vary more (location, fittings).	line unbiased, but SEs and tests unreliable. Transform ($\log y$), robust SEs.
(B) Café	**arc**: $-$, $+$, $-$	**expected value zero**: not linear – in great heat revenue falls again.	line systematically wrong, estimates biased. Add a quadratic term.
(C) Production	**random**: even band	**none** recognisably violated.	line adequate; nothing to change.

Common mistake: judging the assumptions from R^2 – a high R^2 can coexist with an arc or a funnel; only the plots reveal them.

Outliers and leverage values

Squaring makes large deviations count heavily – single unusual points can therefore have a strong influence on the line. Two kinds of unusual points:

Outlier

A point with an unusually large residual $|\hat{u}_i|$ – unusual in y (lamp: the store at 95 euros).

High-leverage point

A point with an unusual x-value, far from $\bar{x}$ – it can pull the line towards itself.

Leverage value h_i

The leverage value measures how **unusual** the x-value of observation i is – and thus how strongly this point can pull the line towards itself:

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^{n}(x_j - \bar{x})^2}, \qquad \frac{1}{n} \leq h_i \leq 1.$$

In words: The further x_i lies from the mean $\bar{x}$, the larger h_i becomes (a long "lever arm").

Leverage values – rule of thumb

Rule of thumb for points with high leverage

Simple regression Values with $h_i > \frac{4}{n}$ are **flagged** as high-leverage points.

Small samples Higher threshold $h_i > \frac{6}{n}$.

(In multiple regression with p explanatory variables: $h_i > 2(p+1)/n$ – see A.7.)

Caution

A flagged point is a reason to **inspect** it (measurement error? special case?) and to compare the fit with and without it – not to delete it automatically. Every removal changes the estimate and must be justified *substantively*.

Leverage values – lamp example

$n = 9$, $\bar{x} = 100$, $S_{xx} = 1500$:

x_i	80	85	90	95	100	105	110	115	120
h_i	0.378	0.261	0.178	0.128	0.111	0.128	0.178	0.261	0.378

Calculation for the first store

$h_1 = \frac{1}{9} + \frac{(80-100)^2}{1500} = 0.1111 + 0.2667 = 0.378$

Assessment

Threshold $4/n = 0.444$: no store has a high leverage – the prices are spread evenly; the extreme prices 80 and 120 have the largest (but still harmless) influence.

In words: The outlier at 95 euros has a small leverage (0.128): without it the slope only moves from -0.447 to -0.459. Without the store at 80 euros (largest leverage, small residual) it moves to -0.424.

Leverage values with Python

From the fitted model (lamp)

```
reg = smf.ols("quantity ~ price", data=df).fit()
h = reg.get_influence().hat_matrix_diag   # leverage value h_i of every store
h.round(3)             # 0.378 0.261 0.178 0.128 0.111 0.128 0.178 0.261 0.378
n = len(df); 4 / n     # threshold 0.444
(h > 4 / n).sum()      # 0: no store is flagged
```

By hand – and the fit without one store

```
x = df["price"]
1 / n + (x - x.mean())**2 / ((x - x.mean())**2).sum()      # the same h_i
smf.ols("quantity ~ price", data=df[df["price"] != 95]).fit().params["price"]
# -0.4590: slope without the store at 95 EUR (with it: -0.4467)
```

▶ **Try it:** notebook, section A.4 (Colab)

Exercise – online shop (C): a month with a big campaign

Problem statement

The online shop records three more months. In month 8 it runs a big campaign (advertising x_i in 1000 EUR):

Month i	1	2	3	4	5	6	7	8
Advertising x_i	1	2	3	4	5	6	7	20

1. Compute $\bar{x}$ and $S_{xx} = \sum (x_i - \bar{x})^2$.
2. Compute the leverage values h_1, h_6 and h_8.
3. Which of these months are flagged by the rule of thumb ($4/n$; small samples $6/n$)?
4. Why are the revenues y_i not needed for this? What should the shop do with a flagged month?

Solution – online shop (C): a month with a big campaign

(1) Mean and sum of squares

$\bar{x} = 48/8 = \mathbf{6}$; $x_i - \bar{x}$: $-5, -4, -3, -2, -1, 0, 1, 14$

$S_{xx} = 25 + 16 + 9 + 4 + 1 + 0 + 1 + 196 = \mathbf{252}$

(2) Rule: $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$

$h_1 = \frac{1}{8} + \frac{25}{252} = 0.125 + 0.0992 = \mathbf{0.2242}$

$h_6 = \frac{1}{8} + \frac{0}{252} = \mathbf{0.125}$ (the minimum $1/n$: $x_6 = \bar{x}$)

$h_8 = \frac{1}{8} + \frac{196}{252} = 0.125 + 0.7778 = \mathbf{0.9028}$

(3) Rule of thumb

$4/n = 0.5$ and $6/n = 0.75$: only **month 8** is flagged (0.9028 exceeds both); months 1 and 6 are harmless.

(4) Interpretation

h_i depends only on the *position of* x_i – the length of the lever arm. Whether month 8 really pulls the line depends on its y_8. Do not delete it: check the campaign month and compare the fit with and without it.

Is the error term normally distributed? The QQ-plot

Starting point

The fourth standard assumption is $u_i \sim N(0, \sigma^2)$. It is needed for **confidence intervals and tests** (Section "Inference") – exactly as in Part 3 for the mean. For computing and interpreting the OLS line it plays no role.

Guiding question

How can we check whether data – here the residuals $\hat{u}_i$ – could come from a normal distribution?

A histogram?

With $n = 9$ or $n = 15$ values the picture depends heavily on the classes: hardly any verdict is possible.

Better: compare quantiles

The **QQ-plot** (quantile-quantile plot) compares the sorted values with those expected under a normal distribution.

The idea: plot quantiles against each other

Construction

1. Determine the same number of quantiles of both distributions (for samples: the **sorted values**).
2. Plot the i-th quantile of one distribution against the i-th quantile of the other as a point.

Reading

Points on a **straight line**: both distributions have the same *shape*. Different location and spread only shift and tilt the line.

Reference line

The line through the pair of 25 % **quantiles** and the pair of 75 % **quantiles**: robust – outliers hardly move it.

Key takeaway

The further the points deviate *systematically* from the line, the more dissimilar the shapes of the two distributions.

Warm-up: prices and quantities of the lamp

Data of the regression example

Price x_i (EUR)	80	85	90	95	100	105	110	115	120
Quantity y_i	20	17	15	9	11	8	4	4	2

Step 1: sort each variable *on its own*

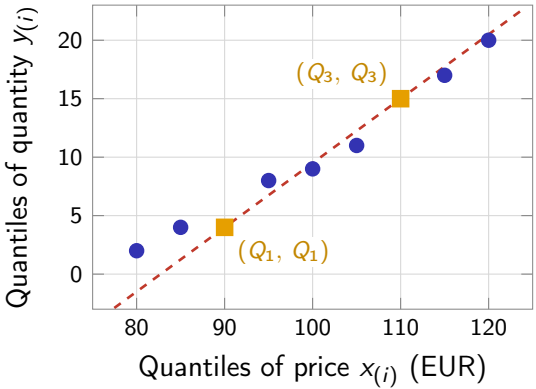
$x_{(i)}$	80	85	90	95	100	105	110	115	120
$y_{(i)}$	2	4	4	8	9	11	15	17	20

Step 2: plot the pairs $\left(x_{(i)}, y_{(i)}\right)$.

Careful: the pairs are broken up

(80, 2) is not an observation! Forbidden in the scatter plot, intended in the QQ-plot: only the *distributions* are compared, not the relationship.

QQ-plot: prices against quantities



Reference line

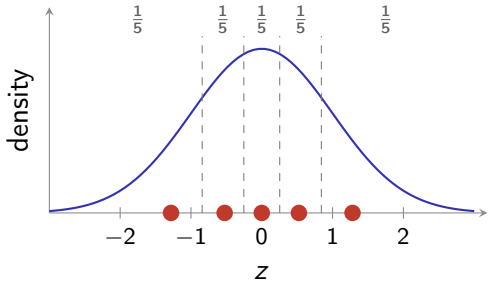
Quartiles: price 90 and 110 euros, quantity 4 and 15. Line through (90, 4) and (110, 15):

$$y = -45.5 + 0.55\,x$$

Reading

The points lie close to the line: prices and quantities are **similarly distributed** – both spread fairly evenly over their range. Location and spread differ (euros vs. units), which only changes the position of the line.

Residuals against the normal distribution: theoretical quantiles



$n = 5$: five strips with probability $\frac{1}{5}$ each; red: z_i in the middle of each strip

Procedure

1. Sort the residuals: $\hat{u}_{(1)} \leq \ldots \leq \hat{u}_{(n)}$.
2. Assign each rank i the **plotting position** $p_i = \frac{i-0.5}{n}$.
3. Theoretical quantile of the standard normal distribution: $z_i = \Phi^{-1}(p_i)$, i.e. $P(Z \leq z_i) = p_i$.
4. Plot the points $\left(z_i,\ \hat{u}_{(i)}\right)$.

In words: z_i is where the i-th smallest of n values "should" lie under a normal distribution. Example $n = 5$: $z_i = -1.28,\ -0.52,\ 0,\ 0.52,\ 1.28$.

Lamp example: computing the quantiles

Rank i	$\hat{u}_{(i)}$	Price	$p_i = \frac{i-0.5}{9}$	$z_i = \Phi^{-1}(p_i)$
1	−3.2333	95	0.0556	−1.5932
2	−1.5333	110	0.1667	−0.9674
3	0.2333	105	0.2778	−0.5895
4	0.3000	85	0.3889	−0.2822
5	0.5333	90	0.5000	0.0000
6	0.7000	115	0.6111	0.2822
7	0.9333	120	0.7222	0.5895
8	1.0000	100	0.8333	0.9674
9	1.0667	80	0.9444	1.5932

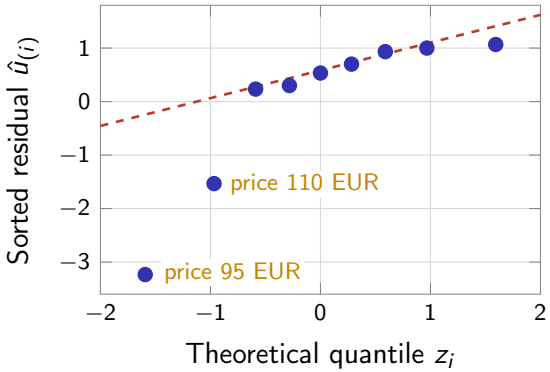
Residuals

from $\hat{y} = 54.67 - 0.4467\,x$, e.g. price 95: $\hat{u} = 9 - 12.23 = -3.23$.

Symmetry

The z_i are symmetric around 0: $z_9 = -z_1$, $z_8 = -z_2$, ... They come from the table of the standard normal distribution (Part 3) or from Python: `stats.norm.ppf(p)`.

Lamp example: QQ-plot of the residuals



Reference line through the quartiles: $0.5833 + 0.5189\,z$

Observation

- **In the middle** the points lie close to the line.
- **At the lower end** the two smallest residuals (95 and 110 euros) lie clearly below it: the lower tail is longer than a normal distribution would suggest.

Conclusion

A hint at a deviation from normality (long lower tail, left-skewed). But $n = 9$ is very few data – **the verdict is uncertain**; tests and confidence intervals for this example should be read with caution.

Exercise – a QQ-plot by hand

Problem statement

A company records the age x (years) and the annual maintenance costs y (in 100 EUR) of $n = 5$ machines:

Age x_i	1	2	3	4	5
Maintenance costs y_i	17	21	25	29	43

Given: $\bar{x} = 3$, $\bar{y} = 27$, $S_{xx} = 10$, $S_{xy} = 60$.

1. Compute $\hat{\beta}_1$, $\hat{\beta}_0$ and all residuals; check their sum.
2. Sort the residuals and compute the plotting positions $p_i = \frac{i-0.5}{n}$ and the normal quantiles $z_i = \Phi^{-1}(p_i)$.
3. Draw the QQ-plot and judge it.
4. Which Python command draws the QQ-plot of the residuals of a fitted model `reg`?

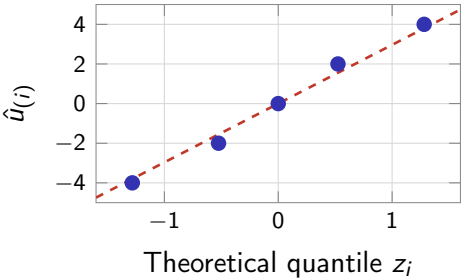
Solution – a QQ-plot by hand

(1) $\hat\beta_1 = 60/10 = \mathbf{6}$, $\hat\beta_0 = 27 - 6\cdot 3 = \mathbf{9}$, $\hat y = 9 + 6x$:

$\hat y_i$	15	21	27	33	39	
$\hat u_i$	2	0	-2	-4	4	(sum 0 ✓)

(2) Sorted residuals, positions and quantiles:

i	1	2	3	4	5
$\hat u_{(i)}$	-4	-2	0	2	4
p_i	0.1	0.3	0.5	0.7	0.9
z_i	-1.28	-0.52	0	0.52	1.28



(3) The points lie roughly on the line; with $n = 5$ no deviation from normality can be detected – and none can be ruled out.

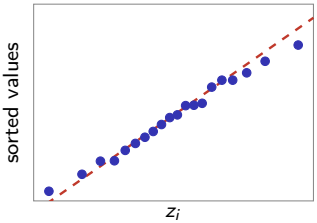
(4) `sm.qqplot(reg.resid, line="q", a=0.5)`

Note: the residual plot $(2, 0, -2, -4, 4$ over $\hat y)$ shows an **arc** – costs rise faster for old machines. A QQ-plot does not check linearity.

▶ **Try it:** notebook, section A.5 (Colab)

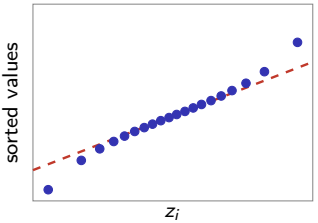
Typical deviations in the QQ-plot

(a) Normal



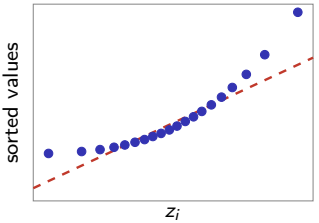
Points close to the line, small random deviations

(b) Heavy tails



S-shape: left end below, right end above the line – more extreme values (outliers)

(c) Right-skewed



Arc: both ends above the line – long upper tail

Key takeaway

Small deviations, especially at the ends, are normal for small n. Take *systematic* shapes such as an S or an arc seriously.

QQ-plots with Python

Two samples: prices against quantities

```
x, y = np.sort(df["price"]), np.sort(df["quantity"])   # empirical quantiles
qx, qy = np.percentile(x, [25, 75]), np.percentile(y, [25, 75])
plt.scatter(x, y); plt.axline((qx[0], qy[0]), (qx[1], qy[1]), ls="--")  # slope 0.55
```

Residuals against the normal distribution – and the table by hand

```
import statsmodels.api as sm; from scipy import stats
sm.qqplot(reg.resid, line="q", a=0.5)    # QQ-plot with line through quartiles
u = np.sort(reg.resid)                   # sorted residuals
p = (np.arange(1, 10) - 0.5) / 9         # plotting positions (i - 0.5)/n
z = stats.norm.ppf(p)                    # -1.5932, -0.9674, ..., 1.5932
```

`sm.qqplot()` computes $p_i = (i - 0.5)/n$ (`a=0.5`; default $i/(n+1)$) and $z_i = \Phi^{-1}(p_i)$ and plots $(z_i, \hat{u}_{(i)})$; `line="q"`: reference line through the (interpolated) quartiles.

▶ **Try it:** notebook, section A.5 (Colab)

Exercise – body heights and shoe sizes: residual plot and QQ-plot

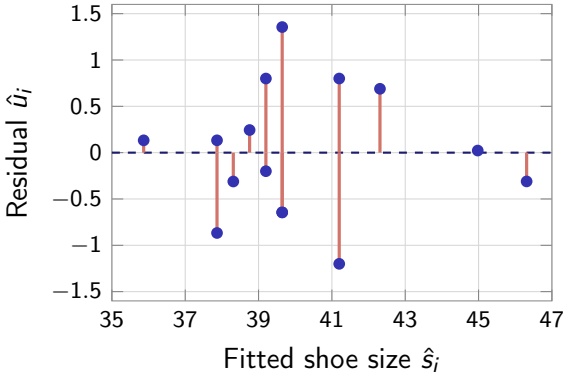
Problem statement

Body heights k_i (cm) and shoe sizes s_i of $n = 15$ persons (data of the exercise at the end of Section 3), OLS line $\hat{s} = 2.3147 + 0.2222\,k$.

1. Compute the fitted values $\hat{s}_i$ and the residuals $\hat{u}_i = s_i - \hat{s}_i$; draw the residual plot and judge it.
2. Sort the residuals; compute the plotting positions $p_i = (i - 0.5)/n$ and the normal quantiles $z_i = \Phi^{-1}(p_i)$.
3. Draw the QQ-plot of the residuals and judge it.

Person i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Body height k_i (cm)	164	162	168	198	166	166	192	160	151	175	180	160	168	175	168
Shoe size s_i	39	38	39	46	40	39	45	37	36	42	43	38	41	40	39

Solution (1/3) – body heights: residual plot



Residuals $\hat{u}_i = s_i - \hat{s}_i$

Between -1.20 (person 14) and 1.36 (person 13) – at most about one shoe size. Example person 13: $\hat{s}_{13} = 39.64$, $\hat{u}_{13} = 41 - 39.64 = 1.36$.

Assessment

No arc, no funnel, no conspicuous single value: **no systematic structure** ✓ – the line is adequate.

Equal body heights $\Rightarrow$ equal $\hat{s}_i$: these points lie vertically above each other.

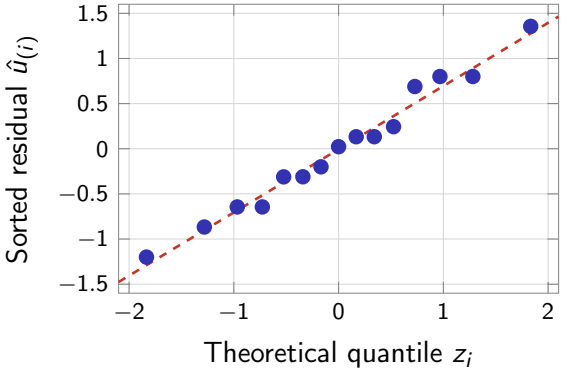
Solution (2/3) – body heights: sorted residuals and normal quantiles

i	$\hat{u}_{(i)}$	$p_i = \frac{i-0.5}{15}$	z_i
1	−1.1999	0.0333	−1.8339
2	−0.8669	0.1000	−1.2816
3	−0.6445	0.1667	−0.9674
4	−0.6445	0.2333	−0.7279
5	−0.3113	0.3000	−0.5244
6	−0.3105	0.3667	−0.3407
7	−0.2001	0.4333	−0.1679
8	0.0227	0.5000	0.0000

i	$\hat{u}_{(i)}$	p_i	z_i
9	0.1329	0.5667	0.1679
10	0.1331	0.6333	0.3407
11	0.2443	0.7000	0.5244
12	0.6891	0.7667	0.7279
13	0.7999	0.8333	0.9674
14	0.8001	0.9000	1.2816
15	1.3555	0.9667	1.8339

Smallest residual: person 14 (175 cm, size 40); largest: person 13 (168 cm, size 41).

Solution (3/3) – body heights: QQ-plot of the residuals



Reference line through the quartiles: $-0.0056 + 0.7002\,z$
(quartiles -0.478, 0.467 interpolated, as in `sm.qqplot`)

Observation

All 15 points lie close to the line; neither an S-shape nor an arc. Small steps arise because shoe sizes are whole numbers.

Conclusion

No indication against normally distributed errors – not a proof, since $n = 15$ only reveals clear deviations. This model is a good candidate for inference (next section).

Can you answer these questions?

1. Why do we check assumptions about u_i with the residuals $\hat{u}_i$?
2. How does the residual plot arise from the scatter plot?
3. Why do the residuals always lie around the zero line?
4. Which assumption is violated by an arc, a funnel, waves?
5. Outlier vs. high-leverage point – and what do you do with a conspicuous single value?
6. How are the z_i of a QQ-plot computed; what does an S-shape mean?

Key takeaway

First compute, then check: the estimates can always be computed, but standard errors, tests and confidence intervals (next section) are only trustworthy if the residuals show no pattern and look roughly normal.

Residual analysis at a glance

Residuals $\hat{u}_i = y_i - \hat{y}_i$ (always $\sum \hat{u}_i = 0$)

- **Residual plot** ($\hat{y}_i$ vs. $\hat{u}_i$): structureless band $\rightarrow$ line adequate; arc $\rightarrow$ wrong functional form; funnel $\rightarrow$ heteroscedasticity; waves (time order) $\rightarrow$ autocorrelation.
- **QQ-plot** ($z_i = \Phi^{-1}\left(\frac{i-0.5}{n}\right)$ vs. $\hat{u}_{(i)}$): points on the line $\rightarrow$ normality plausible; S-shape $\rightarrow$ heavy tails; arc $\rightarrow$ skewness.

Influential points

Outlier: large $|\hat{u}_i|$. Leverage $h_i > 4/n$ (simple regression): unusual x. Inspect, compare the fit with and without the point – do not delete it blindly.

When the effect of one variable depends on another

Interaction model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + u_i$$

The effect of x_1 is now $\beta_1 + \beta_3 x_2$ – it depends on x_2.

Advertising: `sales ~ TV * radio`

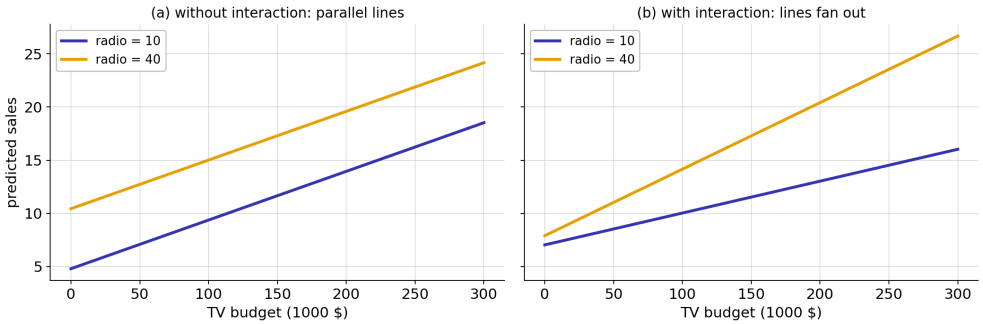
$$\widehat{\texttt{sales}} = 6.750 + 0.0191\,\texttt{TV} + 0.0289\,\texttt{radio} + 0.00109\,\texttt{TV} \cdot \texttt{radio}$$

Interaction: $T = 20.7$, $p < 0.0001$; R^2 rises from 0.897 to 0.968.

Hierarchy principle

If $x_1 x_2$ is in the model, keep x_1 and x_2 as well – even if their own t-tests are not significant.

Interaction, drawn: synergy between TV and radio



- Effect of 1000 \$ more TV budget: $0.0191 + 0.00109 \cdot \texttt{radio}$, i.e. 0.030 at radio $= 10$, but 0.063 at radio $= 40$.
- Parallel lines $=$ no interaction; lines that fan out $=$ synergy.

Exercise – interactions and the hierarchy principle

Problem statement

The interaction model for the advertising data gives

$$\widehat{\texttt{sales}} = 6.750 + 0.0191\,\texttt{TV} + 0.0289\,\texttt{radio} + 0.00109\,\texttt{TV}\cdot\texttt{radio}.$$

1. Interpret the interaction coefficient 0.00109 in words.
2. Compute the effect of 1000 \$ more TV budget when the radio budget is 0 and when it is 50 (in 1000 \$).
3. Suppose the TV main effect were not significant, but the interaction were. Should TV be dropped from the model?

Solution – interactions and the hierarchy principle

(1) The effect of TV on sales is not constant: it grows with the radio budget (and vice versa). Each additional unit of radio budget raises the TV slope by 0.00109 – a **synergy** between the media.

(2) Effect of TV: $0.0191 + 0.00109 \cdot \texttt{radio}$

$$\texttt{radio} = 0: \ 0.0191, \qquad \texttt{radio} = 50: \ 0.0191 + 0.00109 \cdot 50 = 0.0736$$

With a radio budget of 50, TV is almost four times as effective.

(3) No – by the **hierarchy principle**, the main effects stay in the model whenever their interaction does.

Common mistake

Quoting 0.0191 as "the" TV effect. With an interaction, it is only the TV effect at $\texttt{radio} = 0$.

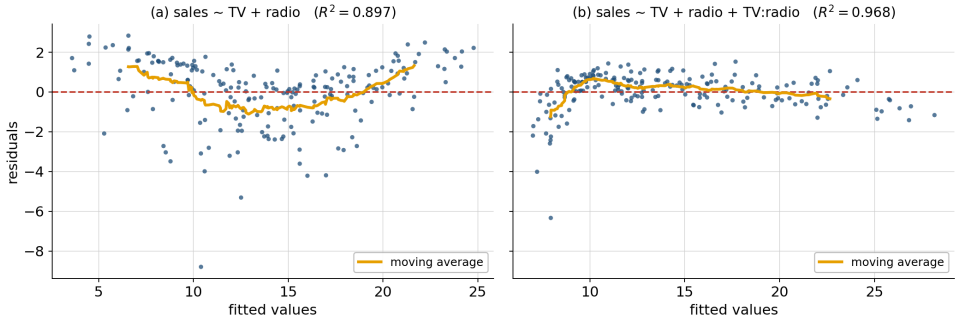
The classical problems – and how to spot them

Problem	Symptom	Tool
Non-linearity / missing term	curve in the residuals	residual plot
Heteroscedasticity	funnel in the residuals	residual plot
Non-normal errors	points off the line	QQ-plot
Outliers, high leverage	single points dominate	h_i, residuals
Collinearity	large SEs, unstable signs	correlations, VIF

Key takeaway

The residual plot and the QQ-plot of the simple regression work unchanged in multiple regression – plot the residuals against the fitted values $\hat{y}_i$.

Residual plot – advertising



- (a) The additive model leaves a **U-shaped** pattern: it over-predicts when one medium dominates and under-predicts when the budget is split.
- (b) With the interaction term the pattern largely disappears – the residuals *told* us that a term was missing.

Outliers and leverage in multiple regression

Recall A.4: an *outlier* is unusual in y (large residual $|\hat{u}_i|$), a *high-leverage point* is unusual in x.

What is new with several explanatory variables

"Unusual in x" now means an unusual **combination** of the explanatory variables (e.g. a young customer with a very high credit limit). Such a point can be invisible in the plots of y against one variable at a time – the leverage h_i detects it anyway.

Rule of thumb for the leverage

The leverages h_i average $(p+1)/n$; flag points with $h_i > 2(p+1)/n$ (simple regression: $4/n$).

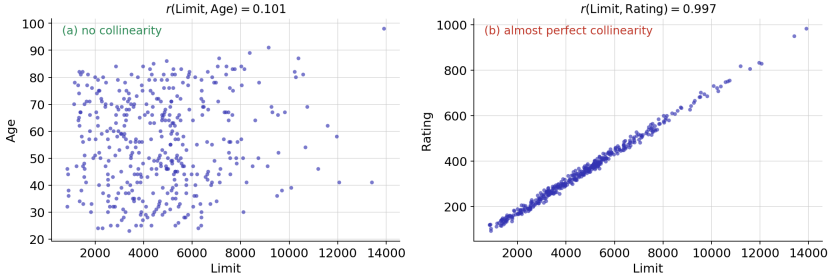
Worst case

A point that is *both* unusual in x and badly fitted in y can dominate the fit – inspect it, compare the fit with and without it.

Collinearity

Definition

Two or more explanatory variables are (almost) linearly dependent – they carry nearly the same information.



In words: When two variables move in lockstep (right), the data cannot separate their individual effects – holding one fixed while the other changes almost never happens.

Collinearity – consequences in the credit data

Model for `Balance`	$\hat{\beta}_{\text{Limit}}$	$\text{SE}(\hat{\beta}_{\text{Limit}})$	T	p
`Age + Limit`	0.1734	0.00503	34.5	< 0.0001
`Age + Limit + Rating`	0.0190	0.06296	0.30	0.76

Symptoms

Adding `Rating` (correlation 0.997 with `Limit`) multiplies the standard error of `Limit` by about 12.5 – the clearly relevant variable suddenly looks "insignificant".

In words: Collinearity does not spoil the *predictions* of the model; it spoils the *attribution* of the effect to individual variables.

Variance inflation factor (VIF)

Definition

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is the R^2 of a regression of x_j on all *other* explanatory variables.

- $\text{VIF}_j = 1$: no collinearity; $\text{VIF}_j > 5$ (some authors: 10): problematic.
- $\sqrt{\text{VIF}_j}$ is the factor by which collinearity inflates $\text{SE}(\hat{\beta}_j)$.
- Remedies: drop one of the redundant variables or combine them into one index.

Credit data (`Age`, `Limit`, `Rating`)

$\text{VIF}_{\text{Age}} = 1.01$, $\text{VIF}_{\text{Limit}} = 160.6$, $\text{VIF}_{\text{Rating}} = 160.7$ ($\sqrt{160.6} = 12.7$ – close to the observed SE ratio $0.06296/0.00503 = 12.5$).

Exercise – a diagnostics case study

Problem statement

House prices are regressed on `size`, `bedrooms`, `age` and `income` of the area ($n = 400$, $p = 4$). The diagnostics show:

- **Residual plot:** a clear U-shaped curve, and the spread of the residuals widens from left to right.
- **Leverage:** one point has $h_i = 0.30$.
- **VIF:** size 8.9, bedrooms 7.5, age 1.3, income 9.2.

Name every problem and suggest a remedy for each.

Solution – a diagnostics case study

1. **U-shape – non-linearity:** the linear model misses curvature. *Remedy:* add e.g. $\texttt{size}^2$ or transform a variable ($\log \texttt{size}$).
2. **Widening spread – heteroscedasticity:** standard errors, t- and F-tests are unreliable. *Remedy:* transform the response, e.g. $\log(\texttt{price})$.
3. **Leverage:** threshold $2(p+1)/n = 10/400 = 0.025$; $h_i = 0.30$ is twelve times as large. *Remedy:* inspect the point, compare the fit with and without it.
4. **Collinearity:** VIFs of 7.5–9.2 for size, bedrooms and income. *Remedy:* drop or combine redundant variables (e.g. keep size, drop bedrooms) and recompute the VIFs.

Common mistake

Deleting flagged points automatically. Diagnostics call for *inspection* and sensitivity checks – not for blind removal.

Appendix – Python commands

Interactions and diagnostics

```
mi = smf.ols("sales ~ TV * radio", data=ad).fit()    # TV + radio + TV:radio
print(mi.summary())
plt.scatter(m2.fittedvalues, m2.resid); plt.axhline(0)    # curvature?
h = m2.get_influence().hat_matrix_diag       # leverages h_i
(h > 2 * (2 + 1) / 200).sum()                # high leverage (p = 2): 2 points
```

Variance inflation factor

```
1 / (1 - smf.ols("Limit ~ Age + Rating", data=cr).fit().rsquared)  # 160.6
from statsmodels.stats.outliers_influence import variance_inflation_factor
X = sm.add_constant(cr[["Age", "Limit", "Rating"]])
[variance_inflation_factor(X.values, j) for j in (1, 2, 3)]  # 1.01 160.6 160.7
```

▶ **Try it:** notebook, section A.7 (Colab)

Appendix summary

- **Derivation**: chain rule, normal equations, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\hat{\beta}_1 = S_{xy}/S_{xx}$; a minimum, as $4n\,S_{xx} > 0$.
- **Least squares**: any other line has $S = \mathrm{RSS} + n(a + b\bar{x})^2 + b^2 S_{xx}$ – the OLS line is the unique minimum.
- **Interactions** $x_1 x_2$: the effect of x_1 is $\beta_1 + \beta_3 x_2$; keep the main effects (hierarchy principle).
- **Leverage** in multiple regression: flag $h_i > 2(p+1)/n$ – inspect, do not delete blindly.
- **Collinearity**: $\mathrm{VIF}_j = 1/(1 - R_j^2) > 5$ inflates $\mathrm{SE}(\hat{\beta}_j)$ by $\sqrt{\mathrm{VIF}_j}$; predictions stay fine, the attribution to single variables does not.